Highlights

A Comprehensive Catalog of Data-Driven Process Indicators for Performance Analysis in Process Mining

- A comprehensive, formal catalog of data-driven process performance indicators.
- Formal foundations ensure unambiguous definitions and catalog extensibility.
- Indicators span several performance dimensions and process granularities.
- The catalog is implemented through a Python library to facilitate adoption.
- Validity is demonstrated by using the library on simulated and real-life event logs.

A Comprehensive Catalog of Data-Driven Process Indicators for Performance Analysis in Process Mining

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Abstract

Measuring process performance is essential for organizations seeking to understand, evaluate, and improve their operations. Process mining enables the analysis of real processes based on data extracted from information systems in the form of event logs. From these data, performance indicators can be automatically derived to assess different dimensions of process performance, such as time, cost, quality, and flexibility. However, existing literature lacks a comprehensive and extensible catalog of process performance indicators that are formally defined, data-driven, comparable, and applicable across different organizational processes. This work addresses this gap by proposing a formalized catalog of generic process performance indicators derived from event logs. The catalog organizes indicators according to different performance dimensions and supports multiple levels of granularity, including activity instances, individual process executions, and groups of such executions. For each indicator, the catalog provides a natural language description, a non-ambiguous formal definition, the associated performance dimension, the applicable granularity level, the event log attributes required for its computation, the additional parameters needed, potential use, and whether the indicator should be minimized or maximized. To facilitate practical adoption, a publicly available Python library was developed to automate the computation of the indicators in accordance with the proposed formal definitions. The validity of the approach is demonstrated through the application of the library to both simulated and publicly available real-life event logs. Taken together, the proposed catalog and tool aim to promote a consistent, reproducible, and extensible use of performance indicators in process analysis, in both academic and organizational contexts.

Keywords: Process mining, Process performance analysis, Process performance indicator, Multidimensional performance analysis

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Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

1. Introduction

Measuring process performance is essential for organizations seeking to understand, evaluate, and improve their operations (Dumas et al., 2013). Process Mining (PM) is a discipline that enables the discovery, monitoring, and analysis of real processes by extracting knowledge from event data recorded in information systems (Van Der Aalst and Carmona, 2022). The processes uncovered through PM are commonly analyzed from two complementary perspectives (Van Der Aalst and Carmona, 2022): the control-flow perspective, which focuses on how activities are ordered and executed, and the performance perspective, which focuses on how well processes perform. Performance analysis is typically conducted through the use of Process Performance Indicators (PPIs), which quantify relevant aspects of process behavior (Estrada-Torres et al., 2023). Leveraging the data contained in event logs, PPIs can be computed automatically (Dumas et al., 2013) to assess multiple performance dimensions, including time, cost, quality, and flexibility (Dumas et al., 2013). Table 1 illustrates an example of a single case from an event log of the serving dishes process in a restaurant.

Case ID	Instance ID	Activity	Lifecycle Type	Timestamp	Resource	Cost	Rating
38	3801	Receive order	complete	01-10-2026 12:37:17	Rose	1 USD	
38	3802	Prepare dish	start	01-10-2026 12:39:43	Peter		
38	3803	Prepare dish	start	01-10-2026 12:41:12	Peter		
38	3803	Prepare dish	complete	01-10-2026 12:57:23	Peter	14 USD	
38	3802	Prepare dish	complete	01-10-2026 13:01:54	Peter	13 USD	
38	3804	Serve order	complete	01-10-2026 13:12:11	Anna	4 USD	5

Table 1: Example of a case from an event log of the serving dishes process in a restaurant.

Processes constitute fundamental building blocks of organizations, as they directly contribute to value creation for both the organization and its customers (Dumas et al., 2013). Consequently, measuring process performance through well-defined PPIs is essential for identifying inefficiencies, monitoring compliance with performance objectives, and supporting continuous improvement initiatives (Popova and Sharpanskykh, 2010; Kis et al., 2017; del Río-Ortega et al., 2018). In practical settings, process analysts are often faced with the challenge of selecting appropriate indicators from a fragmented body of literature, where terminology and definitions vary (Van Looy and Shafagatova, 2016), calculation requirements are not provided explicitly (Milani and Maggi, 2018), and calculation methods are not always formalized (del Río-Ortega et al., 2013). This lack of consistency hinders comparability between studies and limits the reproducibility and reuse of performance analyses.

Ideally, analysts and researchers would have access to a comprehensive and extensible repository of PPIs that (i) provides unambiguous, formally defined indicators, (ii) specifies the data requirements needed for their computation, and (iii) supports analysis at different levels of granularity, such as activity instances, individual process executions, groups of executions, or complete event logs. While prior work has partially addressed these needs by compiling lists of PPIs without formal definitions (Gunasekaran and Kobu, 2007; Muurinen, 2020; van den Ingh, 2016; Van Looy and

Shafagatova, 2016), proposing formalizations tailored to specific domains or contexts (Cho et al., 2020), or focusing on a subset of performance dimensions (Khlif et al., 2019), to the best of the authors' knowledge, existing literature lacks a comprehensive and extensible catalog of process performance indicators that are formally defined, data-driven, comparable, and applicable across different organizational processes.

To address this gap, this work proposes a formalized catalog of generic, data-driven PPIs that can be computed directly from event logs. The catalog is grounded in existing literature since it is compiled based on prior systematic reviews that identify performance indicators used in process mining research (Velásquez et al., 2024; Van Looy and Shafagatova, 2016). Indicators are organized according to performance dimensions and defined at multiple levels of granularity. For each indicator, the catalog provides a natural language description, a precise formal definition, the associated performance dimension, the applicable granularity level, the event log attributes required for computation, any additional parameters that are needed, potential use, and the intended optimization direction (minimization or maximization). Formal definitions of all underlying elements are also provided, ensuring the consistency and extensibility of the catalog.

To support practical adoption in both academic and industrial settings, a publicly available Python library has been developed to automate the computation of the proposed PPIs in accordance with their formal definitions. The applicability and correctness of the approach are validated through the application of the library to both simulated and publicly available real-life event logs. Together, the proposed catalog and library aim to streamline process performance analysis, promote consistency in the definition and use of PPIs, and facilitate reproducible and extensible performance measurement in process mining.

The remainder of this work is structured as follows. Section 2 summarizes existing research on the compilation of PPIs. Section 3 elaborates on the procedure taken for compiling the PPIs cataloged in this work. Section 4 provides the theoretical basis for the calculation of the PPIs, whereas Section 5 elaborates on the catalog and its contents. Section 6 presents the developed Python library and its corresponding validation. Finally, Section 7 provides the conclusions of this work.

2. Related Work

Several studies emphasize the importance of measuring process performance within organizations. For instance, (Popova and Sharpanskykh, 2010) argues that measuring and analyzing organizational performance is important for translating organizational goals into concrete outcomes. Similarly, (Kis et al., 2017) highlights performance measurement as a key component of Business Process Management (BPM) initiatives, which is closely linked to organizational success. Furthermore, (del Río-Ortega et al., 2018) points out that the rapid evolution of technological and social environments has intensified the need to measure organizational performance from multiple perspectives, thus requiring the definition of appropriate PPIs. While these studies underline the relevance of process performance measurement, they do not focus on the explicit definition of PPIs nor on the formalization of the elements required for their computation.

An exception to the above is (Hompes, 2025), which acknowledges the need for a formal definition of process performance. However, its focus primarily lies in the time dimension, without addressing other performance dimensions such as cost, quality, or flexibility. As a result, its applicability is limited when aiming to conduct multidimensional performance analyses.

Several publications compile collections of PPIs, often through literature reviews. For example, (Gunasekaran and Kobu, 2007) identifies PPIs within the logistics and supply chain management domain, whereas (Muurinen, 2020) focuses on PPIs related to the order-to-cash process. A more general compilation is presented in (Khlif et al., 2019), which proposes a framework for evaluating business process performance without restricting the analysis to a specific application context. Although this work provides computation formulas for several PPIs, they are not formally defined, and the analysis is limited to the time and cost dimensions.

Two publications that explicitly consider multiple performance dimensions are (van den Ingh, 2016) and (Van Looy and Shafagatova, 2016). The work presented in (van den Ingh, 2016) adopts a PM approach to evaluate business process performance according to the four dimensions of the Devil's Quadrangle (DQ). The DQ is a framework that highlights inherent trade-offs among four process performance dimensions: time, cost, quality, and flexibility (Dumas et al., 2013), particularly in the context of process redesign (Dumas et al., 2013; Brand and Van der Kolk, 1995). In (van den Ingh, 2016), several PPIs are described in the context of their calculation within Celonis, a commercial process mining tool (Ullrich et al., 2021). However, these PPIs are not presented in formal mathematical definitions, which creates ambiguity about their precise computation and interpretation.

The structured literature review presented in (Van Looy and Shafagatova, 2016) groups PPIs according to multiple performance perspectives. These include coarse-grained perspectives such as financial, customer, supplier, societal, digital innovation, and employee performance, as well as fine-grained, process-oriented perspectives aligned with the DQ dimensions: general process performance, time-related performance, cost-related performance, internal quality, and flexibility. While this work identifies a broad set of PPIs and provides operationalizations when available in the original works, these operationalizations are typically expressed as textual descriptions or simple formulas, which limit reproducibility and hinder cross-study comparison.

A more formal approach to defining PPIs is presented in (Cho et al., 2020), which proposes formally defined indicators at the process instance level for the four DQ dimensions. These PPIs are applied to measure the performance of emergency room processes in the healthcare domain. Although this work provides precise definitions, its domain-specific scope restricts its applicability to other organizational contexts.

In contrast to the aforementioned studies, the catalog proposed in this work aims to provide a comprehensive repository of generic, data-driven PPIs applicable across diverse domains and organizational processes. The indicators in the catalog have been derived from existing literature reviews that identify publications in which PPIs are proposed (Velásquez et al., 2024; Van Looy and Shafagatova, 2016). To ensure extensibility and avoid ambiguity, the catalog follows a formal mathematical approach, supported by an explicit definition of the theoretical basis required for the consistent specification and computation of PPIs.

3. Methodology

This work builds upon existing literature reviews to identify and compile the PPIs included in the proposed catalog. The first source is a structured literature review that reports a collection of indicators for measuring business process performance (Van Looy and Shafagatova, 2016). In that review, indicators are grouped according to the Balanced Scorecard (BSC) model (Norton and Kaplan, 2018), which distinguishes between financial performance, customer-related performance, internal business process performance, and learning and growth performance. As the present work focuses specifically on process performance, only the indicators belonging to the internal business process performance perspective were considered. This perspective is further subdivided in (Van Looy and Shafagatova, 2016) into the four dimensions of the Devil's Quadrangle (DQ). Although operationalizations are provided for several indicators, these are not formally defined and are reported as presented in the original sources, without standardization across publications.

The second source is a more recent systematic literature review that examined publications addressing costs and other performance dimensions within process-oriented disciplines (Velásquez et al., 2024). While this review does not explicitly provide a consolidated list of PPIs, it identifies a subset of publications in which either PPIs are proposed or the DQ framework is adopted. By analyzing this subset of publications, additional PPIs related to the time, cost, quality, and flexibility dimensions were identified, covering varying granularity levels.

Considering both the PPIs reported in (Van Looy and Shafagatova, 2016) and those identified from the publications highlighted in (Velásquez et al., 2024), a standardization procedure was carried out. This step was necessary because several PPIs were either defined for specific domains or referred to using different terminology in different publications. For example, the terms lead time (e.g., in (Bakx, 2010)) and cycle time (e.g., in (Lehnert et al., 2014)) have both been used to denote the total elapsed time from the start of a process execution to its completion. To address this, the definitions provided by each publication for every identified PPI were analyzed, and generic, standardized names were given to each indicator to consistently represent equivalent definitions. During this process, five additional PPIs not explicitly identified in the literature were incorporated, as they were complementary to existing indicators. For instance, a PPI measuring the elapsed time from the start of a process execution to the first occurrence of a specific activity was included as a complement to a PPI measuring the elapsed time from the first occurrence of that activity to the completion of the process execution (Balaban et al., 2011; Cho et al., 2020). The standardization resulted in a consolidated list of 91 unique PPIs.

The standardized PPIs were then analyzed to determine both the performance dimensions to which they relate and the process granularities at which they can be computed. In addition to the four DQ dimensions, a general category was introduced to include PPIs that are not directly associated with a specific performance dimension, but that either contribute to understanding process behavior or are used for the computation of other PPIs. Some indicators, such as the count of human resources (e.g., in (Berk, 2010) for the cost dimension, in (Cho et al., 2020) for the quality dimension, and in (van den Ingh, 2016) for the flexibility dimension), were associated with more than one dimension. Four granularities were considered:

1. Activity instance, corresponding to the performance of a single instantiation of an activity within a case. A formal definition of an activity instance is provided in Definition 3 in Section 4.
2. Activity, corresponding to the aggregated performance of all instantiations of a given activity within the event log.
3. Case, corresponding to the performance of a single process execution recorded in the event log.
4. Group of cases, which encompasses two types of calculations:
   (a) The performance of all process executions belonging to a specific group of cases.
   (b) The expected performance of a case belonging to a given group of cases.

It should be noted that when a group of cases includes all process executions in the event log, this granularity is equivalent to considering an event log granularity. Applying the above procedure resulted in a final set of 310 PPIs. Table 2 summarizes the number of PPIs defined for each combination of performance dimension and granularity.

	General	Time	Cost	Quality	Flexibility
Activity Instance	0	4	14	4	0
Activity	3	4	14	12	4
Case	5	19	22	23	9
Group of Cases	6	14	23	38	12
Expected Value	5	19	22	25	9

Table 2: PPIs that have been defined for each dimension/granularity pair.

Once the final set of PPIs was established, a formalization procedure was followed to provide precise mathematical definitions for each PPI. However, before defining the PPIs themselves, all relevant elements required for their computation were formally specified. These elements characterize the data used to calculate PPIs and primarily relate to the structure and content of event logs representing process executions. Providing explicit mathematical definitions for these foundational elements ensures the extensibility of the catalog, as any future PPI can be defined consistently using the same theoretical basis. The formal definitions of these elements are presented in Section 4.

Building upon these formal foundations, mathematical definitions are then provided for each PPI. This formalization of PPIs is detailed in Section 5.

4. Definitions

This section presents the formal definitions of the elements characterizing the underlying data used to compute the PPIs. It must be noted that, due to the length of the proofs of the theorems, lemmas, and propositions that are stated over the definitions, these are provided separately from the main body, in Appendix A.

We adopt the notation of sequences, functions and partial functions from (Van Der Aalst and Carmona, 2022): $\sigma = \langle a_1, a_2, \dots a_n \rangle \in X^*$ denotes a sequence over X of length $|\sigma| = n$; $\sigma_i = a_i$ for $1 \leq i \leq |\sigma|$; $f \in X \rightarrow Y$ is a total function, i.e., $f(x) \in Y$ for any $x \in X$; $f \in X \not\rightarrow Y$ is a partial function with domain $dom(f) \subseteq X$, and if $x \notin dom(f)$, then $f(x) = \perp$, i.e., the function is not defined for x.

Definition 1 (Universes, adapted from (Van Der Aalst and Carmona, 2022))**.** *U_{event} is the universe of events, U_{act} is the universe of activities, U_{case} is the universe of cases, U_{time} is the universe of timestamps, $U_{actinst}$ is the universe of activity instances, U_{res} is the universe of resources, U_{role} is the universe of roles, U_{att} is the universe of attributes, U_{val} is the universe of values, and $U_{map} = U_{att} \not\rightarrow U_{val}$ is the universe of attribute-value mappings. $U_{act} \cup U_{case} \cup U_{time} \cup U_{actinst} \cup U_{res} \cup U_{role} \subseteq U_{val}$, $\perp \notin U_{val}$. U_{res} is the union of the (disjoint) universes of human and non-human resources, i.e., $U_{res} = U_{hres} \cup U_{nhres}$ and $U_{hres} \cap U_{nhres} = \varnothing$.*

$f \in U_{map}$ is a function mapping any subset of attributes onto values (Van Der Aalst and Carmona, 2022). For any $f \in U_{map}$: $f(act) \in U_{act} \cup \{\perp\}$, $f(event) \in U_{event} \cup \{\perp\}$, $f(case) \in U_{case} \cup \{\perp\}$, $f(time) \in U_{time} \cup \{\perp\}$, $f(actinst) \in U_{actinst} \cup \{\perp\}$, $f(res) \in U_{res} \cup \{\perp\}$, and $f(role) \in U_{role} \cup \{\perp\}$.

For the calculation of certain PPIs, subsets of U_{act} can be defined: $Autl \subseteq U_{act}$, the set of automatic activities; $Desl \subseteq U_{act}$, the set of desirable activities; $DCl \subseteq U_{act}$, the set of direct cost activities; $Unwl \subseteq U_{act}$, the set of unwanted activities.

Definition 2 (Event log, adapted from (Van Der Aalst and Carmona, 2022) to include cases)**.** *An event log is a tuple $L = (E, C, \#, \prec)$ consisting of a set of events $E \subseteq U_{event}$, a set of cases $C \subseteq U_{case}$, a mapping $\# \in E \cup C \to U_{map}$, and a strict partial ordering $\prec \subseteq E \times E$ on events.*

For any $e \in E$ and $att \in dom(\#(e))$: $\#_{att}(e) = \#(e)(att)$ is the value of attribute att for event e (Van Der Aalst and Carmona, 2022). An event e can have any number of attributes (Van Der Aalst and Carmona, 2022). The attributes $\#_{case}(e) \in U_{case}$, $\#_{act}(e) \in U_{act}$, and $\#_{time}(e) \in U_{time}$, are considered required event attributes due to their use in most of the defined PPI (Van Der Aalst and Carmona, 2022). The set of cases C should equal all cases referred to by events: $C = \{\#_{case}(e) \mid e \in E\}$.

Besides the mandatory attributes, there are a number of desirable and optional event attributes. Desirable event attributes that are required for the calculation of several PPIs are: $\#_{hres}(e)$, the human resource associated with event e; and $\#_{tc}(e)$, the total cost of executing event e.

There are also some optional event attributes used for the calculation of some PPIs: $\#_{role}(e)$, the role of the human resource associated with event e; $\#_{res}(e)$, the non-human resource associated with event e, e.g., machinery or tools; $\#_{unt}(e)$, the outcome units associated with event e, e.g., the produced or sold units; $\#_{uns}(e)$, the outcome units associated with event e that were not successful, e.g., units produced with errors; $\#_{fc}(e)$, the fixed cost of executing event e; $\#_{vc}(e)$, the variable cost of executing event e; $\#_{lc}(e)$, the labor cost of executing event e; and $\#_{ic}(e)$, the inventory cost of executing event e.

The ordering of events can not go against the order imposed by the time attributes, i.e., for all $e, e' \in E$, if $\#_{time}(e) < \#_{time}(e')$ then $e' \not\prec e$ (Van Der Aalst and Carmona, 2022).

Cases, as events, can have attributes. We adopt the same notation used for accessing event attributes for accessing case attributes, i.e., for any $c \in C$ and $att \in dom(\#(c))$: $\#_{att}(c) = \#(c)(att)$ is the value of attribute att for case c.

The following are case attributes that can be used for the calculation of specific PPIs: $\#_{cli}(c)$, the client associated with case c; $\#_{mainc}(c)$, the maintenance cost associated with the execution of case c; $\#_{mdc}(c)$, the missed deadline cost associated with the execution of case c; $\#_{transc}(c)$, the transportation cost associated with the execution of case c; $\#_{warec}(c)$, the warehousing cost associated with the execution of case c; and $\#_{qual}(c)$, a quality score associated with the execution of c, such as the value from a user satisfaction score.

An activity instance is a collection of related events that together represent the execution of an activity for a case (Van Der Aalst and Carmona, 2022).

The attribute type $\#_{type}$ may have values from the set $\{$*schedule*, *assign*, *withdraw*, *reassign*, *start*, *suspend*, *resume*, *abort*, *complete*, *autoskip*, *manualskip*$\} \cup \{\perp\}$ and specifies the lifecycle type of events. $\#_{actinst}$ records the activity instance for each event.

Definition 3 (Activity instances)**.** *For an event log $L = (E, C, \#, \prec)$, the set of activity instances $I \subseteq U_{actinst}$ is defined as $I = \{\#_{actinst}(e) \mid e \in E\}$. Activity instances are not shared across cases, i.e.,*

$$\neg\exists e, e' \in E[\#_{actinst}(e) = \#_{actinst}(e') \wedge \#_{case}(e) \neq \#_{case}(e')].$$

All events within an activity instance correspond to the same activity:

$$\neg\exists e, e' \in E[\#_{actinst}(e) = \#_{actinst}(e') \wedge \#_{act}(e) \neq \#_{act}(e')].$$

We will assume that event logs are homogeneous, i.e., if a lifecycle type of event is defined for an event, then a lifecycle type of event is defined for all events:

$$\exists e \in E[\#_{type}(e) \neq \perp] \implies \forall e \in E[\#_{type}(e) \neq \perp].$$

This also holds for activity instances:

$$\exists e \in E[\#_{actinst}(e) \neq \perp] \implies \forall e \in E[\#_{actinst}(e) \neq \perp].$$

Definition 4 (Sets of start and complete events)**.** *For an event log $L = (E, C, \#, \prec)$, let E_s be the set of start events, i.e., $E_s = \{e \in E \mid \#_{type}(e) = start\}$, and let E_c be the set of complete events, i.e., $E_c = \{e \in E \mid \#_{type}(e) = complete\}$.*

Definition 5 (Sets of activities and activity instances)**.** *For an event log $L = (E, C, \#, \prec)$, let A be the set of activities, $A = \{\#_{act}(e) \mid e \in E\}$, and let I be the set of activity instances, $I = \{\#_{actinst}(e) \mid e \in E\}$.*

Event logs may differ in the lifecycle types of events they record and in whether these are combined with recorded activity instances. Many different kinds of event logs can thus be defined, but here we restrict ourselves to three types of event logs.

Definition 6 (Atomic event log)**.** *An event log $L^a = (E, C, \#, \prec)$ is an atomic event log if and only if it contains no lifecycle information, i.e.,*

$$\forall e \in E[\#_{type}(e) = \perp].$$

Definition 7 (Derivable-interval event log)**.** *An event log $L^{di} = (E, C, \#, \prec)$ is a derivable-interval event log if and only if it contains start and complete lifecycle information, but no activity instance information, i.e.,*

$$\forall e \in E[\#_{type}(e) \in \{start, complete\} \wedge \#_{actinst}(e) = \perp].$$

Definition 8 (Explicit-interval event log)**.** *An event log $L^{ei} = (E, C, \#, \prec)$ is an explicit-interval event log if and only if it contains start and complete lifecycle information as well as activity instance information, i.e.,*

$$\forall e \in E[\#_{type}(e) \in \{start, complete\} \wedge \#_{actinst}(e) \neq \perp].$$

In addition, no two events referring to the same activity instance (note that these will be in the same case) can refer to the same lifecycle type of events:

$$\neg\exists e, e' \in E[e \neq e' \wedge \#_{actinst}(e) = \#_{actinst}(e') \wedge \#_{type}(e) = \#_{type}(e')].$$

Fig. 1 illustrates the three defined types of event logs using the process depicted through the event log in Table 1. Note that the derivable-interval event log and explicit-interval event log examples illustrate that it is possible to match related events based on, e.g., their $\#_{actinst}$ attribute.

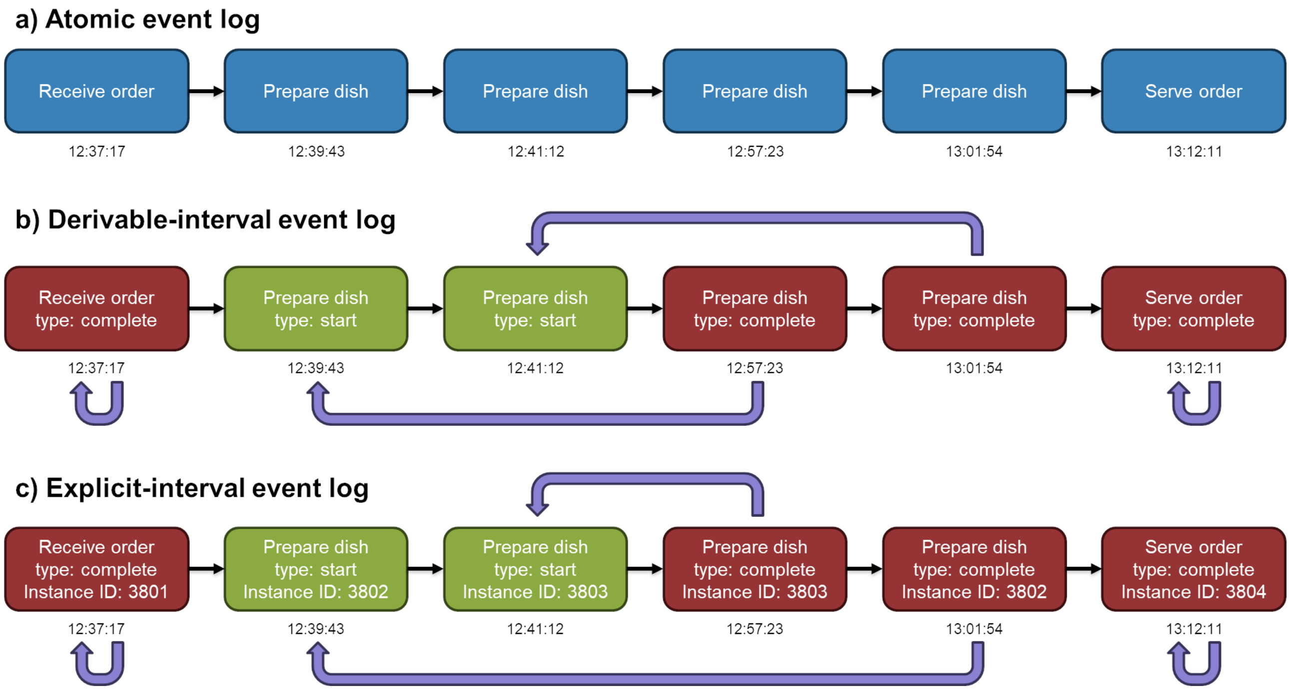

Figure 1: Examples of the different types of event logs that have been defined.

For an event log $L = (E, C, \#, \prec)$, let E_s be the set of start events and E_c be the set of complete events. For derivable-interval event logs and explicit-interval event logs we require that it is possible to define an injective function $match\colon E_c \to E_s \cup E_c$ such that this function matches a complete event to a start event that it does *not precede*, whenever that is possible; otherwise it maps it to the same complete event (which then indicates that the instance was completed instantly at creation time). Hence, i.e.,

$$\forall e \in E_c[(e \not\prec match(e) \wedge match(e) \in E_s) \vee e = match(e)].$$

For setting the requirements of the *match* function, we define functions that indicate whether two events are compatible, based on their attributes.

Definition 9 (Event compatibility). *Let L be an event log, $L = (E, C, \#, \prec)$. Relations $actinstcmp \subseteq E \times E$ and $cmp \subseteq E \times E$ hold when two events are compatible, given their case, activity, and activity instance attributes:*

$$\begin{aligned} actinstcmp(e, e') \quad &\textit{iff} \quad (\#_{actinst}(e) \neq \bot \vee \#_{actinst}(e') \neq \bot) \implies \#_{actinst}(e) = \#_{actinst}(e'), \\ cmp(e, e') \quad &\textit{iff} \quad \#_{case}(e) = \#_{case}(e') \wedge \#_{act}(e) = \#_{act}(e') \wedge actinstcmp(e, e'). \end{aligned}$$

Proposition 1 (Compatibility is an equivalence relation). *All properties of an equivalence relation hold for cmp, i.e., let L be an event log, $L = (E, C, \#, \prec)$, then:*

(a) cmp is reflexive, i.e., for all $e \in E$, $cmp(e, e)$.

(b) cmp is symmetric, i.e., for all $e, e' \in E$, if $cmp(e, e')$, then $cmp(e', e)$.

(c) cmp is transitive, i.e., for all $e, e', e'' \in E$, if $cmp(e, e')$ and $cmp(e', e'')$, then $cmp(e, e'')$.

The proof that cmp is transitive is provided in Appendix A (see Proof of Proposition 1 (c)).

A requirement for the *match* function, then, is that matching events should be compatible.

$$\forall e \in E_c \forall e' \in E_c \cup E_s[match(e) = e' \implies cmp(e, e')]$$

Although several approaches for the matching of events, i.e., event abstraction, have been proposed in literature (van Zelst et al., 2021), we distinguish the above as the minimal requirements that a *match* function should possess. However, we impose additional requirements for the *match* function using a First In, First Out (FIFO) approach to guarantee that PPIs are uniquely defined for most event logs. Different approaches, such as a Last In, First Out (LIFO) one or more complex heuristics, could also be considered, but this work does not formalize them.

For a FIFO *match* function, we require that events be matched respecting their partial ordering and lifecycle information, i.e.,

$$\forall e, e' \in E_c[(e \prec e' \land cmp(e, e')) \implies match(e') \not\prec match(e)].$$

Moreover, if an event is the match of a complete event, then all compatible start events that it does not precede have been matched with another complete event, i.e.,

$$\forall e \in E \forall e' \in E_c \forall e'' \in E_s[(match(e') = e \land cmp(e', e'') \land e \not\prec e'') \implies \\ (e \in E_s \land \#_{time}(e) = \#_{time}(e'')) \lor \exists e''' \in E_c[match(e''') = e'' \land e' \not\prec e''']].$$

Fig. 2 provides graphical representations that summarize the requirements of the *match* function for an example based on the serving dishes process introduced in Table 1. The arched arrows in the pictures denote a *match* between two events. A red cross indicates that a *match* is not valid as it does not fulfill the corresponding requirement.

It must be noted that, by considering the above requirements for the *match* function, it is possible that some start events will not be matched to any complete event. In this case, such start events will be ignored for the calculation of the PPIs. Additionally, even after considering the FIFO requirements, there are cases where the *match* function could not be uniquely defined for a derivable-interval event log, as shown in theorem 1.

Theorem 1 (Non-uniqueness of the *match* function for derivable-interval event logs)**.** *Let L be a derivable-interval event log. There exist at least two non-identical match functions for L if and only if:*

$$\exists e, e', e'' \in E[e \not\prec e' \land e' \not\prec e \land cmp(e, e') \land cmp(e, e'') \land \\ ((e, e' \in E_s \land e'' \in E_c \land e'' \not\prec e) \lor (e, e' \in E_c \land e'' \in E_s \land e \not\prec e''))].$$

The proof is provided in Appendix A (see Proof of Theorem 1).

We can define a uniquely-matched derivable-interval event log by imposing the restriction that no three events exist such that the conditions of theorem 1 hold, i.e.:

Definition 10 (Uniquely-matched derivable-interval event log)**.** *Let L be a derivable-interval event log. If it holds that:*

$$\neg\exists e, e', e'' \in E[e \not\prec e' \land e' \not\prec e \land cmp(e, e') \land cmp(e, e'') \land \\ ((e, e' \in E_s \land e'' \in E_c \land e'' \not\prec e) \lor (e, e' \in E_c \land e'' \in E_s \land e \not\prec e''))].$$

then we refer to L as a uniquely-matched derivable-interval event log.

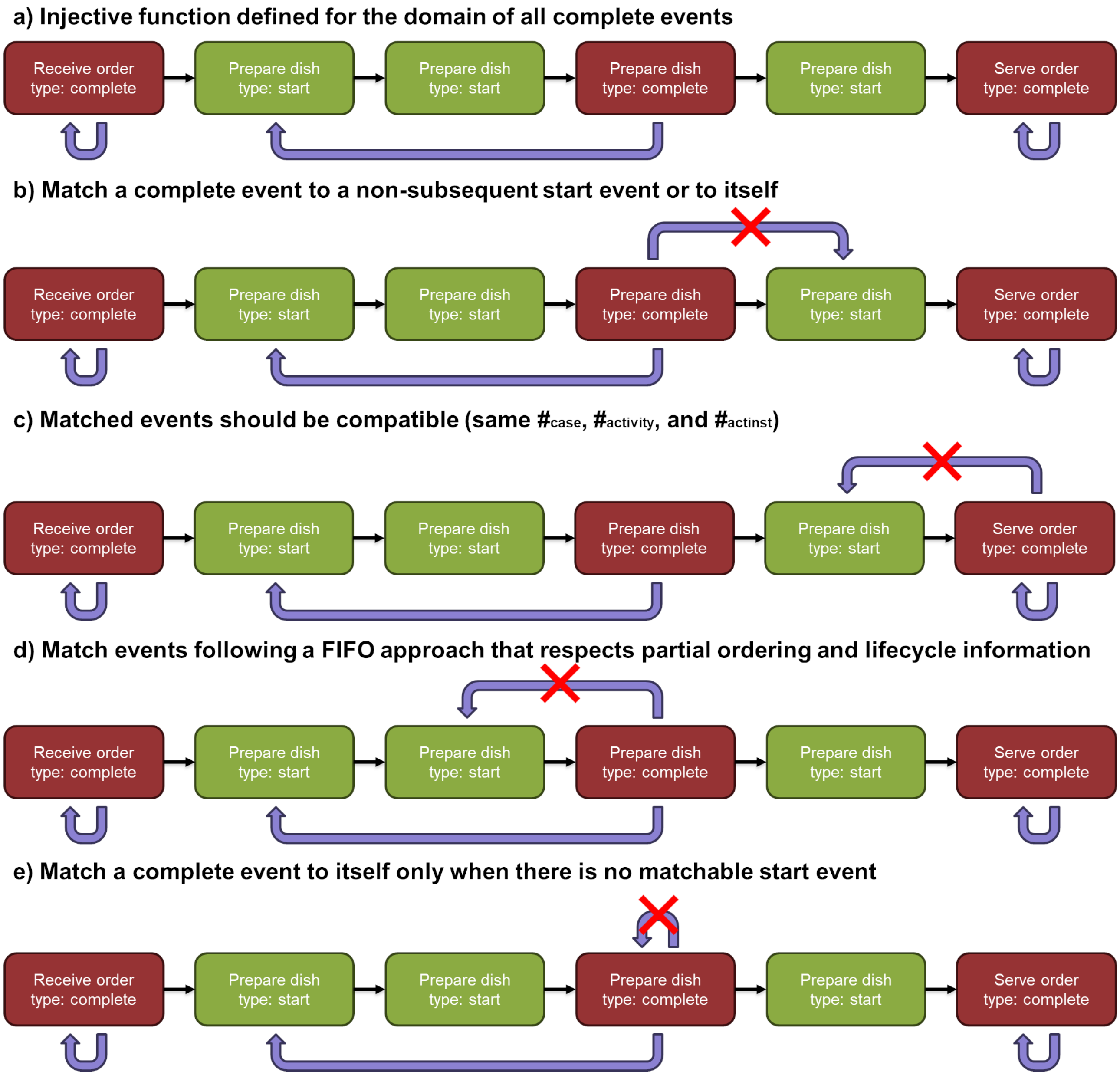

Figure 2: Graphic depiction of the requirements of the *match* function.

Theorem 2 (For uniquely-matched derivable-interval event logs *match* functions are unique)**.** *Let L be a uniquely-matched derivable-interval event log. For any pair match1 and match2 of match functions, it holds that match1 = match2.*

The proof of theorem 2 is equivalent to that of case 2 in the proof of theorem 1 (see Proof of Theorem 1).

It must be noted that, although the *match* function is not uniquely defined for derivable-interval event logs, the requirements that are imposed on it ensure that the sum of services times (difference between the timestamp of a complete event and its matched event) for any set of complete events that share the same case and activity is the same regardless of the *match* function, i.e.:

Definition 11 (Set of complete events that share a case and activity)**.** *For an event log $L = (E, C, \#, \prec)$, let $E_{c,a}$ be the set of complete events that possess the same case and activity attributes,* *i.e.,* $E_{c,a} = \{e \in E_c \mid \#_{case}(e) = c \wedge \#_{act}(e) = a\}$.

Lemma 1 (For derivable-interval event logs the sum of service times of compatible events is the same regardless of the match function)**.** *Let L be a derivable-interval event log. For any pair match1 and match2 of match functions, it holds that, for any set of complete events that possess the same case and activity attributes $E_{c,a}$:*

$$\sum_{e \in E_{c,a}} \#_{time}(e) - \#_{time}(match1(e)) = \sum_{e \in E_{c,a}} \#_{time}(e) - \#_{time}(match2(e)).$$

The proof is provided in Appendix A (see Proof of Lemma 1).

While the above minimizes the effects of different *match* functions that require the $\#_{time}$ attribute of events for derivable-interval event logs, it must be mentioned that PPIs that rely on other attributes such as $\#_{fc}$, $\#_{ic}$, $\#_{lc}$, $\#_{tc}$, $\#_{vc}$, and $\#_{unt}$, could yield different results depending on the chosen *match* function. For these cases, it is possible to consider additional requirements for the *match* function, such as prioritizing the matching of events with a higher $\#_{tc}$. However, several possibilities exist that could be considered for such additional requirements (e.g., should we prioritize $\#_{tc}$ or $\#_{unt}$?). We instead make use of a unique event identifier attribute $\#_{eid} \in \mathbb{N}$ that allows giving an order to events that have the same $\#_{time}$ attribute. When this occurs, the event with the lower $\#_{eid}$ is regarded as the preceding event.

Definition 12 (Totally-ordered derivable-interval event log)**.** *An event log $L^{tdi} = (E, C, \#, \prec)$ is a derivable-interval event log with a strict total ordering $\prec: E \times E \to \{true, false\}$ on events and with event identifier information, i.e.,*

$$\forall e \in E[\#_{eid}(e) \in \mathbb{N}].$$

These event identifiers are unique, i.e.,

$$\forall e, e' \in E[e \neq e' \implies \#_{eid}(e) \neq \#_{eid}(e')]$$

For totally-ordered derivable-interval event logs, the ordering of events can not go against the order imposed by the time attributes. Moreover, if two events possess the same $\#_{time}$ attribute value, their ordering must not go against the order imposed by their event identifiers, i.e., for all $e, e' \in E$, if $\#_{time}(e) < \#_{time}(e') \vee (\#_{time}(e) = \#_{time}(e') \wedge \#_{eid}(e) < \#_{eid}(e'))$ then $e \prec e'$.

Definition 13 (Conversion from derivable-interval event log to totally-ordered derivable-interval event log)**.** *Let $L^{di} = (E, C, \#, \prec)$ be a derivable-interval event log. The associated totally-ordered derivable-interval event log,* ***TotDer***$(L^{tdi}) = (E, C, \#', \prec')$ *is defined as follows:*

$$\begin{aligned} \#' &= \# \oplus \{(e, (eid, fresheventid(e))) \mid e \in E\}, \\ \prec' &= \{(e \prec e' \vee (e \not\prec e' \wedge e' \not\prec e \wedge \#_{eid}(e) < \#_{eid}(e'))) \mid e, e' \in E\}. \end{aligned}$$

The function fresheventid: $E \to \mathbb{N}$ is any injective function from E to $\mathbb{N}$.

The *fresheventid* function assigns a unique event identifier to each event, allowing sorting events with the same timestamp by such identifier, fulfilling the requirements of a totally-ordered derivable-interval event log.

Theorem 3 (For totally-ordered derivable-interval event logs *match* functions are unique)**.** *Let L be a totally-ordered derivable-interval event log. For any pair match1 and match2 of match functions, it holds that match1 = match2.*

The proof of theorem 3 is equivalent to that of case 2 in the proof of theorem 1 (see Proof of Theorem 1).

The non-uniqueness of the *match* function does not apply for explicit-interval event logs, as possessing explicit activity instances ensures that the function is always uniquely defined, as shown in theorem 4.

Theorem 4 (For explicit-interval event logs *match* functions are unique)**.** *Let L be an explicit-interval event log. For any pair match1 and match2 of match functions, it holds that match1 = match2.*

The proof is provided in Appendix A (see Proof of Theorem 4).

Definition 14 (Reassignment of matched events)**.** *Let L be an event log,* $L = (E, C, \#, \prec)$. *Let* $e \in E_c$, $e' \in E_s \cup E_c$. *Function* $reassign(e, e')$ *reassigns the currently matched event of e to* e', *i.e:*

$$match(e) \triangleq e'.$$

Definition 15 (Badness of matched events)**.** *Let L be an event log,* $L = (E, C, \#, \prec)$. *Functions* $sbadness\colon 2^{E_s} \to \mathbb{N}$ *and* $cbadness\colon 2^{E_c} \to \mathbb{N}$ *yield the number of wrongly matched start and complete events, respectively, in L:*

$$\begin{aligned}
sbadness(E_s) \quad &\equiv \quad |\{e \in E_s \mid \exists e' \in E \exists e'' \in E_c[(match(e'') = e' \wedge cmp(e'', e) \wedge e' \not\prec e) \wedge \\
&\qquad (e' \in E_s \wedge \#_{time}(e') \neq \#_{time}(e)) \wedge \neg \exists e''' \in E_c[match(e''') = e \wedge e'' \not\prec e''']]\}|, \\
cbadness(E_c) \quad &\equiv \quad |\{e \in E_c \mid \exists e' \in E_c[e \prec e' \wedge cmp(e, e') \wedge match(e') \prec match(e)]\}|.
\end{aligned}$$

Theorem 5 (For derivable-interval event logs and explicit-interval event logs a *match* function always exists)**.** *For any L that is a derivable-interval event log or an explicit-interval event log, it holds that, for all complete events of L, i.e., for all* $e \in E_c$, *there exists a match function.*

The proof is provided in Appendix A (see Proof of Theorem 5).

Additional event log types could include more complex transitional information, such as scheduling and assignment of activities, in addition to their start and complete events. There are a number of additional constraints that would apply to these event logs, such as preserving the scheduled-assigned-start-complete lifecycle, but we do not formalize these.

We define conversion functions that allow transforming atomic event logs into derivable-interval event logs, and derivable-interval event logs into explicit-interval event logs. This is achieved by introducing values for the $\#_{type}$ and $\#_{actinst}$ attributes, respectively.

To convert a atomic event log into a derivable-interval event log, we assign *complete* to the type attribute of each event.

Definition 16 (Conversion from atomic event log to derivable-interval event log)**.** *Let* $L^a = (E, C, \#, \prec)$ *be an atomic event log. The associated derivable-interval event log,* $\boldsymbol{Der}(L^a) = (E, C, \#', \prec)$ *is defined as follows:*

$$\#' = \# \oplus \{(e, (type, complete)) \mid e \in E\}.$$

To convert derivable-interval event log into a explicit-interval event log, we assign to each complete event a unique activity instance (which one does not really matter), and we assign to each start event the activity instance corresponding to the matching complete event. As the *match* function is not uniquely defined for derivable-interval event logs, different conversion results can result depending on the used *match* function. It must be noted that the conversion should always be in the context of a given *match* function.

Definition 17 (Conversion from derivable-interval event log to explicit-interval event log)**.** *Let $L^{di} = (E, C, \#, \prec)$ be a derivable-interval event log. The associated explicit-interval event log,* $\boldsymbol{Expl}(L^{ei}) = (E, C, \#', \prec)$ *is defined as follows:*

$$\begin{aligned} \#' = \# \quad \oplus \quad & (\{(e, (actinst, freshactinst(e))) \mid e \in E_c\} \cup \\ & \{(e, (actinst, freshactinst(e'))) \mid e \in E_s \wedge e' \in E_c \wedge match(e') = e\} \cup \\ & \{(e, (actinst, freshactinst(e))) \mid e \in E_s \wedge \neg\exists e' \in E_c[match(e') = e]\}). \end{aligned}$$

The function $freshactinst \colon E \to U_{actinst}$ *is any injective function from E to $U_{actinst}$.*

Functions defined from this point onward are defined for explicit-interval event logs, so we assume that atomic event logs have been converted to derivable-interval event logs and derivable-interval event logs to explicit-interval event logs.

4.1. Functions that Support the Calculation of PPIs

To facilitate the definition of the PPIs, several auxiliary functions that provide access to event and case attributes, conduct calculations over the data, or obtain particular sets of elements have been defined. These auxiliary functions support the calculation of the defined PPIs. Some functions may be defined in different ways due to different possible interpretations of their calculation. As the aim of this work is not to impose specific interpretations of indicators, we have opted to define several variations of functions when this situation arises. These variations are denoted by a superscript in the function's name.

Next, the auxiliary functions that are later used in the calculation of the PPIs are defined:

Definition 18 (Accessing events and their attributes in cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$. Function* $events \colon C \to 2^{U_{event}}$ *yields the events of cases, while the functions* $act \colon C \to 2^{U_{act}}$, $res \colon C \to 2^{U_{res}}$, $hres \colon C \to 2^{U_{hres}}$, $role \colon C \to 2^{U_{role}}$, *and* $inst \colon C \to 2^{U_{actinst}}$ *yield the activities, the resources, the human resources, the roles, and the activity instances used in cases, respectively:*

$$\begin{aligned} events(c) &\equiv \{e \in E \mid \#_{case}(e) = c\}, \\ act(c) &\equiv \{\#_{act}(e) \mid e \in events(c)\}, \\ res(c) &\equiv \{\#_{res}(e) \mid e \in events(c)\}, \\ hres(c) &\equiv \{\#_{hres}(e) \mid e \in events(c)\}, \\ role(c) &\equiv \{\#_{role}(e) \mid e \in events(c)\}, \\ inst(c) &\equiv \{\#_{actinst}(e) \mid e \in events(c)\}. \end{aligned}$$

Definition 19 (Accessing events and their attributes in activities)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let A be the set of activities. Function* $events \colon A \to 2^{U_{event}}$ *yields the events associated with activities, while the functions* $res \colon A \to 2^{U_{res}}$, $hres \colon A \to 2^{U_{hres}}$, $role \colon A \to 2^{U_{role}}$, *and* $inst \colon A \to 2^{U_{actinst}}$ *yield the the resources, the human resources, the roles, and the activity* *instances associated with activities:*

$$\begin{aligned}
events(a) &\equiv \{e \in E \mid \#_{act}(e) = a\},\\
res(a) &\equiv \{\#_{res}(e) \mid e \in events(a)\},\\
hres(a) &\equiv \{\#_{hres}(e) \mid e \in events(a)\},\\
role(a) &\equiv \{\#_{role}(e) \mid e \in events(a)\},\\
inst(a) &\equiv \{\#_{actinst}(e) \mid e \in events(a)\}.
\end{aligned}$$

Definition 20 (Accessing events and their attributes in activity instances)**.** *Let L be an event log, $L = (E, C, \#, \prec)$, let I be the set of activity instances, and let E_c be the set of complete events. Functions $str: I \to U_{event}$ and $cpl: I \to U_{event}$ yield the uniquely defined events that start and complete activity instances, respectively, while the functions $stime: I \to U_{time}$ and $ctime: I \to U_{time}$ yield the timestamps of start and complete events of activity instances, the functions $case: I \to U_{case}$, $act: I \to U_{act}$, $res: I \to U_{res}$, $hres: I \to U_{hres}$, and $role: I \to U_{role}$ yield the case, the activity, the resource, the human resource, and the role associated to activity instances, respectively:*

$$\begin{aligned}
str(i) &\equiv match(cpl(i)),\\
cpl(i) &\equiv e \in E_c \text{ where } \#_{actinst}(e) = i,\\
stime(i) &\equiv \#_{time}(str(i)),\\
ctime(i) &\equiv \#_{time}(cpl(i)),\\
case(i) &\equiv \#_{case}(cpl(i)),\\
act(i) &\equiv \#_{act}(cpl(i)),\\
res(i) &\equiv \#_{res}(cpl(i)),\\
hres(i) &\equiv \#_{hres}(cpl(i)),\\
role(i) &\equiv \#_{role}(cpl(i)).
\end{aligned}$$

Definition 21 (Start and end activity instances of cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$. Functions $strin: C \to U_{actinst}$ and $endin: C \to U_{actinst}$ yield the start and end activity instances of cases, respectively:*

$$\begin{aligned}
strin(c) &\equiv \{i \in inst(c) \mid \neg\exists i' \in inst(c)[str(i') \prec str(i)]\},\\
endin(c) &\equiv \{i \in inst(c) \mid \neg\exists i' \in inst(c)[cpl(i) \prec cpl(i')]\}.
\end{aligned}$$

Proposition 2 (Properties of sets produced by functions that yield the start and end activity instances of cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$. Let c be a case of the event log, $c \in C$. The following properties hold:*

(a) For all $x, x' \in strin(c)$, $stime(x) = stime(x')$.

(b) For all $x, x' \in endin(c)$, $ctime(x) = ctime(x')$.

Definition 22 (Start and end timestamps of cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$. Functions $startt: C \to U_{time}$ and $endt: C \to U_{time}$ yield the start and end timestamps of cases, respectively:*

$$\begin{aligned}
startt(c) &\equiv \#_{time}(str(x)) \text{ for any } x \in strin(c)\\
endt(c) &\equiv \#_{time}(cpl(x)) \text{ for any } x \in endin(c).
\end{aligned}$$

Definition 23 (Calculations based on cases and activities)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let A be the set of activities. Function $inst\colon C \times A \to 2^{U_{actinst}}$ yields the activity instances of cases where a specific activity occurs, $count\colon C \times A \to \mathbb{N}$ yields the number of occurrences of activities in cases, while functions $fi^s\colon C \times A \to 2^{U_{actinst}}$ and $fi^c\colon C \times A \to 2^{U_{actinst}}$ yield the first occurrences of activities in cases considering their start or complete events, respectively, and function $fi^{sc}\colon C \times A \times A \to 2^{U_{actinst}}$ yields the first occurrences of an activity after the first occurrence of another one in cases:*

$$\begin{aligned}
inst(c,a) &\equiv \{i \in inst(c) \mid act(i) = a\},\\
count(c,a) &\equiv |inst(c,a)|,\\
fi^s(c,a) &\equiv \{i \in inst(c,a) \mid \neg\exists i' \in inst(c,a)[str(i') \prec str(i)]\},\\
fi^c(c,a) &\equiv \{i \in inst(c,a) \mid \neg\exists i' \in inst(c,a)[cpl(i') \prec cpl(i)]\},\\
fi^{sc}(c,a,b) &\equiv \{i \in inst(c,b) \mid fi^s(c,a) \neq \varnothing \wedge \forall i' \in fi^s(c,a)[str(i') \prec str(i)] \wedge\\
&\qquad \neg\exists i'' \in inst(c,b)[str(i') \prec str(i'') \wedge cpl(i'') \prec cpl(i)]\}.
\end{aligned}$$

Proposition 3 (Properties of sets produced by functions that yield the first instances of activities in cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let A be the set of activities. Let c be a case of the event log, $c \in C$. Let a, b be activities contained in the event log, $a, b \in A$. The following properties hold:*

(a) For all $x, x' \in fi^s(c,a)$, $stime(x) = stime(x')$.

(b) For all $x, x' \in fi^c(c,a)$, $ctime(x) = ctime(x')$.

(c) For all $x, x' \in fi^{sc}(c,a,b)$, $stime(x) = stime(x')$.

Definition 24 (Calculations based on first instances of activities in cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let A be the set of activities. Functions $fitc\colon C \times A \to \mathbb{R}$ and $filt\colon C \times A \to \mathbb{R}$ yield the average cost and the average time, respectively, of first occurrences of activities in cases:*

$$\begin{aligned}
fitc^{\times}(c,a) &\equiv \begin{cases} \frac{\sum_{i \in fi^s(c,a)} TC^{\times}(i)}{|\{i \in fi^s(c,a)\}|} & \text{if } fi^s(c,a) \neq \varnothing\\ 0 & \text{if } fi^s(c,a) = \varnothing \end{cases},\\
filt(c,a) &\equiv \begin{cases} \frac{\sum_{i \in fi^s(c,a)} LT(i)}{|\{i \in fi^s(c,a)\}|} & \text{if } fi^s(c,a) \neq \varnothing\\ 0 & \text{if } fi^s(c,a) = \varnothing \end{cases}.
\end{aligned}$$

The $\times$ in $fitc^{\times}$ indicates that the function can take multiple forms; in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.

Definition 25 (Preceding and subsequent activity instances)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let I be the set of activity instances. Functions $prev\colon I \to 2^{U_{actinst}}$ and $next\colon I \to 2^{U_{actinst}}$ yield the activity instances that occurred right before and right after activity instances, respectively, while function $prevstr\colon I \to 2^{U_{actinst}}$ yields the activity instances that start before but finish after activity instances, and $concstr\colon I \to 2^{U_{actinst}}$ yields the activity instances that start at the same* *time as activity instances:*

$$\begin{aligned}
prev(i) &\equiv \{i' \in inst(case(i)) \mid \\
&\quad cpl(i') \prec str(i) \land \neg\exists i'' \in inst(case(i))[cpl(i') \prec cpl(i'') \land cpl(i'') \prec str(i)]\}, \\
next(i) &\equiv \{i' \in inst(case(i)) \mid \\
&\quad cpl(i) \prec str(i') \land \neg\exists i'' \in inst(case(i))[cpl(i) \prec str(i'') \land str(i'') \prec str(i')]\}, \\
prevstr(i) &\equiv \{i' \in inst(case(i)) \mid str(i') \prec str(i) \land cpl(i') \not\prec str(i)\}, \\
concstr(i) &\equiv \{i' \in inst(case(i)) \mid stime(i) = stime(i')\}.
\end{aligned}$$

Proposition 4 (Properties of sets produced by functions that yield the preceding and subsequent activity instances)**.** *Let L be an event log, $L = (E, C, \#, \prec)$, let I be the set of activity instances. Let i be an activity instance, $i \in I$. The following properties hold:*

(a) For all $x, x' \in prev(i)$, $ctime(x) = ctime(x')$.

(b) For all $x, x' \in next(i)$, $stime(x) = stime(x')$.

Definition 26 (Calculations based on preceding and subsequent activity instances)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let I be the set of activity instances. Function $dres\colon I \to \mathbb{N}$ yields the delegation of work for activity instances (note that, if all activity instances that were completed right before the activity instance were associated with the same human resource as the activity instance, then the function returns 0, and if none of the activity instances that were completed right before the activity instance were associated with that human resource, then the function returns 1), while function $dfrel\colon C \to 2^{U_{act}} \times 2^{U_{act}}$ yields the directly-follows relations in cases:*

$$\begin{aligned}
dres(i) &\equiv \begin{cases} \frac{|\{i' \in prev(i) \mid hres(i') \neq hres(i)\}|}{|prev(i)|} & \text{if } prev(i) \neq \varnothing \\ 0 & \text{if } prev(i) = \varnothing \end{cases}, \\
dfrel(c) &\equiv \{(act(i), act(i')) \in act(c) \times act(c) \mid i \in inst(c) \land i' \in inst(c) \land i' \in next(i)\}.
\end{aligned}$$

Definition 27 (Time calculations based on intervals of activity instances)**.** *Let L be an event log, $L = (E, C, \#, \prec)$ and let I be the set of activity instances. Functions $instbetween^s\colon I \times I \to 2^{U_{actinst}}$, $instbetween^c\colon I \times I \to 2^{U_{actinst}}$, $instbetween^{sc}\colon I \times I \to 2^{U_{actinst}}$, and $instbetween^w\colon I \times I \to 2^{U_{actinst}}$ yield the activity instances that were started, were completed, were either started or completed, or were active, respectively, within pairs of activity instances, while function $timew\colon E \times E \times E \times E \to \mathbb{R}$ yields the time within a pair of events of two events, function $lt\colon I \times I \to \mathbb{R}$ yields the lead time between pairs of activity instances, function $st\colon I \times I \to \mathbb{R}$ yields the service time between pairs of activity instances (note that different sets of activity instances can be considered for the calculation), and $wt\colon I \times I \to \mathbb{R}$ yields the waiting time between pairs of activity instances (note* *that different sets of activity instances can also be considered in this):*

$$
\begin{aligned}
instbetween^{s}(i,i') &\equiv \{i'' \in inst(case(i)) \mid case(i'') = case(i') \wedge \\
&\quad\; str(i'') \not\prec str(i) \wedge cpl(i') \not\prec str(i'')\},\\
instbetween^{c}(i,i') &\equiv \{i'' \in inst(case(i)) \mid case(i'') = case(i') \wedge \\
&\quad\; cpl(i'') \not\prec str(i) \wedge cpl(i') \not\prec cpl(i'')\},\\
instbetween^{sc}(i,i') &\equiv instbetween^{s}(i,i') \cup instbetween^{c}(i,i'),\\
instbetween^{w}(i,i') &\equiv instbetween^{sc}(i,i') \cup \{i'' \in inst(case(i)) \mid case(i'') = case(i') \wedge \\
&\quad\; str(i) \not\prec str(i'') \wedge cpl(i'') \not\prec cpl(i')\},\\
timew(x,x',y,z) &\equiv min(\#_{time}(x'), \#_{time}(z)) - max(\#_{time}(x), \#_{time}(y)),\\
lt(i,i') &\equiv ctime(i') - stime(i),\\
st^{\times}(i,i') &\equiv \sum_{i'' \in instbetween^{\times}(i,i')} timew(str(i), cpl(i'), str(i''), cpl(i'')),\\
wt^{\times}(i,i') &\equiv \sum_{i'' \in instbetween^{\times}(i,i')} \begin{cases} timew(str(i), cpl(i'), cpl(x), str(i'')) & for\ any\ x \in prev(i'') \\ 0 & if\ prev(i'') = \varnothing \end{cases}.
\end{aligned}
$$

The $\times$ in $st^{\times}$ and $wt^{\times}$ indicates that the functions can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses $instbetween^{s}$; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses $instbetween^{c}$; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses $instbetween^{sc}$; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses $instbetween^{w}$.

Definition 28 (Traces of cases and variants of groups of cases)**.** *Let L be an event log, $L = (E, C, \#, \prec)$. Function $seq\colon C \to 2^{U^{*}_{actinst}}$ yields for each case their activity instance sequences in order of their start times (as activity instances may share the same start time, this may yield multiple sequences), while function $trace\colon C \to 2^{U^{*}_{act}}$ yields the sequences of activities in cases, and function $variants\colon 2^{U_{case}} \to 2^{U^{*}_{act}}$ yields all variants of traces in groups of cases:*

$$
\begin{aligned}
seq(c) &\equiv \{\langle i_1, \ldots, i_n\rangle \mid inst(c) = \{i_1, \ldots, i_n\} \wedge |inst(c)| = n \wedge \forall_{1 \le p < q \le n}[str(i_q) \not\prec str(i_p)]\},\\
trace(c) &\equiv \{\langle act(i_1), \ldots, act(i_n)\rangle \mid \langle i_1, \ldots, i_n\rangle \in seq(c)\},\\
variants(g) &\equiv \cup_{c \in g}\ trace(c).
\end{aligned}
$$

5. Process Performance Indicators

Tables 3, 4, 5, 6, and 7 show the list of PPIs that have been defined for the "General" category and for the time, cost, quality, and flexibility dimensions, respectively. These tables also show for which granularities each PPI can be defined, and the references in which its use was observed. As explained above, during the formalization process, some PPIs were added for completeness; therefore, no references are provided for them. An example of the above is Lead Time from activity A (time dimension), which was defined in the likeness of Lead Time to activity A (Balaban et al., 2011; Cho et al., 2020) due to their similar usage. Additionally, more than one variation has been defined for some PPIs at the activity instance granularity. These cases are marked with an asterisk (*) in the tables.

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Activity count			✓	✓	✓	Added for completeness.
Activity Instance count		✓	✓	✓	✓	(Longo and Motta, 2005)
Case count				✓		(del Río-Ortega et al., 2013; Jagdev et al., 1997; Walsh, 1996)
Human Resource count		✓	✓	✓	✓	Added for completeness.
Resource count		✓	✓	✓	✓	Added for completeness.
Role count			✓	✓	✓	Added for completeness.

Table 3: List of general PPIs

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Active time			✓		✓	(Meier et al., 2015; Fogarty, 1992)
Activity count			✓	✓	✓	(van den Ingh, 2016)
activity Instance count			✓	✓	✓	(van den Ingh, 2016)
Automated Activity count			✓	✓	✓	(van den Ingh, 2016)
Automated activity Instance count			✓	✓	✓	(van den Ingh, 2016)
Automated Activity Service Time			✓	✓	✓	(Longo and Motta, 2005)
Case count and Lead Time Ratio				✓		(Chimhamhiwa et al., 2009)
Case Count where Lead Time over Value				✓		(van den Ingh, 2016)
Case Percentage where Lead Time over Value				✓		(van den Ingh, 2016)
Case Percentage with Missed Deadline				✓		(Bosilj-Vuksic et al., 2008; Glavan, 2011; Kutucuoglu et al., 2002; Vernadat et al., 2013; Wetzstein et al., 2008)
Handover count			✓		✓	(van den Ingh, 2016)
Idle Time			✓		✓	(Chimhamhiwa et al., 2009; de Kort, 2015; Korherr and List, 2007)

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Lead Time	✓	✓	✓	✓	✓	(Longo and Motta, 2005; del Río-Ortega et al., 2013; Walsh, 1996; Fogarty, 1992; van den Ingh, 2016; Chimhamhiwa et al., 2009; Glavan, 2011; Vernadat et al., 2013; de Kort, 2015; Korherr and List, 2007; Bakx, 2010; Bhagwat and Sharma, 2007; Gunasekaran and Kobu, 2007; Han et al., 2010; Lehnert et al., 2014; Pourshahid et al., 2009; van Heck et al., 2010)
Lead Time and Case count Ratio				✓		(Wetzstein et al., 2008)
Lead Time Deviation from time Limit			✓		✓	(van den Ingh, 2016; Oman et al., 2017)
Lead Time Deviation from Expectation			✓		✓	(van den Ingh, 2016; Oman et al., 2017)
Lead Time from activity A			✓		✓	Added for completeness.
Lead Time from activity A to activity B			✓		✓	(Longo and Motta, 2005; del Río-Ortega et al., 2013; Walsh, 1996; Bhagwat and Sharma, 2007; Pourshahid et al., 2009; Balaban et al., 2011; Cho et al., 2020)
Lead Time to activity A			✓		✓	(Balaban et al., 2011; Cho et al., 2020)
Service and Lead Time Ratio	✓	✓	✓	✓	✓	(Fogarty, 1992; Kis et al., 2017)
Service Time	✓	✓	✓	✓	✓	(Meier et al., 2015; Fogarty, 1992; van den Ingh, 2016; Chimhamhiwa et al., 2009; de Kort, 2015; Korherr and List, 2007; Bhagwat and Sharma, 2007; van Heck et al., 2010; Oman et al., 2017; Balaban et al., 2011; Cho et al., 2020; Haberberger, 2019)
Service Time from activity A to activity B			✓	✓	✓	Added for completeness.
Waiting Time	✓	✓	✓		✓	(Walsh, 1996; Fogarty, 1992; van den Ingh, 2016; Chimhamhiwa et al., 2009; Lehnert et al., 2014; Pourshahid et al., 2009; Rinaldi et al., 2015)
Waiting Time from activity A to activity B			✓		✓	(Cho et al., 2020)

Table 4: List of time dimension PPIs

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Automated Activity Cost			✓	✓	✓	(Longo and Motta, 2005)
Desired Activity Count			✓	✓	✓	(van den Ingh, 2016)
Direct Cost			✓	✓	✓	(Karadgi, 2014)
Fixed Cost	✓*	✓	✓	✓	✓	(Haberberger, 2019; Karadgi, 2014)
Human Resource and Case count Ratio				✓		(Longo and Motta, 2005)
Human Resource count		✓	✓	✓	✓	(Cho et al., 2020; Berk, 2010)
Inventory Cost	✓*	✓	✓	✓	✓	(Bhagwat and Sharma, 2007; Gunasekaran and Kobu, 2007; Kis et al., 2017; Herzog, 2006)
Labor and Total Cost Ratio	✓	✓	✓	✓	✓	(Kis et al., 2017)
Labor Cost	✓*	✓	✓	✓	✓	(van den Ingh, 2016; Kis et al., 2017; Haberberger, 2019; Berk, 2010)
Maintenance Cost			✓	✓	✓	(Kutucuoglu et al., 2002)
Missed Deadline Cost			✓	✓	✓	(van den Ingh, 2016)
Overhead Cost			✓	✓	✓	(Gunasekaran and Kobu, 2007; Karadgi, 2014)
Resource count		✓	✓	✓	✓	(Cho et al., 2020; Berk, 2010)
Rework Cost		✓	✓	✓	✓	(Oman et al., 2017)
Rework Count		✓	✓	✓	✓	(van den Ingh, 2016)
Rework Percentage		✓	✓	✓	✓	(van den Ingh, 2016)
Total Cost	✓*	✓	✓	✓	✓	(Longo and Motta, 2005; Jagdev et al., 1997; Meier et al., 2015; van den Ingh, 2016; Chimhamhiwa et al., 2009; Bosilj-Vuksic et al., 2008; Glavan, 2011; Kutucuoglu et al., 2002; Vernadat et al., 2013; Korherr and List, 2007; Han et al., 2010; Lehnert et al., 2014; Oman et al., 2017; Cho et al., 2020; Herzog, 2006; Lima Junior et al., 2016)
Total Cost and Lead Time ratio	✓	✓	✓	✓	✓	(Glavan, 2011; Bhagwat and Sharma, 2007)
Total Cost and outcome Unit Ratio	✓	✓	✓	✓	✓	(Oman et al., 2017; Karadgi, 2014)
Total Cost and Service Time ratio	✓	✓	✓	✓	✓	(Glavan, 2011; Bhagwat and Sharma, 2007)
Transportation Cost			✓	✓	✓	(Gunasekaran and Kobu, 2007)
Variable Cost	✓*	✓	✓	✓	✓	(Haberberger, 2019; Karadgi, 2014)
Warehousing Cost			✓	✓	✓	(Oman et al., 2017)

Table 5: List of cost dimension PPIs

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
activity Instance count by Human Resource		✓	✓	✓	✓	(van den Ingh, 2016)
activity Instance count by Role			✓	✓	✓	(van den Ingh, 2016)
Automated Activity count			✓	✓	✓	(van den Ingh, 2016)
Automated activity Instance count			✓	✓	✓	(van den Ingh, 2016)
Case and Client count Ratio				✓		(van den Ingh, 2016)
Case Count where Activity After timeframe				✓		(van den Ingh, 2016)
Case Count where Activity Before timeframe				✓		(van den Ingh, 2016)
Case Count where Activity During timeframe				✓		(van den Ingh, 2016)
Case Count where End Activity is A				✓		(Longo and Motta, 2005; Chimhamhiwa et al., 2009; Bosilj-Vuksic et al., 2008; Glavan, 2011; Vernadat et al., 2013; Korherr and List, 2007; Bhagwat and Sharma, 2007; Gunasekaran and Kobu, 2007; Lehnert et al., 2014; Pourshahid et al., 2009; Cho et al., 2020; Herzog, 2006; Spremic et al., 2008)
Case Count where Start Activity is A				✓		Added for completeness.
Case Count with Rework				✓		(van den Ingh, 2016)
Case Percentage where Activity After timeframe				✓		(van den Ingh, 2016)
Case Percentage where Activity Before timeframe				✓		(van den Ingh, 2016)
Case Percentage where Activity During timeframe				✓		(van den Ingh, 2016)
Case Percentage where End Activity is A				✓		(Longo and Motta, 2005; Chimhamhiwa et al., 2009; Bosilj-Vuksic et al., 2008; Glavan, 2011; Vernadat et al., 2013; Korherr and List, 2007; Bhagwat and Sharma, 2007; Gunasekaran and Kobu, 2007; Lehnert et al., 2014; Pourshahid et al., 2009; Balaban et al., 2011; Cho et al., 2020; Herzog, 2006; Spremic et al., 2008)

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Case Percentage where Start Activity is A				✓		Added for completeness.
Case percentage with Missed Deadline				✓		(Bosilj-Vuksic et al., 2008; Glavan, 2011; Kutucuoglu et al., 2002; Vernadat et al., 2013; Wetzstein et al., 2008)
Case Percentage with rework				✓		(van den Ingh, 2016)
Client count and Total Cost Ratio		✓		✓	✓	(van den Ingh, 2016)
Desired Activity Count			✓	✓	✓	(van den Ingh, 2016)
Human Resource count		✓	✓	✓	✓	(van den Ingh, 2016)
Non-Automated Activity count			✓	✓	✓	(van den Ingh, 2016)
Non-Automated Activity instance count			✓	✓	✓	(van den Ingh, 2016)
outcome Unit count	✓*	✓	✓	✓	✓	(Vernadat et al., 2013)
Overall Quality			✓		✓	(Longo and Motta, 2005; Chimhamhiwa et al., 2009; Bosilj-Vuksic et al., 2008; Glavan, 2011; Vernadat et al., 2013; de Kort, 2015; Korherr and List, 2007; Bhagwat and Sharma, 2007; Gunasekaran and Kobu, 2007; Lehnert et al., 2014; Pourshahid et al., 2009; Cho et al., 2020; Kis et al., 2017; Haberberger, 2019; Herzog, 2006; Lima Junior et al., 2016; Spremic et al., 2008)
Repeatability			✓	✓	✓	(van den Ingh, 2016)
Rework Count		✓	✓	✓	✓	(Cho et al., 2020)
Rework Count by Value		✓	✓	✓	✓	(van den Ingh, 2016)
Rework of activities Subset			✓	✓	✓	(Cho et al., 2020)
Rework Percentage		✓	✓	✓	✓	(Cho et al., 2020)
Rework Percentage by Value		✓	✓	✓	✓	(van den Ingh, 2016)
Rework Time		✓	✓	✓	✓	(Vernadat et al., 2013; Mirsu, 2013; Wu, 2012)
Successful outcome Unit Count	✓	✓	✓	✓	✓	(Vernadat et al., 2013)
Successful outcome Unit Percentage	✓	✓	✓	✓	✓	(Vernadat et al., 2013)
Total Cost and Client count Ratio		✓		✓	✓	(van den Ingh, 2016)
Unwanted Activity Count			✓	✓	✓	(van den Ingh, 2016; Kis et al., 2017)
Unwanted Activity instance Count			✓	✓	✓	(van den Ingh, 2016; Kis et al., 2017)

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Unwanted Activity instance Percentage			✓	✓	✓	(van den Ingh, 2016; Kis et al., 2017)
Unwanted Activity Percentage			✓	✓	✓	(van den Ingh, 2016; Kis et al., 2017)

Table 6: List of quality dimension PPIs

PPI	Activity Instance	Activity	Case	Group of Cases	Expected Value	References
Activity and Role count Ratio			✓	✓	✓	(de Kort, 2015)
Activity instance and Human Resource count Ratio		✓	✓	✓	✓	(de Kort, 2015)
Client count		✓		✓	✓	(van den Ingh, 2016)
Directly-Follows Relations and Activity count Ratio			✓	✓	✓	(Cho et al., 2020)
Directly-Follows Relations count		✓	✓	✓	✓	(Cho et al., 2020)
Human Resource count		✓	✓	✓	✓	(van den Ingh, 2016)
Optional Activity count			✓	✓	✓	Added for completeness.
Optionality			✓	✓	✓	(van den Ingh, 2016; de Kort, 2015; Haberberger, 2019)
Role and Variant count Ratio				✓		(van den Ingh, 2016)
Role count			✓	✓	✓	(van den Ingh, 2016)
Variant Case Coverage			✓	✓		(van den Ingh, 2016)
Variant count				✓		(van den Ingh, 2016; Gunasekaran and Kobu, 2007; Cho et al., 2020)

Table 7: List of flexibility dimension PPIs

Each PPI, at its various granularities, requires different attributes for its calculation. These attributes can be grouped according to their relevance and need for calculating the PPIs, as follows:

- Minimum attributes: those attributes that are assumed to always be included in an event log ($\#_{case}$, $\#_{act}$, and $\#_{time}$), and those that can be inferred using the definitions provided in this work ($\#_{type}$ and $\#_{actinst}$).
- Total cost and resources attributes: those attributes that indicate the total cost of events (i.e., $\#_{tc}$), and resource-related attributes ($\#_{hres}$, $\#_{res}$, and $\#_{role}$).

- Other event attributes: those attributes that are less commonly included in event logs, but can be used to calculate certain PPIs, namely fine-grained cost information ($\#_{fc}$, $\#_{vc}$, $\#_{lc}$, and $\#_{ic}$) and attributes associated to the outcome units of the process ($\#_{unt}$ and $\#_{uns}$).
- Case attributes: those attributes that are related to case-granularity costs ($\#_{mainc}$, $\#_{mdc}$, $\#_{transc}$, $\#_{warec}$), the overall quality of its execution (i.e., $\#_{qual}$), and the client associated with the case (i.e., $\#_{cli}$).

Fig. 3 shows the number of PPIs that can be calculated when considering event logs with different numbers of attributes. It is highlighted that over half of the PPIs can be calculated even when only the minimum attributes are included in the event log.

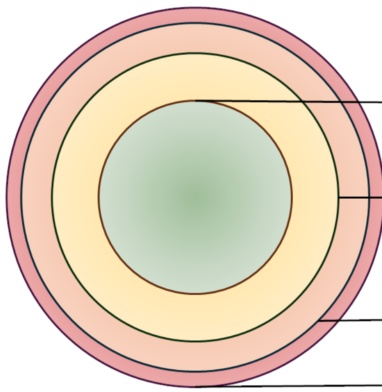

Figure 3: Percentage of PPIs that can be calculated according to the availability of attributes in the event log.

For every PPI included in the catalog, the following information is provided:

- The dimension and granularity to which it is related.
- The function name given to the PPI, as well as the formal parameters that it requires.
- The mathematical formalization of the PPI.
- A natural language explanation of the PPI, as well as its potential use.
- The attributes that are required in the event log for the calculation of the PPI, as well as any additional input (e.g., a reference value, such as a deadline).
- Whether the PPI should be maximized or minimized, and the assumptions that support this reasoning.

Due to space limitations, it is not feasible to include the entire PPI catalog in the main body of this work. However, to illustrate the information provided by the catalog, Tables 8, 9, 10, and 11 show the information of the PPI (at its finest granularity) with the most references for each dimension. Refer to Appendix B for the natural language descriptions of every PPI, and to Appendix C for their formal mathematical definitions.

Indicator	Lead Time (activity instance granularity)
Dimension	Time
Granularity	Activity Instance

Formal Parameters	$i \in I$
Function Name	$LT(i)$
Formal Definition	$ST(i) + WT(i)$
Explanation	The total elapsed time of the activity instance, measured as the sum of the elapsed time between the start and complete events of the activity instance, and the elapsed time between the complete event of the activity instance that precedes the current activity instance, and the start event of the current activity instance. Also known as throughput time or cycle time.
Potential Use	Event logs associated with processes in which the total elapsed time of activity instances is relevant.
Required Event Log Attributes	Event attribute: activity instance ($\#_{\boldsymbol{actinst}}$) Event attribute: case id ($\#_{\boldsymbol{case}}$) Event attribute: timestamp ($\#_{\boldsymbol{time}}$) Event attribute: lifecycle type ($\#_{\boldsymbol{type}}$)
Extra Input	-
Assumptions	Minimizing Lead Time is desirable.
Desired Value	MIN

Table 8: Lead Time (time dimension, activity instance granularity).

Indicator	Total Cost considering single events of activity instances (activity instance granularity)
Dimension	Cost
Granularity	Activity Instance
Formal Parameters	$i \in I$
Function Name	$TC^{sgl}(i)$
Formal Definition	$\begin{cases} \#_{tc}(cpl(i)) & \textit{if } \#_{tc}(cpl(i)) \neq \bot \\ \#_{tc}(str(i)) & \textit{if } \#_{tc}(cpl(i)) = \bot \wedge \#_{tc}(str(i)) \neq \bot \\ \bot & \textit{if } \#_{tc}(cpl(i)) = \bot \wedge \#_{tc}(str(i)) = \bot \end{cases}$
Explanation	The total cost associated with an activity instance, measured as the latest recorded value among the events of the activity instance.
Potential Use	Event logs where total cost information has been recorded.
Required Event Log Attributes	Event attribute: activity instance ($\#_{\boldsymbol{actinst}}$) Event attribute: total cost ($\#_{\boldsymbol{tc}}$) Event attribute: lifecycle type ($\#_{\boldsymbol{type}}$)
Extra Input	-
Assumptions	Minimizing Total Cost is desirable.
Desired Value	MIN

Table 9: Total Cost considering single events of activity instances (cost dimension, activity instance granularity).

Indicator	Overall Quality (case granularity)
Dimension	Quality
Granularity	Case
Formal Parameters	$c \in C$
Function Name	$OQ(c)$
Formal Definition	$\#_{qual}(c)$
Explanation	The overall quality associated with the outcome of the case.

Potential Use	Event logs containing information that allows measuring overall quality, such as customer satisfaction or a quantifier of the process outcome.
Required Event Log Attributes	Event attribute: case id ($\#_{case}$) Case attribute: quality indicator ($\#_{qual}$)
Extra Input	-
Assumptions	Maximizing the quality of the process is desirable.
Desired Value	MAX

Table 10: Overall Quality (quality dimension, case granularity).

Indicator	Variant count (group of cases granularity)
Dimension	Flexibility
Granularity	Group of Cases
Formal Parameters	$g \subseteq C$: $g \neq \varnothing$
Function Name	$V(g)$
Formal Definition	$\lvert variants(g) \rvert$
Explanation	The number of variants that are observed in the group of cases.
Potential Use	Processes with several possible execution variants.
Required Event Log Attributes	Event attribute: activity ($\#_{act}$) Event attribute: activity instance ($\#_{actinst}$) Event attribute: case id ($\#_{case}$) Event attribute: lifecycle type ($\#_{type}$)
Extra Input	-
Assumptions	A greater number of variants implies greater flexibility.
Desired Value	MAX

Table 11: Variant count (flexibility dimension, group of cases granularity).

6. Implementation and Validation

To facilitate the use of the PPI catalog, a Python library has been developed. Following the formal definitions of every PPI, the library provides implementations for all of them. It has been developed in a modular way, enabling the catalog's extensibility. The library is available at `https://bit.ly/ppi-library-pypi`. Extended documentation for its use is also available at `https://bit.ly/ppi-library-docs`. Like other Python libraries, it can be installed using the Python package installer. An overview of the library is provided next.

Our library uses similar object types to those of the main process mining library for Python, PM4Py (Berti et al., 2023), which allows loading and processing event logs with PM4Py before PPI calculation. Any file format is compatible as long as it can be processed into tabular data.

Our library provides implementations of the conversion functions in Definitions 16 and 17, allowing the use of the different kinds of event logs defined in this work, i.e., atomic event logs, derivable-interval event logs, and explicit-interval event logs. It must be noted that, since the library works on tabular data, an implicit conversion from derivable-interval event logs to totally-ordered derivable-interval event logs occurs (see Definition 13). Another special case to note is the existence of event logs that do not conform to the standard of recording a single event per tabular data row, as commonly observed in practical process mining applications. We follow the description provided by bupaR, a library in R for process mining (Janssenswillen et al., 2019), where these logs are called activity logs and are described as dataframes in which each row represents a single activity instance that can have multiple timestamps stored in different columns.

Our library provides an *event_log_formatter* function that, given the mapping of the event log's columns to the attributes defined throughout Section 4, the event log type is detected automatically, and the corresponding conversion functions are applied to convert it into an explicit-interval event log. Fig. 4 shows an example of using this function to load an event log like that of Table 1. Additional parameters can be used to map additional attributes from the event log, but the *case_id_key*, *activity_key*, and *timestamp_key* parameters are required to map the $\#_{case}$, $\#_{act}$, and $\#_{time}$ attributes of the event log, respectively.

```python
from process_performance_indicators.formatting import
    (StandardColumnMapping, event_log_formatter)

mapping = StandardColumnMapping(
    case_id_key = "Case ID",
    instance_key = "Instance ID",
    activity_key = "Activity",
    timestamp_key = "Timestamp",
    lifecycle_type_key = "Lifecycle Type",
    human_resource_key = "Resource",
    total_cost_key = "Cost",
    quality_key = "Rating"
)

formatted_log = event_log_formatter(event_log, mapping)
```

Figure 4: Formatting an event log using the library's attribute mapping.

Fig. 5 shows the outline followed by the library for grouping PPIs. At a first level, they are split among the different dimensions, whereas at a second level, they are split among the different granularities. Fig. 6 shows example code for calculating two indicators belonging to distinct dimensions and granularities.

```
process_performance_indicators/
  indicators/
    cost/
      instances.py | activities.py | cases.py | groups.py
    time/
      instances.py | activities.py | cases.py | groups.py
    quality/
      instances.py | activities.py | cases.py | groups.py
    flexibility/
      activities.py | cases.py | groups.py
    general/
      activities.py | cases.py | groups.py
```

Figure 5: Structure of the PPIs in the library.

```python
import process_performance_indicators.indicators as ppis

# Time dimension, activitiy granularity:
lt = ppis.time.activities.lead_time(formatted_log, activity_name = "A")

# Cost dimension, case granularity:
mdc = ppis.cost.cases.missed_deadline_cost(formatted_log, case_id = "1")
```

Figure 6: Using the library to calculate distinct PPIs.

Additional functionalities have also been implemented to allow the calculation of multiple PPIs simultaneously. Fig. 7 illustrates the use of the *run_indicators* function to calculate all PPIs associated with a specific dimension and granularity. Additional parameters can be provided to this function to allow the calculation of PPIs that require additional input, such as the name of a specific human resource.

```
from process_performance_indicators.execution.runner import run_indicators
from process_performance_indicators.execution.models import IndicatorArguments

args = IndicatorArguments(
    case_id = "1",
    human_resource_name = "John"
)

results = run_indicators(
    formatted_log, args, dimension = ["quality"], granularity = ["cases"]
)
```

Figure 7: Calculating several PPIs using a single function.

Using the library, the applicability of the PPIs in the catalog over multiple real-life event logs with different relevant attributes has been evaluated. An attribute is relevant if we can map it to any of the attributes required by the PPIs ($\#_{act}$, $\#_{actinst}$, $\#_{case}$, $\#_{cli}$, $\#_{fc}$, $\#_{hres}$, $\#_{ic}$, $\#_{lc}$, $\#_{mainc}$, $\#_{mdc}$, $\#_{res}$, $\#_{role}$, $\#_{qual}$, $\#_{tc}$, $\#_{time}$, $\#_{transc}$, $\#_{type}$, $\#_{uns}$, $\#_{unt}$, $\#_{warec}$). It must be noted that, to the authors' knowledge, there are no publicly available real-life event logs that contain cost information. Due to this limitation, no publicly available real-life event log exists where 100% of the PPIs can be calculated, especially for the cost dimension. To address this, simulations of the different types of event logs defined in this work (atomic event log, derivable-interval event log, uniquely-matched derivable-interval event log, explicit-interval event log) have been used to validate the capability of the library to calculate all defined PPIs.

Table 12 shows the results of this evaluation, indicating the number of calculated PPIs for each dimension for every considered event log, whereas Table 13 shows the same results, but indicating the number of calculated PPIs for each granularity for every considered event log. It must be noted that, for the simulated atomic event logs, the library could not calculate 100% of the PPIs because some of the PPIs consider the ratio between another PPI and Service Time. Since all activity instances of an atomic event log match complete events to themselves, Service Time is always 0, which causes ratios where Service Time is used as the denominator to become undefined.

Event Log	Relevant Attributes	General	Time Dimension	Cost Dimension	Quality Dimension	Flexibility Dimension	Overall
BPI Challenge 2013 - Incidents (Steeman, 2013)	7/21	15/19 (78.9%)	59/60 (98.3%)	16/95 (16.8%)	77/102 (75.5%)	31/34 (91.2%)	198/310 (63.9%)
BPI Challenge 2017 (van Dongen, 2017)	6/21	12/19 (63.2%)	57/60 (95.0%)	15/95 (15.8%)	66/102 (64.7%)	16/34 (47.1%)	166/310 (53.5%)
IT Incident Log (Shifat, 2020)	8/21	15/19 (78.9%)	59/60 (98.3%)	17/95 (17.9%)	78/102 (76.5%)	34/34 (100.0%)	203/310 (65.5%)
Italian Help Desk (Polato, 2017)	8/21	15/19 (78.9%)	59/60 (98.3%)	16/95 (16.8%)	78/102 (76.5%)	34/34 (100.0%)	202/310 (65.2%)
Production (Levy, 2014)	9/21	16/19 (84.2%)	60/60 (100.0%)	20/95 (21.1%)	90/102 (88.2%)	24/34 (70.6%)	210/310 (67.7%)

Event Log	Relevant Attributes	General	Time Dimension	Cost Dimension	Quality Dimension	Flexibility Dimension	Overall
Simulated Atomic Log	21/21	19/19 (100.0%)	59/60 (98.3%)	89/95 (93.7%)	102/102 (100.0%)	34/34 (100.0%)	303/310 (97.7%)
Simulated derivable interval event log	21/21	19/19 (100.0%)	60/60 (100.0%)	95/95 (100.0%)	102/102 (100.0%)	34/34 (100.0%)	310/310 (100.0%)
Simulated uniquely matched derivable interval event log	21/21	19/19 (100.0%)	60/60 (100.0%)	95/95 (100.0%)	102/102 (100.0%)	34/34 (100.0%)	310/310 (100.0%)
Simulated explicit interval event log	21/21	19/19 (100.0%)	60/60 (100.0%)	95/95 (100.0%)	102/102 (100.0%)	34/34 (100.0%)	310/310 (100.0%)

Table 12: Calculated PPIs for each dimension for every considered event log

Event Log	Relevant Attributes	Activity Instance Granularity	Activity Granularity	Cases Granularity	Group of Cases Granularity	Overall
BPI Challenge 2013 - Incidents (Steeman, 2013)	7/21	3/22 (13.6%)	19/37 (51.4%)	55/78 (70.5%)	121/173 (69.9%)	198/310 (63.9%)
BPI Challenge 2017 (van Dongen, 2017)	6/21	3/22 (13.6%)	15/37 (40.5%)	46/78 (59.0%)	102/173 (59.0%)	166/310 (53.5%)
IT Incident Log (Shifat, 2020)	8/21	3/22 (13.6%)	20/37 (54.1%)	56/78 (71.8%)	124/173 (71.7%)	203/310 (65.5%)
Italian Help Desk (Polato, 2017)	8/21	3/22 (13.6%)	20/37 (54.1%)	55/78 (70.5%)	124/173 (71.7%)	202/310 (65.2%)
Production (Levy, 2014)	9/21	8/22 (36.4%)	24/37 (64.9%)	56/78 (71.8%)	122/173 (70.5%)	210/310 (67.7%)
Simulated Atomic Log	21/21	19/22 (86.4%)	36/37 (97.3%)	77/78 (98.7%)	171/173 (98.8%)	303/310 (97.7%)
Simulated derivable interval event log	21/21	22/22 (100.0%)	37/37 (100.0%)	78/78 (100.0%)	173/173 (100.0%)	310/310 (100.0%)
Simulated uniquely matched derivable interval event log	21/21	22/22 (100.0%)	37/37 (100.0%)	78/78 (100.0%)	173/173 (100.0%)	310/310 (100.0%)
Simulated explicit interval event log	21/21	22/22 (100.0%)	37/37 (100.0%)	78/78 (100.0%)	173/173 (100.0%)	310/310 (100.0%)

Table 13: Calculated PPIs for each granularity for every considered event log

The number of PPIs that can be calculated for each event log is mainly determined by the number of relevant attributes that it contains. Since the four simulated event logs contained all relevant attributes, it was possible to calculate 100% of the PPIs on them (with the exception of atomic event logs, as described above). Since real-life event logs do not usually contain every single attribute considered in this work, it is not possible to calculate all PPIs for them. This mainly affects the cost dimension, as an average of only 17,66% of the PPIs in this dimension could be calculated for the publicly available real-life event logs. Nevertheless, the overall number of PPIs that can be calculated on these real-life event logs is of 63,16%, which increases to 82,26% when not considering the cost dimension. This coverage indicates that, even in the context of real-life event logs with a limited number of relevant attributes, the catalog enables the calculation of several indicators that can be used to measure processes performance.

An interesting observation is that, when grouping the PPIs by granularity, the number of PPIs that can be calculated for the activity instance and activity granularities tends to be lower than the number of PPIs that can be calculated for the case and group of cases granularities, which indicates that measuring performance at coarse granularity levels requires less specialized event attributes than what is required for measuring performance at the finer granularity levels.

7. Conclusions

This work addresses the need for a comprehensive and extensible catalog of PPIs that can be systematically accessed and reused by both practitioners and researchers engaged in data-driven business process management. This objective is achieved through a rigorous mathematical approach that provides formal definitions not only for the PPIs themselves, but also for all elements characterizing the input data required for their computation, namely event logs. By explicitly formalizing these foundational elements, the proposed catalog minimizes interpretation bias and establishes a shared, unambiguous basis for analyzing process performance across different contexts and applications.

While completeness cannot be guaranteed, the comprehensiveness of the catalog is supported by its grounding in existing literature that describes PPIs for the different dimensions of the DQ. Moreover, the formal basis underlying the catalog enables extensibility, allowing additional PPIs to be incorporated in a consistent manner as new indicators are identified or required. In this sense, the catalog is intended not as a static artifact, but as a structured and evolving repository of formally defined, data-driven performance indicators.

To facilitate practical adoption, a publicly available Python library has been developed to automate the computation of all PPIs included in the catalog. This library can be readily integrated into process analysis pipelines and serves as an executable counterpart to the formal definitions provided in the article. Its application to publicly available real-life event logs demonstrates the feasibility of computing the proposed PPIs using commonly available event data, which validates the practical applicability of the approach.

Several lines of future work are considered for this research. First, to further support adoption in industrial and applied settings, additional dissemination mechanisms may be explored, such as a dedicated website offering interactive access to the PPI catalog, or the diffusion of the catalog in relevant workshops and conferences of the field. Second, extending the formalizations to support the calculation of PPIs for object-centric event logs (Ghahfarokhi et al., 2021) is of interest. Preliminarily, this could be achieved by treating each object as a separate case, enabling PPI computation at the object level. Adapting existing approaches for defining cases in object-centric contexts (e.g., (Adams et al., 2022)) is also considered. Third, the inclusion of additional performance dimensions, such as sustainability (Wolf et al., 2025), would broaden the scope and utility of the catalog. Finally, future research could focus on providing decision-support guidelines to help practitioners select appropriate PPIs for specific process improvement initiatives, for example by identifying recurring PPI usage patterns in existing case studies and relating them to their specific contexts.

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Appendix A. Proofs

Proof of Proposition 1 (c). Let $e, e', e'' \in E$, $\#_{act}(e) = a$, and $\#_{case}(e) = c$. We know that $cmp(e, e')$ and $cmp(e', e'')$, and we want to prove that $cmp(e, e'')$. For any $x, x' \in E$, we have

$$cmp(x, x') \equiv \#_{act}(x) = \#_{act}(x') \wedge \#_{case}(x) = \#_{case}(x') \wedge actinstcmp(x, x').$$

Since $cmp(e, e')$, $\#_{act}(e) = a$, and $\#_{case}(e) = c$, we have that $\#_{act}(e') = a$ and $\#_{case}(e') = c$. As we also know that $cmp(e', e'')$, then $\#_{act}(e'') = a$ and $\#_{case}(e'') = c$.

From $cmp(e, e')$ and $cmp(e', e'')$, we also have that $actinstcmp(e, e')$ and $actinstcmp(e', e'')$. For any $x, x' \in E$, this means that

$$actinstcmp(x, x') \equiv (\#_{actinst}(x) \neq \perp \vee \#_{actinst}(x') \neq \perp) \implies \#_{actinst}(x) = \#_{actinst}(x').$$

We distinguish two cases:

Case 1. $\#_{actinst}(e) \neq \perp$, i.e., $\#_{actinst}(e)$ equals to some $i \in U_{actinst}$. In that case, since $actinstcmp(e, e')$, then $\#_{actinst}(e') = i$. As we also know that $actinstcmp(e', e'')$, then $\#_{actinst}(e'') = i$. Since $\#_{actinst}(e) \neq \perp$, $\#_{actinst}(e'') \neq \perp$, and $\#_{actinst}(e) = \#_{actinst}(e'')$, then $actinstcmp(e, e'')$.

Case 2. $\#_{actinst}(e) = \perp$. In that case, since $actinstcmp(e, e')$, then $\#_{actinst}(e') = \perp$. As we also know that $actinstcmp(e', e'')$, then $\#_{actinst}(e'') = \perp$. Since $\#_{actinst}(e) = \perp$ and $\#_{actinst}(e'') = \perp$, then $actinstcmp(e, e'')$.

Finally, given that $\#_{act}(e) = a$, $\#_{act}(e'') = a$, $\#_{case}(e) = c$, $\#_{case}(e'') = c$, and $actinstcmp(e, e'')$, it stands that $cmp(e, e'')$ □

Proof of Theorem 1. Let L be a derivable-interval event log. To prove theorem 1, it is necessary to show that:

Case 1. If there exist three events $e, e', e'' \in E$ such that:

$$e \not\prec e' \wedge e' \not\prec e \wedge cmp(e, e') \wedge cmp(e, e'') \wedge \\ ((e, e' \in E_s \wedge e'' \in E_c \wedge e'' \not\prec e) \vee (e, e' \in E_c \wedge e'' \in E_s \wedge e \not\prec e'')),$$

then there exists at least one pair of *match* functions, *match1* and *match2*, such that *match1* $\neq$ *match2*. Here, we distinguish the following cases:

Case 1.1. $\exists e, e', e'' \in E[e \not\prec e' \wedge e' \not\prec e \wedge cmp(e, e') \wedge cmp(e, e'') \wedge e, e' \in E_s \wedge e'' \in E_c \wedge e'' \not\prec e]$. In this case, $match1(e'') = e$ and $match2(e'') = e'$.

Case 1.2. $\exists e, e', e'' \in E[e \not\prec e' \wedge e' \not\prec e \wedge cmp(e, e') \wedge cmp(e, e'') \wedge e, e' \in E_c \wedge e'' \in E_s \wedge e \not\prec e'']$. In this case, $match1(e) = e''$, $match1(e') = e'$, $match2(e) = e$, and $match2(e') = e''$.

Case 2. If there not exist three events $e, e', e'' \in E$ such that:

$$e \not\prec e' \wedge e' \not\prec e \wedge cmp(e, e') \wedge cmp(e, e'') \wedge \\ ((e, e' \in E_s \wedge e'' \in E_c \wedge e'' \not\prec e) \vee (e, e' \in E_c \wedge e'' \in E_s \wedge e \not\prec e'')),$$

then it holds that, for any pair *match1* and *match2* of *match* functions, *match1* = *match2*. By contradiction, let *match1* and *match2* two distinct *match* functions for L, for the theorem to not hold, there must exist some event $e \in E_c$ such that $match1(e) \neq match2(e)$. We distinguish the following cases:

Case 2.1. $match1(e) = e$. In that case, $match2(e) = e'$ with $e \neq e'$.

Case 2.2. $match1(e) = e'$ (with $e \neq e'$). In that case, either $match2(e) = e$ (case 2.2a) or $match2(e) = e''$ (with $e'' \neq e'$ and $e'' \neq e$) (case 2.2b). Mutatis mutandis, case 2.2a is equal to case 2.1, so we only need to consider case 2.2b.

Case 1.1). The *match* function requires that, for all complete events x:

$$(x \not\prec match(x) \land match(x) \in E_s) \lor x = match(x).$$

Since $match1(e'') = e$, $e'' \not\prec e$, and $e \in E_s$, this requirement is fulfilled for *match1*. Similarly, since $match2(e'') = e'$, $e'' \not\prec e'$, and $e' \in E_s$, this requirement is also fulfilled for *match2*.

The second requirement is that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

We have that $match1(e'') = e$ and $cmp(e'', e)$, which fulfills the requirement for *match1*. The same stands for *match2*, as we have that $match2(e'') = e'$ and $cmp(e'', e')$.

The third requirement is that, for all pairs of complete events x, x':

$$(x \prec x' \land cmp(x, x')) \implies match(x') \not\prec match(x).$$

If we consider $E_c = \{e''\}$, this requirement is fulfilled for both *match1* and *match2*.

The last requirement is that, for all events x, for all complete events x', and all start events x'':

$$(match(x') = x \land cmp(x', x'') \land x \not\prec x'') \implies \begin{array}{l}(x \in E_s \land \#_{time}(x) = \#_{time}(x'')) \lor \\ \exists x''' \in E_c[match(x''') = x'' \land x' \not\prec x'''].\end{array}$$

If we consider $E = \{e, e', e''\}$, since $e, e' \in E_s$ and $e'' \in E_c$, with $match1(e'') = e$, the case of $x = e$, $x' = e''$, and $x'' = e'$ should be considered for *match1*. We have that $cmp(e'', e')$, $e \not\prec e'$, and $e, e' \in E_s$, so the implication stands true and the requirement is fulfilled for *match1*. For *match2*, as $match2(e'') = e'$, the case of $x = e'$, $x' = e''$, and $x'' = e$ should be considered. We have that $cmp(e'', e)$, $e' \not\prec e$, and $e, e' \in E_s$, so the implication stands true and the requirement is also fulfilled for *match2*.

Case 1.2). The *match* function requires that, for all complete events x:

$$(x \not\prec match(x) \land match(x) \in E_s) \lor x = match(x).$$

When considering *match1*, since $match1(e) = e''$, $e \not\prec e''$, and $e'' \in E_s$, this requirement is fulfilled for e. Since $e' = match1(e')$, this requirement is also fulfilled for e'. Following the same approach for *match2*, since $e = match2(e)$, this requirement is fulfilled for e. Since $match2(e') = e''$, $e' \not\prec e''$, and $e'' \in E_s$, this requirement is also fulfilled for e'.

The second requirement is that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

We have that $match1(e) = e''$, $cmp(e, e'')$, $match1(e') = e'$, and $cmp(e', e')$ (since *cmp* is reflexive), which fulfills the requirement for *match1*. The same stands for *match2*, as we have that $match2(e) = e$, $cmp(e, e)$ (since *cmp* is reflexive), $match2(e') = e''$, and $cmp(e', e'')$.

The third requirement is that, for all pairs of complete events x, x':

$$(x \prec x' \land cmp(x, x')) \implies match(x') \not\prec match(x).$$

If we consider $E_c = \{e, e'\}$, we have that $cmp(e, e')$, but $\#_{time}(e) = \#_{time}(e')$ (hence, $e \not\prec e'$ and $e' \not\prec e$), so the implication stands true and this requirement is fulfilled for both *match1* and *match2*.

The last requirement is that, for all events x, for all complete events x', and all start events x'':

$$(match(x') = x \land cmp(x', x'') \land x \not\prec x'') \implies \quad (x \in E_s \land \#_{time}(x) = \#_{time}(x'')) \lor \exists x''' \in E_c[match(x''') = x'' \land x' \not\prec x'''].$$

If we consider $E = \{e, e', e''\}$, since $e'' \in E_s$ and $e, e' \in E_c$, with $match1(e') = e'$, the case of $x = e'$, $x' = e'$, and $x'' = e''$ should be considered for *match1*. Since we have that $cmp(e', e'')$, $e' \not\prec e''$, and $e' \notin E_s$, we require that:

$$\exists x''' \in E_c[match(x''') = e'' \land e' \not\prec x'''].$$

We have that $e \in E_c$, $match1(e) = e''$, and $e' \not\prec e$, so the requirement is fulfilled for *match1*. For *match2*, as $match2(e) = e$, the case of $x = e$, $x' = e$, and $x'' = e''$ should be considered. Since we have that $cmp(e, e'')$, $e \not\prec e''$, and $e \notin E_s$, we require that:

$$\exists x''' \in E_c[match(x''') = e'' \land e \not\prec x'''].$$

We have that $e' \in E_c$, $match2(e') = e''$, and $e \not\prec e'$, so the requirement is fulfilled for *match2*.

Case 2.1). Given that for all complete events x we have that

$$(x \not\prec match(x) \land match(x) \in E_s) \lor x = match(x),$$

it follows from $e \in E_c$ and $e \neq match2(e)$, that $e \not\prec e'$ and $e' \in E_s$.

In addition,

$$\forall x \in E_c \forall x' \in E_c \cup E_s[match(x) = x' \implies cmp(x, x')],$$

Hence, as $match2(e) = e'$, we have that $cmp(e', e)$.

Another requirement of the *match* function is that, if an event is the match of a complete event, then all compatible start events that it does not precede have been matched with another complete event, i.e.,

$$\forall x \in E \forall x' \in E_c \forall x'' \in E_s[(match(x') = x \land cmp(x', x'') \land x \not\prec x'') \implies (x \in E_s \land \#_{time}(x) = \#_{time}(x'')) \lor \exists x''' \in E_c[match(x''') = x'' \land x' \not\prec x''']].$$

As we have that $e \in E_c$, $E_c \subseteq E$, $e' \in E_s$, $match1(e) = e$, $cmp(e, e')$, and $e \not\prec e'$, there should exist an $e'' \in E_c$ such that $match1(e'') = e'$ and $e \not\prec e''$. For this e'', it therefore holds that $e'' \not\prec e'$, $cmp(e', e'')$, and by transitivity $cmp(e, e'')$.

For this case, it is required that:

$$\neg \exists x, x', x'' \in E[x \not\prec x' \land x' \not\prec x \land cmp(x, x') \land cmp(x, x'') \land ((x, x' \in E_s \land x'' \in E_c \land x'' \not\prec x) \lor (x, x' \in E_c \land x'' \in E_s \land x \not\prec x''))].$$

Since $cmp(e, e'')$, $cmp(e, e')$, $e, e'' \in E_c$, $e' \in E_s$, $e \not\prec e'$, and $e \not\prec e''$, it should also hold that $e'' \prec e$.

Another requirement for *match* functions is that

$$\forall x, x' \in E_c[(x \prec x' \land cmp(x, x')) \implies match(x') \not\prec match(x)].$$

As $e'' \prec e$ and $cmp(e'', e)$, it follows that $match1(e) \not\prec match1(e'')$ and $match2(e) \not\prec match2(e'')$. Since $match1(e'') = e'$, $match1(e) = e$, $e'' \prec e$, and $e \not\prec e'$, the requirements of the *match* function hold for *match1*. We distinguish two cases for *match2*: $match2(e'') = e''$ (case 2.1a) and $match2(e'') = e'''$, with $e'' \neq e'''$ (case 2.1b).

For case 2.1a, we consider the requirement of the *match* function that, if an event is the match of a complete event, then all compatible start events that it does not precede have been matched with another complete event, i.e.,

$$\forall x \in E \forall x' \in E_c \forall x'' \in E_s[(\mathit{match}(x') = x \wedge \mathit{cmp}(x', x'') \wedge x \not\prec x'') \implies (x \in E_s \wedge \#_{\mathit{time}}(x) = \#_{\mathit{time}}(x'')) \vee \exists x''' \in E_c[\mathit{match}(x''') = x'' \wedge x' \not\prec x''']].$$

As we have that $e'' \in E_c$, $E_c \subseteq E$, $e' \in E_s$, $\mathit{match2}(e'') = e''$, $\mathit{cmp}(e'', e')$, and $e'' \not\prec e'$, there should exist an $e''' \in E_c$ such that $\mathit{match2}(e''') = e'$, $e''' \neq e''$, and $e'' \not\prec e'''$. However, we know that $\mathit{match2}(e) = e'$, and $e'' \prec e$, so $\mathit{match2}(e'') \neq e''$.

For case 2.1b, since $\mathit{match2}(e'') = e'''$, $\mathit{match2}(e) = e'$, and $\mathit{match2}(e) \not\prec \mathit{match2}(e'')$, it follows that $e'' \not\prec e'''$, $e' \not\prec e'''$, $\mathit{cmp}(e''', e'')$ and by transitivity $\mathit{cmp}(e''', e')$ and $\mathit{cmp}(e''', e)$.

For this case, it is required that:

$$\neg\exists x, x', x'' \in E[x \not\prec x' \wedge x' \not\prec x \wedge \mathit{cmp}(x, x') \wedge \mathit{cmp}(x, x'') \wedge ((x, x' \in E_s \wedge x'' \in E_c \wedge x'' \not\prec x) \vee (x, x' \in E_c \wedge x'' \in E_s \wedge x \not\prec x''))].$$

Since $\mathit{cmp}(e', e''')$, $\mathit{cmp}(e', e'')$, $e', e''' \in E_s$, $e'' \in E_c$, and $e'' \not\prec e'''$, it also holds that $\#_{\mathit{time}}(e') \neq \#_{\mathit{time}}(e''')$.

We also consider the requirement for *match* functions that

$$\forall x \in E \forall x' \in E_c \forall x'' \in E_s[(\mathit{match}(x') = x \wedge \mathit{cmp}(x', x'') \wedge x \not\prec x'') \implies (x \in E_s \wedge \#_{\mathit{time}}(x) = \#_{\mathit{time}}(x'')) \vee \exists x''' \in E_c[\mathit{match}(x''') = x'' \wedge x' \not\prec x''']].$$

As we have that $e' \in E_s$, $E_s \subseteq E$, $e'' \in E_c$, $e''' \in E_s$, $\mathit{match1}(e'') = e'$, $\mathit{cmp}(e'', e''')$, $e' \not\prec e'''$, and $\#_{\mathit{time}}(e') \neq \#_{\mathit{time}}(e''')$, there should exist an $e'''' \in E_c$ such that $\mathit{match1}(e'''') = e'''$, $e'''' \neq e''$, and $e'' \not\prec e''''$. For this e'''', it therefore holds that $e'''' \not\prec e'''$, $\mathit{cmp}(e''', e'''')$, and by transitivity $\mathit{cmp}(e'', e'''')$.

For this case, it is required that:

$$\neg\exists x, x', x'' \in E[x \not\prec x' \wedge x' \not\prec x \wedge \mathit{cmp}(x, x') \wedge \mathit{cmp}(x, x'') \wedge ((x, x' \in E_s \wedge x'' \in E_c \wedge x'' \not\prec x) \vee (x, x' \in E_c \wedge x'' \in E_s \wedge x \not\prec x''))].$$

Since $\mathit{cmp}(e'', e'''')$, $\mathit{cmp}(e'', e''')$, $e'', e'''' \in E_c$, $e''' \in E_s$, $e'' \not\prec e'''$, and $e'' \not\prec e''''$, it should also hold that $e'''' \prec e''$.

If we consider the requirement for *match* functions that

$$\forall x, x' \in E_c[(x \prec x' \wedge \mathit{cmp}(x, x')) \implies \mathit{match}(x') \not\prec \mathit{match}(x)].$$

As $e'''' \prec e''$ and $\mathit{cmp}(e'''', e'')$, it follows that $\mathit{match1}(e'') \not\prec \mathit{match1}(e'''')$ and $\mathit{match2}(e'') \not\prec \mathit{match2}(e'''')$. Since $\mathit{match1}(e'''') = e'''$, $\mathit{match1}(e'') = e'$, $e'''' \prec e''$, and $e' \not\prec e'''$, the requirements of the *match* function hold for *match1*. However, this leads us to equivalent cases as case 2.1a and case 2.1b for *match2*: $\mathit{match2}(e'''') = e''''$ (case 2.1a) and $\mathit{match2}(e'''') = e'''''$, with $e'''' \neq e'''''$ (case 2.1b).

Therefore, for both *match1* and *match2* to be valid *match* functions, there should exist a start event e''''' such that $\mathit{match2}(e'''') = e'''''$, which in turn would require that a complete event e'''''' exists such that $\mathit{match1}(e'''''') = e'''''$, ad infinitum. But this is impossible, as E is a finite set. Therefore, either *match1* or *match2* cannot be a valid *match* function.

Case 2.2b). Given that for all complete events x we have that

$$(x \not\prec \mathit{match}(x) \wedge \mathit{match}(x) \in E_s) \vee x = \mathit{match}(x).$$

it follows from $e \in E_c$, $e \neq \mathit{match1}(e)$, and $e \neq \mathit{match2}(e)$, that $e \not\prec e'$, $e \not\prec e''$, and $e', e'' \in E_s$.

In addition,

$$\forall x \in E_c \forall x' \in E_c \cup E_s[\mathit{match}(x) = x' \implies \mathit{cmp}(x, x')],$$

Hence, as $\mathit{match1}(e) = e'$ and $\mathit{match2}(e) = e''$, we have that $\mathit{cmp}(e', e)$, $\mathit{cmp}(e'', e)$, and by transitivity $\mathit{cmp}(e', e'')$.

For this case, it is required that:

$$\neg\exists x, x', x'' \in E[\#_{\mathit{time}}(x) = \#_{\mathit{time}}(x') \wedge \mathit{cmp}(x, x') \wedge \mathit{cmp}(x, x'') \wedge \\ ((x, x' \in E_s \wedge x'' \in E_c \wedge x'' \not\prec x) \vee (x, x' \in E_c \wedge x'' \in E_s \wedge x \not\prec x''))].$$

Since $\mathit{cmp}(e', e'')$, $\mathit{cmp}(e', e)$, $e', e'' \in E_s$, $e \in E_c$, and $e \not\prec e'$, it also holds that $\#_{\mathit{time}}(e') \neq \#_{\mathit{time}}(e'')$. We distinguish two cases, $e'' \prec e'$ and $e' \prec e''$. Mutatis mutandis, both cases are equal, so we only need to consider one. We will assume that $e' \prec e''$.

We consider the requirement of the *match* function that, if an event is the match of a complete event, then all compatible start events that it does not precede have been matched with another complete event, i.e.,

$$\forall x \in E \forall x' \in E_c \forall x'' \in E_s[(\mathit{match}(x') = x \wedge \mathit{cmp}(x', x'') \wedge x \not\prec x'') \implies \\ (x \in E_s \wedge \#_{\mathit{time}}(x) = \#_{\mathit{time}}(x'')) \vee \exists x''' \in E_c[\mathit{match}(x''') = x'' \wedge x' \not\prec x''']].$$

As we have that $e'' \in E_s$, $E_s \subseteq E$, $e \in E_c$, $e' \in E_s$, $\mathit{match2}(e) = e''$, $\mathit{cmp}(e, e')$, and $e' \prec e''$, there should exist an $e''' \in E_c$ such that $\mathit{match2}(e''') = e'$, $e''' \neq e$, and $e \not\prec e'''$. For this e''', it therefore holds that $e''' \not\prec e'$, $\mathit{cmp}(e', e''')$, and by transitivity $\mathit{cmp}(e, e''')$ and $\mathit{cmp}(e'', e''')$.

For this case, it is required that:

$$\neg\exists x, x', x'' \in E[x \not\prec x' \wedge x' \not\prec x \wedge \mathit{cmp}(x, x') \wedge \mathit{cmp}(x, x'') \wedge \\ ((x, x' \in E_s \wedge x'' \in E_c \wedge x'' \not\prec x) \vee (x, x' \in E_c \wedge x'' \in E_s \wedge x \not\prec x''))].$$

Since $\mathit{cmp}(e, e''')$, $\mathit{cmp}(e, e')$, $e, e''' \in E_c$, $e' \in E_s$, $e \not\prec e'$, and $e \not\prec e'''$, it should also hold that $e''' \prec e$.

We now consider the requirement for *match* functions that

$$\forall x, x' \in E_c[(x \prec x' \wedge \mathit{cmp}(x, x')) \implies \mathit{match}(x') \not\prec \mathit{match}(x)].$$

As $e''' \prec e$ and $\mathit{cmp}(e''', e)$, it follows that $\mathit{match1}(e) \not\prec \mathit{match1}(e''')$ and $\mathit{match2}(e) \not\prec \mathit{match2}(e''')$. Since $\mathit{match2}(e''') = e'$, $\mathit{match2}(e) = e''$, $e''' \prec e$, and $e'' \not\prec e'$, the requirements of the *match* function hold for *match1*. We distinguish three cases for *match1*: $\mathit{match1}(e''') = e''$ (case 2.2b.1), $\mathit{match2}(e''') = e'''$ (case 2.2b.2), and $\mathit{match2}(e''') = e''''$, with $e''' \neq e''''$ and $e'''' \neq e''$ (case 2.2b.3).

For case 2.2b.1, since $\mathit{match1}(e''') \prec \mathit{match1}(e)$, $\mathit{match1}(e''') = e''$, and $\mathit{match1}(e) = e'$, it should follow that $e' \not\prec e''$. However, we know that $e'' \not\prec e'$, so for $\mathit{match2}(e''') = e''$ to hold, we need that $e' \not\prec e'' \wedge e'' \not\prec e'$, but, for this case, it is required that:

$$\neg\exists x, x', x'' \in E[x \not\prec x' \wedge x' \not\prec x \wedge \mathit{cmp}(x, x') \wedge \mathit{cmp}(x, x'') \wedge \\ ((x, x' \in E_s \wedge x'' \in E_c \wedge x'' \not\prec x) \vee (x, x' \in E_c \wedge x'' \in E_s \wedge x \not\prec x''))].$$

As $\mathit{cmp}(e', e'')$, $\mathit{cmp}(e', e''')$, $e', e'' \in E_s$, $e''' \in E_c$, and $e''' \not\prec e'$, it should also hold that $e' \prec e'' \vee e'' \prec e'$, so $\mathit{match2}(e''') \neq e''$.

Mutatis mutandis, cases 2.2b.2 and 2.2b.3 lead to the same situation as Cases 2.1a and 2.1b. Therefore, for both *match1* and *match2* to be valid *match* functions, there should exist a start event e'''' such that $\mathit{match2}(e''') = e''''$, which in turn would require that a complete event e''''' exists such that $\mathit{match1}(e''''') = e''''$, ad infinitum. But this is impossible, as E is a finite set. Therefore, either *match1* or *match2* cannot be a valid *match* function. □

Proof of Lemma 1. Let L be a derivable-interval event log and let *match1* and *match2* two distinct *match* functions for L. For all sets of complete events that possess the same case and activity attributes $E_{c,a}$, it should hold that:

$$\sum_{e \in E_{c,a}} \#_{time}(e) - \#_{time}(match1(e)) = \sum_{e \in E_{c,a}} \#_{time}(e) - \#_{time}(match2(e)).$$

We distinguish the following cases:

Case 1. For all cases $c \in C$ and for all activities $a \in A$, $\neg \exists e, e' \in E_{c,a}[\#_{type}(e) = \#_{type}(e') \wedge e \not\prec e' \wedge e' \not\prec e]$. In that case, L is a uniquely-matched derivable-interval event log and, as proven for Theorem 2, it holds that, for all $e \in E_c$, $match1(e) = match2(e)$.

Case 2. For some case $c \in C$ and for some activity $a \in A$, $\exists e, e' \in E_{c,a}[\#_{type}(e) = \#_{type}(e') \wedge e \not\prec e' \wedge e' \not\prec e]$.

Case 1). Since it holds that, for all $e \in E_c$, $match1(e) = match2(e)$, then it also holds that, for all $e \in E_c$, $\#_{time}(e) - \#_{time}(match1(e)) = \#_{time}(e) - \#_{time}(match2(e))$, i.e., the sum of service times is the same.

Case 2). If we zoom in on a single complete event $e \in E_{c,a}$, we distinguish two possibilities, $match1(e) = e$ (case 2a), and $match1(e) = e'$, with $e' \in E_s$, $e \not\prec e'$ (case 2b).

For case 2a, it is possible that other complete events $e' \in E_{c,a}$ exist such that $e \not\prec e' \wedge e' \not\prec e$. If, for all of these events, $match1(e') = e'$, then no start event $e'' \in E_s$ exists that can be matched to them and, thus, for all $e' \in E_{c,a}[e \not\prec e' \wedge e' \not\prec e]$, $match2(e') = e'$. It holds, then, that for all $e \in E_{c,a}$, $match1(e) = match2(e)$ and $\#_{time}(e) - \#_{time}(match1(e)) = \#_{time}(e) - \#_{time}(match2(e))$, i.e., the sum of service times is the same.

If a complete event $e' \in E_{c,a}$ does exist such that $e \not\prec e' \wedge e' \not\prec e$ and $match1(e') = e''$, with $e'' \in E_s$ and $e' \not\prec e''$, then we can have that $match2(e) = e''$ and $match2(e') = e'$. In that case, it is certainly possible that

$$\#_{time}(e) - \#_{time}(match1(e)) \neq \#_{time}(e) - \#_{time}(match2(e))$$

and

$$\#_{time}(e') - \#_{time}(match1(e')) \neq \#_{time}(e') - \#_{time}(match2(e')).$$

However, since $\#_{time}(e) = \#_{time}(e')$, if we look at the sum of service times of both events, it holds that:

$$(\#_{time}(e) - \#_{time}(e)) + (\#_{time}(e') - \#_{time}(e'')) = (\#_{time}(e) - \#_{time}(e'')) + (\#_{time}(e') - \#_{time}(e')).$$

For case 2b, it is possible that other complete events $e'' \in E_{c,a}$ exist such that $e \not\prec e'' \wedge e'' \not\prec e$. If, for all of these events, $match1(e'') = e''$, then we have the same situation as case 2a. If a complete event $e'' \in E_{c,a}$ does exist such that $e \not\prec e'' \wedge e'' \not\prec e$ and $match1(e'') = e'''$, with $e''' \in E_s$, $e'' \not\prec e'''$, and $\#_{time}(e') \neq \#_{time}(e''')$, then we can have that $match2(e) = e'''$ and $match2(e'') = e'$. In that case, it is certainly possible that

$$\#_{time}(e) - \#_{time}(match1(e)) \neq \#_{time}(e) - \#_{time}(match2(e))$$

and

$$\#_{time}(e'') - \#_{time}(match1(e'')) \neq \#_{time}(e'') - \#_{time}(match2(e'')).$$

However, since $e \not\prec e'' \wedge e'' \not\prec e$, if we look at the sum of service times of both events, it holds that:

$$(\#_{time}(e) - \#_{time}(e')) + (\#_{time}(e'') - \#_{time}(e''')) = (\#_{time}(e) - \#_{time}(e''')) + (\#_{time}(e'') - \#_{time}(e')).$$

For case 2b, it is also possible that another start event $e'' \in E_s$ exists such that $\#_{case}(e') = \#_{case}(e'')$, $\#_{act}(e') = \#_{act}(e'')$, and $e' \not\prec e'' \wedge e'' \not\prec e'$. In that case, we can have that $match2(e) = e''$. However, since $e' \not\prec e'' \wedge e'' \not\prec e'$, it holds that:

$$\#_{time}(e) - \#_{time}(e') = \#_{time}(e) - \#_{time}(e'').$$

It is noted that, if additional events are considered, we arrive at the same situations as cases 2a and 2b. Thus, it holds that the sum of service times remains the same, regardless of the *match* function. □

Proof of Theorem 4. By contradiction. Let L be a explicit-interval event log and let *match1* and *match2* two distinct *match* functions for L. Hence, $match1(e) \neq match2(e)$ for some $e \in E_c$.

We distinguish the following cases:

Case 1. $match1(e) = e$. In that case, $match2(e) = e'$ with $e \neq e'$.

Case 2. $match1(e) = e'$ (with $e \neq e'$). In that case, either $match2(e) = e$ (case 2a) or $match2(e) = e''$ (with $e'' \neq e'$ and $e'' \neq e$) (case 2b). Mutatis mutandis, case 2a is equal to case 1, so we only need to consider case 2b.

Case 1). Given that for all complete events x we have that

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x),$$

it follows from $e \in E_c$ and $e \neq match2(e)$, that $e \not\prec e'$ and $e' \in E_s$.

In addition,

$$\forall x \in E_c \forall x' \in E_c \cup E_s[match(x) = x' \implies cmp(x, x')],$$

Hence, as $match2(e) = e'$, we have that $cmp(e', e)$.

Another requirement of the *match* function is that, if an event is the match of a complete event, then all compatible start events that it does not precede have been matched with another complete event, i.e.,

$$\forall x \in E \forall x' \in E_c \forall x'' \in E_s[(match(x') = x \wedge cmp(x', x'') \wedge x \not\prec x'') \implies (x \in E_s \wedge \#_{time}(x) = \#_{time}(x'')) \vee \exists x''' \in E_c[match(x''') = x'' \wedge x' \not\prec x''']].$$

As we have that $e \in E_c$, $E_c \subseteq E$, $e' \in E_s$, $match1(e) = e$, $cmp(e, e')$, and $e \not\prec e'$, there should exist an $e'' \in E_c$ such that $match1(e'') = e'$, $e'' \neq e$, and $e \not\prec e''$. For this e'', it therefore holds that $e'' \not\prec e'$, $cmp(e', e'')$, and by transitivity $cmp(e, e'')$.

A requirement of explicit-interval event logs is that no two events referring to the same activity instance can refer to the same lifecycle type of event, i.e.:

$$\neg \exists x, x' \in E[x \neq x' \wedge \#_{actinst}(x) = \#_{actinst}(x') \wedge \#_{type}(x) = \#_{type}(x')].$$

As $e, e'' \in E$, it must hold that:

$$e = e'' \vee \#_{actinst}(e) \neq \#_{actinst}(e'') \vee \#_{type}(e) \neq \#_{type}(e'')$$

However, we know that $e \neq e''$, $cmp(e, e'')$ (hence, $\#_{actinst}(e) = \#_{actinst}(e'')$), and $e, e'' \in E_c$ (hence, $\#_{type}(e) = \#_{type}(e'')$). Therefore. a valid e'' that satisfies the requirements for the *match* function does not exist for *match1*.

Case 2b). Given that for all complete events x we have that

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x),$$

it follows from $e \in E_c$, $e \neq match1(e)$, and $e \neq match2(e)$, that $e \not\prec e'$, $e \not\prec e''$, and $e', e'' \in E_s$.

In addition,

$$\forall x \in E_c \forall x' \in E_c \cup E_s [match(x) = x' \implies cmp(x, x')],$$

Hence, as $match1(e) = e'$ and $match2(e) = e''$, we have that $cmp(e', e)$, $cmp(e'', e)$, and by transitivity $cmp(e', e'')$.

We consider the requirement of explicit-interval event logs that no two events referring to the same activity instance can refer to the same lifecycle type of event, i.e.:

$$\neg \exists x, x' \in E[x \neq x' \land \#_{actinst}(x) = \#_{actinst}(x') \land \#_{type}(x) = \#_{type}(x')].$$

As $e', e'' \in E$, it must hold that:

$$e' = e'' \lor \#_{actinst}(e') \neq \#_{actinst}(e'') \lor \#_{type}(e') \neq \#_{type}(e'')$$

However, we know that $e' \neq e''$, $cmp(e', e'')$ (hence, $\#_{actinst}(e') = \#_{actinst}(e''))$, and $e', e'' \in E_s$ (hence, $\#_{type}(e') = \#_{type}(e''))$. Therefore. an event log that contains both e' and e'' would not be a explicit-interval event log. □

Proof of Theorem 5. For a *match* function to not exist, then, for a given event log L, it should not be possible to fulfill at least one of its requirements. First, *match* should be an injective function. As we have that $match \colon E_c \to E_s \cup E_c$, there will always be enough elements in the function's range ($E_s \cup E_c$) to define an injective function for all elements in the domain (E_c), i.e., $|E_c| \leq |E_s \cup E_c|$.

The *match* function requires that, for all complete events x:

$$(x \not\prec match(x) \land match(x) \in E_s) \lor x = match(x).$$

This requirement does not prevent the definition of a *match* function, either. If, for any complete event $e \in E_c$, no start event $e' \in E_s$ exists such that $e \not\prec e' \land match(e) = e'$, it is always possible to match e to itself, i.e., $match(e) = e$.

It is also necessary for the *match* function that, for all complete events x and all events x', that:

$$match(x) = x' \implies cmp(x, x').$$

This requirement does not prevent the definition of a *match* function, either. Since *cmp* is reflexive, it holds for any event $e \in E$ that $cmp(e, e)$.

Following the two previous requirements, it is indeed possible for a complete event $e \in E_c$ to be matched to some start event $e' \in E_s$, but only if $e \not\prec e' \land cmp(e, e')$. Otherwise, it will always be possible for $match(e) = e$ to satisfy both requirements.

Another requirement of the *match* function is that. for all pairs of complete events x, x':

$$(x \prec x' \land cmp(x, x')) \implies match(x') \not\prec match(x).$$

This requirement would not be fulfilled if a pair of compatible complete events e, e' exists such that $e \prec e' \land match(e') \prec match(e)$. When all events are matched to themselves, the above does not occur and this requirement holds. However, when there are complete events matched to start events, the above may occur. We distinguish the following cases:

Case 1.1. $match(e) = e''$, $match(e') = e'''$, with $e'', e''' \in E_s$, $e'' \neq e'''$.

Case 1.2. $match(e) = e$, $match(e') = e''$, with $e'' \in E_s$.

Case 1.3. $match(e) = e''$, $match(e') = e'$, with $e'' \in E_s$. Mutatis mutandis, this is the same situation as case 1.2.

Case 1.1). In this case, by reassigning the matched events through $reassign(e, e''')$ and $reassign(e', e'')$, it is possible to satisfy the requirement that $match(e') \not\prec match(e)$. We consider the requirement that, for all complete events x:

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x).$$

Since $e'', e''' \in E_s$, if we consider the original matches, we have that $e \not\prec e''$ and $e' \not\prec e'''$. As we also know that $e \prec e'$ and $e''' \prec e''$, then $e''' \prec e$ and $e'' \prec e'$, which satisfies this requirement for $match(e) = e'''$ and $match(e') = e''$.

Now we consider the requirement that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

If we consider the original matches, we have that $cmp(e, e'')$ and $cmp(e', e''')$. We also know that $cmp(e, e')$. Since cmp is transitive, then $cmp(e, e''')$ and $cmp(e', e'')$, which satisfies this requirement for $match(e) = e'''$ and $match(e') = e''$.

Case 1.2). In this case, by reassigning the matched events through $reassign(e, e'')$ and $reassign(e', e')$, it is possible to satisfy the requirement that $match(e') \not\prec match(e)$. We consider the requirement that, for all complete events x:

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x).$$

As we know that $e'' \prec e$, then this requirement is satisfied for $match(e) = e''$. Similarly, the requirement is also satisfied for $match(e') = e'$, as e' is matched to itself.

Now we consider the requirement that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

If we consider the original matches, we have that $cmp(e', e'')$. We also know that $cmp(e, e')$. Since cmp is both transitive and reflective, then $cmp(e, e'')$ and $cmp(e', e')$, which satisfies this requirement for $match(e) = e''$ and $match(e') = e'$.

Assume the existence of another complete event $y \in E_c$, $cmp(e, y)$. It is possible that, by performing the above reassignments of matched events for e and e', additional violations occur such that:

$$((y \prec e \wedge cmp(y, e)) \wedge match(e) \prec match(y)) \vee ((e \prec y \wedge cmp(e, y)) \wedge match(y) \prec match(e)) \vee$$
$$((y \prec e' \wedge cmp(y, e')) \wedge match(e') \prec match(y)) \vee ((e' \prec y \wedge cmp(e', y)) \wedge match(y) \prec match(e')).$$

However, it is noted that this would lead to the same situation as cases 1.1, 1.2, and 1.3. Let cb_0 be the result of $cbadness(E_c)$ before performing any reassignments over wrongly matched complete events. Let $cb_1, cb_2, ..., cb_n$ be the result of $cbadness(E_c)$ after performing $1, 2, ..., n$ reassignments. For a reassignment to be meaningful, it is required that, whenever a reassignment is performed, it holds that $cb_n < cb_{n-1}$. Since E_c is a finite set, by performing a finite number of meaningful reassignments, it is always possible to obtain a set of $match$ function assignments such that $cbadness(E_c) = 0$.

The final requirement of the $match$ function is that, for all events x, for all complete events x', and all start events x'', that:

$$(match(x') = x \wedge cmp(x', x'') \wedge x \not\prec x'') \implies \begin{array}{l}(x \in E_s \wedge \#_{time}(x) = \#_{time}(x'')) \vee \\ \exists x''' \in E_c[match(x''') = x'' \wedge x' \not\prec x'''].\end{array}$$

This requirement would not be fulfilled if, for any event e and any complete event e', a start event e'' exists such that all the requirements of the implication are met, but e is not a start event with the same timestamp as e'', and no suitable complete event exists that matches e'', i.e.:

$$(match(e') = e \wedge cmp(e', e'') \wedge e \not\prec e'')\wedge \\ (e \in E_s \wedge \#_{time}(e) \neq \#_{time}(e''))\wedge \quad \neg\exists x''' \in E_c[match(x''') = e'' \wedge e' \not\prec x'''].$$

We distinguish the following cases:

Case 2.1. $e \in E_s$, with $\#_{time}(e) \neq \#_{time}(e'')$.

Case 2.2. $e \in E_c$, i.e., $e = e'$.

Case 2.1). In this case, it is possible to reassign the matched events through $reassign(e', e'')$. As $e \not\prec e''$ and $\#_{time}(e) \neq \#_{time}(e'')$ (i.e., $e'' \prec e$), the implication is always true, thus satisfying the requirement. We consider the requirement that, for all complete events x:

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x).$$

Since $e \in E_s$, if we consider the original match, we have that $e' \not\prec e$. As we also know that $e'' \prec e$, then $e'' \prec e'$, which satisfies this requirement for $match(e') = e''$.

Now we consider the requirement that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

We know that $cmp(e', e'')$, which satisfies this requirement for $match(e') = e''$.

Case 2.2). In this case, it is possible to reassign the matched events through $reassign(e', e'')$. As $e \not\prec e''$ and $\#_{time}(e) \neq \#_{time}(e'')$ (i.e., $e'' \prec e$), the implication is always true, thus satisfying the requirement. We consider the requirement that, for all complete events x:

$$(x \not\prec match(x) \wedge match(x) \in E_s) \vee x = match(x).$$

Since $e \in E_c$ and $e = e'$, the requirement is satisfied for $match(e') = e$ as e' is matched to itself.

Now we consider the requirement that, for all complete events x and all events x':

$$match(x) = x' \implies cmp(x, x').$$

Since cmp is reflective and $e = e'$, then $cmp(e', e)$, which satisfies this requirement for $match(e') = e$.

Assume the existence of another complete event $y \in E_c$, $cmp(e', y)$. It is possible that, by performing the above reassignment of matched events for e', additional violations occur such that:

$$((y \prec e' \wedge cmp(y, e')) \wedge match(e') \prec match(y)) \vee ((e' \prec y \wedge cmp(e', y)) \wedge match(y) \prec match(e')).$$

However, it is noted that this would lead to the same situation as cases 1.1, 1.2, and 1.3. Let cb_0 be the result of $cbadness(E_c)$ before performing any reassignments over wrongly matched complete events. Let $cb_1, cb_2, ..., cb_n$ be the result of $cbadness(E_c)$ after performing $1, 2, ..., n$ reassignments. For a reassignment to be meaningful, it is required that, whenever a reassignment is performed, it holds that $cb_n < cb_{n-1}$. Since E_c is a finite set, by performing a finite number of meaningful reassignments, it is always possible to obtain a set of $match$ function assignments such that $cbadness(E_c) = 0$.

Similarly, assume the existence of another start event $z \in E_s$, $cmp(e', z)$. It is possible that, by performing the above reassignment of matched events for e', a new violation occurs such that:

$$\begin{aligned}(match(e') = e'' \wedge cmp(e', z'') \wedge e'' \not\prec z) \wedge \\ (e'' \in E_s \wedge \#_{time}(e'') \neq \#_{time}(z)) \wedge \quad \neg \exists x''' \in E_c[match(x''') = z \wedge e' \not\prec x'''].\end{aligned}$$

However, it is noted that this would lead to the same situation as cases 2.1 and 2.2. Let sb_0 be the result of $sbadness(E_s)$ before performing any reassignments over wrongly matched start events. Let $sb_1, sb_2, ..., sb_n$ be the result of $cbadness(E_s)$ after performing $1, 2, ..., n$ reassignments. For a reassignment to be meaningful, it is required that, whenever a reassignment is performed, it holds that $sb_n < sb_{n-1}$. Since E_s is a finite set, by performing a finite number of meaningful reassignments, it is always possible to obtain a set of *match* function assignments such that $cbadness(E_s) = 0$.

It must be noted that, by performing meaningful reassignments to wrongly matched start events, $cbadness(E_c)$ might increase. However, performing meaningful reassignments to wrongly matched complete events does not affect $sbadness(E_s)$. I.e., once enough meaningful reassignments of start events have been performed such that $sbadness(E_s) = 0$, it is always possible to perform a finite number of meaningful reassignments of complete events such that $cbadness(E_c) = 0$. □

Appendix B. Natural Language Descriptions of PPIs

Tables B.14, B.15, B.16, B.17, and B.18 show a natural language description for all PPIs that have been identified for different performance dimensions and granularities. Specifically, the PPIs are defined at the granularity of activity instances (denoted by I), activities (denoted by A), individual cases (denoted by C), or groups of cases (denoted by G). Each table corresponds to a specific performance dimension, namely General (G), Time (T), Cost (C), Quality (Q), and Flexibility (F). Each table row shows the PPI name, its performance dimension (D.), its granularity (G.), its description, potential use, attributes required for its computation, as well as any required additional input, and its desired optimization value (D. Value, whether the PPI must be minimized or maximized), with its associated assumptions.

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Activity count (case granularity)	G	C	Number of activities that occur in a case.	Calculation of several other PPIs.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	-	N/A
Activity count (group of cases granularity)	G	G	Number of activities that occur in a group of cases.	Calculation of several other PPIs.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	-	N/A
Expected Activity count	G	G	Expected number of activities that occur in a case belonging to a group of cases.	Calculation of several other PPIs.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	-	N/A
activity Instance count (activity granularity)	G	A	Number of times that an activity is instantiated in the event log.	Calculation of several other PPIs.	***act*** – Event attribute: activity	-	-	N/A
activity Instance count (case granularity)	G	C	Number of times that any activity is instantiated in a case.	Calculation of several other PPIs.	***case*** – Event attribute: case id	-	-	N/A
activity Instance count (group of cases granularity)	G	G	Number of times that any activity is instantiated in a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id	-	-	N/A
Expected activity Instance count	G	G	Expected number of times that any activity is instantiated in a case belonging to a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id	-	-	N/A

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case count (group of cases granularity)	G	G	Number of cases belonging to a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id	-	-	N/A
Human Resource count (activity granularity)	G	A	Number of human resources that are involved in the execution of an activity in the event log.	Calculation of several other PPIs.	***act*** – Event attribute: activity ***hres*** – Event attribute: human resource	-	-	N/A
Human Resource count (case granularity)	G	C	Number of human resources that are involved in the execution of a case.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	-	N/A
Human Resource count (group of cases granularity)	G	G	Number of human resources that are involved in the execution of a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	-	N/A
Expected Human Resource count	G	G	Expected number of human resources that are involved in the execution of a case belonging to a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	-	N/A
Resource count (activity granularity)	G	A	Number of resources that are involved in the execution of an activity in the event log.	Calculation of several other PPIs.	***act*** – Event attribute: activity ***res*** – Event attribute: resource	-	-	N/A
Resource count (case granularity)	G	C	Number of resources that are involved in the execution of a case.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	-	N/A
Resource count (group of cases granularity)	G	G	Number of resources that are involved in the execution of a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	-	N/A

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Resource count	G	G	Expected number of resources that are involved in the execution of a case belonging to a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	-	N/A
Role count (case granularity)	G	C	Number of human resource roles that are involved in the execution of a case.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	-	N/A
Role count (group of cases granularity)	G	G	Number of human resource roles that are involved in the execution of a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	-	N/A
Expected Role count	G	G	Expected number of human resource roles that are involved in the execution of a case belonging to a group of cases.	Calculation of several other PPIs.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	-	N/A

Table B.14: General PPI descriptions

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Active Time (case granularity)	T	C	Difference between the lead time and the idle time of a case.	Event log that contains lifecycle information for activity instances where the actual time that the case is active is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Maximizing active time is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Active Time	T	G	Expected difference between the lead time and the idle time of a case belonging to a group of cases.	Event log that contains lifecycle information for activity instances where the actual time that the case is active is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Maximizing active time is desirable.	MAX
Activity count (case granularity)	T	C	Number of activities that occur in a case.	If the event log does not contain timestamps, Activity count can be used as a proxy for the time performance of a case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of activities implies that the case takes longer to complete.	MIN
Activity count (group of cases granularity)	T	G	Number of activities that occur in a group of cases.	If the event log does not contain timestamps, Activity count can be used as a proxy for the time performance of a group of cases.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of activities implies that the cases in a group of cases take longer to complete.	MIN
Expected Activity count	T	G	Expected number of activities that occur in a case belonging to a group of cases.	If the event log does not contain timestamps, Expected Activity count can be used as a proxy for the time performance of a group of cases.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of activities implies that a case takes longer to complete.	MIN
activity Instance count (case granularity)	T	C	Number of times that any activity is instantiated in a case.	If the event log does not contain timestamps, activity Instance count can be used as a proxy for the time performance of a case.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	A greater number of activity instances implies that the case takes longer to complete.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
activity Instance count (group of cases granularity)	T	G	Number of times that any activity is instantiated in a group of cases.	If the event log does not contain timestamps, activity Instance count can be used as a proxy for the time performance of a group of cases.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	A greater number of activity instances implies that the cases in a group of cases take longer to complete.	MIN
Expected activity Instance count	T	G	Expected number of times that any activity is instantiated in a case belonging to a group of cases.	If the event log does not contain timestamps, Expected activity Instance count can be used as a proxy for the time performance of a group of cases.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	A greater number of activity instances implies that a case takes longer to complete.	MIN
Automated Activity count (case granularity)	T	C	Number of automated activities that occur in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX
Automated Activity count (group of cases granularity)	T	G	Number of automated activities that occur in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX
Expected Automated Activity count	T	G	Expected number of automated activities that occur in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX
Automated activity Instance count (case granularity)	T	C	Number of times that an automated activity is instantiated in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Automated activity Instance count (group of cases granularity)	T	G	Number of times that an automated activity is instantiated in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX
Expected Automated activity Instance count	T	G	Expected number of times that an automated activity is instantiated in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities take less time to be executed.	MAX
Automated Activity Service Time (case granularity)	T	C	Sum of the service time of all instantiations of automated activities in a case.	Event log that contains automated activities and lifecycle information for activity instances.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing service time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Automated Activity Service Time (group of cases granularity)	T	G	Sum of the service time of all instantiations of automated activities in a group of cases.	Event log that contains automated activities and lifecycle information for activity instances.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing service time is desirable.	MIN
Expected Automated Activity Service Time	T	G	Expected sum of the service time of all instantiations of automated activities in a case belonging to a group of cases.	Event log that contains automated activities and lifecycle information for activity instances.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing service time is desirable.	MIN
Case count and Lead Time Ratio (group of cases granularity)	T	G	Ratio between the number of cases belonging to a group of cases and the lead time of a group of cases.	Event log is associated with a process for which the number of cases executed within a specific period is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Maximizing the number of cases performed in a specific period is desirable.	MAX

Indicator	D. G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case Count where Lead Time Over value (group of cases granularity)	T G	Number of cases belonging to a group of cases whose lead time is greater than a given value.	Event log is associated with a process in which the lead time of a case should not exceed a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the elapsed time value that should not be exceeded	Exceeding the lead time value of interest is undesired.	MIN
Case Percentage where Lead Time Over value (group of cases granularity)	T G	Percentage of cases belonging to a group of cases whose lead time is greater than a given value.	Event log is associated with a process in which the lead time of a case should not exceed a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the elapsed time value that should not be exceeded	Exceeding the lead time value of interest is undesired.	MIN
Case Percentage with Missed Deadline (group of cases granularity)	T G	Percentage of cases belonging to a group of cases whose latest event occurs after a given deadline.	Event log is associated with a process in which cases should be completed before a defined deadline.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A timestamp indicating the deadline that should not be exceeded	Exceeding the deadline of interest is undesired.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Handover count (case granularity)	T	C	Number of times that the human resource associated with an activity instance differs from the human resource associated with the preceding activity instance within a case.	If actual time spent on a case is not of interest, Handover count can be used as a workload estimation within the time dimension.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	-	Reducing the times that work handover occurs is desirable.	MIN
Expected Handover count	T	G	Expected number of times that the human resource associated with an activity instance differs from the human resource associated with the preceding activity instance within a case belonging to a group of cases.	If actual time spent on a case is not of interest, Expected Handover count can be used as a workload estimation within the time dimension.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	-	Reducing the times that work handover occurs is desirable.	MIN
Idle Time (case granularity)	T	C	Sum of time periods in which no activity instance is executed during a case.	Event log that contains lifecycle information for activity instances in which the time the case is inactive is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing idle time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Idle Time	T	G	Expected sum of time periods in which no activity instance is executed during a case belonging to a group of cases.	Event log that contains lifecycle information for activity instances in which the time the case is inactive is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing idle time is desirable.	MIN
Lead Time (activity instance granularity)	T	I	Sum of the waiting time and the service time of an activity instance. Also known as cycle time.	Event log is associated with a process in which the total elapsed time of activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing lead time is desirable.	MIN
Lead Time (activity granularity)	T	A	Sum of the total elapsed time for all instantiations of an activity in the event log. Also known as cycle time.	Event log is associated with a process in which the total elapsed time of activity instances is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing lead time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Lead Time (case granularity)	T	C	Total elapsed time between the earliest and latest timestamps in a case. Also known as cycle time.	Event log is associated with a process in which the total elapsed time of a case is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing lead time is desirable.	MIN
Lead Time (group of cases granularity)	T	G	Total elapsed time between the earliest and latest events in a group of cases. Also known as cycle time.	Event log is associated with a process in which the total elapsed time of a group of cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing lead time is desirable.	MIN
Expected Lead Time	T	G	Expected total elapsed time between the earliest and latest timestamps in a case belonging to a group of cases. Also known as cycle time.	Event log is associated with a process in which the total elapsed time of a case is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing lead time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Lead Time and Case count Ratio (group of cases granularity)	T	G	Ratio between the lead time of a group of cases and the number of cases belonging to a group of cases.	Event log is associated with a process for which the number of cases executed within a specific period is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Maximizing the number of cases performed during a specific period is desirable.	MIN
Lead Time Deviation from time Limit (case granularity)	T	C	Difference between the time that a case is expected to take and its lead time.	Event log is associated with a process in which the lead time of a case should not exceed a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the time limit value that the case should not exceed	Exceeding the lead time limit value is undesired.	MAX
Expected Lead Time Deviation from time Limit	T	G	Difference between the time that a case in a group of cases is expected to take and its lead time.	Event log is associated with a process in which the lead time of a case should not exceed a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the time limit value that the case should not exceed	Exceeding the lead time limit value is undesired.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Lead Time Deviation from Expectation (case granularity)	T	C	Absolute value of the difference between the time that a case is expected to take and its lead time.	Event log is associated with a process in which the lead time of a case should be as close as possible to a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the time value that a case is expected to take	A lead time that is closer to the expected value is desirable.	MIN
Expected Lead Time Deviation from Expectation	T	G	Absolute value of the difference between the time that a case in a group of cases is expected to take and its lead time.	Event log is associated with a process in which the lead time of a case should be as close as possible to a specific value.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A number indicating the time value that a case is expected to take	A lead time that is closer to the expected value is desirable.	MIN
Lead Time from activity A (case granularity)	T	C	Total elapsed time between the earliest instantiation of a specific activity and the latest event of a case.	Event log is associated with a process in which the total elapsed time of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Minimizing lead time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Lead Time from activity A	T	G	Expected total elapsed time between the earliest instantiation of a specific activity and the latest activity event of a case belonging to a group of cases.	Event log is associated with a process in which the total elapsed time of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Minimizing lead time is desirable.	MIN
Lead Time from activity A to activity B (case granularity)	T	C	Total elapsed time between the earliest instantiation of a specific activity and the earliest instantiation of another specific activity that occurs after the former one in a case.	Event log is associated with a process in which the total elapsed time of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing lead time is desirable.	MIN
Expected Lead Time from activity A to activity B	T	G	Expected total elapsed time between the earliest instantiation of a specific activity and the earliest instantiation of another specific activity that occurs after the former one in a case belonging to a group of cases.	Event log is associated with a process in which the total elapsed time of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing lead time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Lead Time to activity A (case granularity)	T	C	Total elapsed time between the earliest event and the earliest instantiation of a specific activity in a case.	Event log is associated with a process in which the total elapsed time of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Minimizing lead time is desirable.	MIN
Expected Lead Time to activity A	T	G	Expected total elapsed time between the earliest event and the earliest instantiation of a specific activity in a case belonging to a group of cases.	Event log is associated with a process in which the total elapsed time of a specific is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Minimizing lead time is desirable.	MIN
Service and Lead Time Ratio (activity instance granularity)	T	I	Ratio between the service time and the lead time of an activity instance.	Event log in which the difference between service time and lead time is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between service time and lead time should be as large as possible.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Service and Lead Time Ratio (activity granularity)	T	A	Ratio between the service time and the lead time of all instantiations of an activity in the event log.	Event log in which the difference between service time and lead time is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between service time and lead time should be as large as possible.	MAX
Service and Lead Time Ratio (case granularity)	T	C	Ratio between the service time and the lead time of a case.	Event log in which the difference between service time and lead time is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between service time and lead time should be as large as possible.	MAX
Service and Lead Time Ratio (group of cases granularity)	T	G	Ratio between the service time and the lead time of all activity instances in a group of cases.	Event log in which the difference between service time and lead time is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between service time and lead time should be as large as possible.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Service and Lead Time Ratio	T	G	Expected ratio between the service time and the lead time of a case belonging to a group of cases.	Event log in which the difference between service time and lead time is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between service time and lead time should be as large as possible.	MAX
Service Time (activity instance granularity)	T	I	Elapsed time between the start and complete events of an activity instance.	Event log that contains lifecycle information for activity instances where the actual time spent executing them is of interest.	***actinst*** – Event attribute: activity instance ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing service time is desirable.	MIN
Service Time (activity granularity)	T	A	Sum of elapsed time between the start and complete events of all instantiations of an activity in the event log.	Event log that contains lifecycle information for activity instances where the actual time spent executing them is of interest.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing service time is desirable.	MIN
Service Time (case granularity)	T	C	Sum of elapsed time between the start and complete events of all activity instances of a case.	Event log that contains lifecycle information for activity instances where the actual time spent executing them is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing service time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Service Time (group of cases granularity)	T	G	Sum of elapsed time between the start and complete events of all activity instances of a group of cases.	Event log that contains lifecycle information for activity instances where the actual time spent executing them is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing service time is desirable.	MIN
Expected Service Time	T	G	Expected sum of elapsed time between the start and complete events of all activity instances of a case belonging to a group of cases.	Event log that contains lifecycle information for activity instances where the actual time spent executing them is of interest.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing service time is desirable.	MIN
Service Time from activity A to activity B (case granularity)	T	C	Sum of elapsed time between the start and complete events of all activity instances of a case, which occur between the earliest instantiation of a specific activity, and the earliest instantiation of another specific activity that occurs after the former one, including both.	Event log is associated with a process in which the actual time spent executing a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing service time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Service Time from activity A to activity B (group of cases granularity)	T	G	Sum of elapsed time between the start and complete events of all activity instances of every case belonging to a group of cases, which occur between the earliest instantiation of a specific activity, and the earliest instantiation of another specific activity that occurs after the former one, including both, within each case.	Event log is associated with a process in which the actual time spent executing a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing service time is desirable.	MIN
Expected Service Time from activity A to activity B	T	G	Expected sum of elapsed time between the start and complete events of all activity instances of a case belonging to a group of cases, which occur between the earliest instantiation of a specific activity, and the earliest instantiation of another specific activity that occurs after the former one, including both.	Event log is associated with a process in which the actual time spent executing a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing service time is desirable.	MIN
Waiting Time (activity instance granularity)	T	I	Elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it.	Event log in which the time spent between every activity instance is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing waiting time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Waiting Time (activity granularity)	T	A	Sum of elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it, for all the instantiations of a specific activity in the event log.	Event log in which the time spent between every activity instance is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing waiting time is desirable.	MIN
Waiting Time (case granularity)	T	C	Sum of elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it, for all activity instances in a case.	Event log in which the waiting time between every activity instance is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing waiting time is desirable.	MIN
Expected Waiting Time	T	G	Expected sum of elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it, for all activity instances in a case belonging to a group of cases.	Event log in which the time spent between every activity instance is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing waiting time is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Waiting Time from activity A to activity B (case granularity)	T	C	Sum of elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it, for all activity instances in a case that occur between the earliest instantiation of a specific activity and the earliest instantiation of another specific activity that occurs after the former one, including both.	Event log in which the time spent between every activity instance of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing waiting time is desirable.	MIN
Expected Waiting Time from activity A to activity B	T	G	Expected sum of elapsed time between the start event of an activity instance and the complete event of the activity instance that precedes it, for all activity instances in a case that occurs between the earliest instantiation of a specific activity and the earliest instantiation of another specific activity that occurs after the former one, including both, for all cases belonging to a group of cases.	Event log in which the time spent between every activity instance of a specific subprocess is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the first specific activity ***b*** – the second specific activity	Minimizing waiting time is desirable.	MIN

Table B.15: Time PPI descriptions

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Automated Activity Cost (case granularity)	C	C	Total cost associated with all instantiations of automated activities in a case.	Event log that contains automated activities and cost information.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing total cost is desirable.	MIN
Automated Activity Cost (group of cases granularity)	C	G	Total cost associated with all instantiations of automated activities in a group of cases.	Event log that contains automated activities and cost information.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing total cost is desirable.	MIN
Expected Automated Activity Cost	C	G	Expected total cost associated with all instantiations of automated activities in a case belonging to a group of cases.	Event log that contains automated activities and cost information.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Minimizing total cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Desired Activity count (case granularity)	C	C	Number of instantiated activities whose occurrence is desirable in a case.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Desired Activity count (group of cases granularity)	C	G	Number of instantiated activities whose occurrence is desirable in a group of cases.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Expected Desired Activity count	C	G	Expected number of instantiated activities whose occurrence is desirable in a case belonging to a group of cases.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Direct Cost (case granularity)	C	C	Total cost associated with all instantiations of activities that have a direct effect on the outcome of a case.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN
Direct Cost (group of cases granularity)	C	G	Total cost associated with all instantiations of activities that have a direct effect on the outcome of cases in a group of cases.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Direct Cost	C	G	Expected total cost associated with all instantiations of activities that have a direct effect on the outcome of a case belonging to a group of cases.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN
Fixed Cost considering single events of activity instances (activity instance granularity)	C	I	Fixed cost associated with an activity instance, measured as the latest recorded value among the events of the activity instance.	Event log in which fixed cost information is recorded.	***actinst*** – Event attribute: activity instance ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN
Fixed Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	Fixed cost associated with an activity instance, measured as the sum of all values among the events of the activity instance.	Event log in which fixed cost information is recorded.	***actinst*** – Event attribute: activity instance ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN
Fixed Cost (activity granularity)	C	A	Sum of fixed cost of all instantiations of an activity in the event log.	Event log in which fixed cost information is recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Fixed Cost (case granularity)	C	C	Sum of fixed cost of all activity instances of a case.	Event log in which fixed cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN
Fixed Cost (group of cases granularity)	C	G	Sum of fixed cost of all activity instances of a group of cases.	Event log in which fixed cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN
Expected Fixed Cost	C	G	Expected fixed cost of a case belonging to a group of cases.	Event log in which fixed cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***fc*** – Event attribute: fixed cost ***type*** – Event attribute: lifecycle type	-	Minimizing fixed cost is desirable.	MIN
Human Resource and Case count Ratio (group of cases granularity)	C	G	Ratio between the number of human resources that are involved in the execution of cases in a group of cases, and the number of cases belonging to a group of cases.	Event log in which the optimal allocation of human resources to cases allows an improvement in cost performance.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources per case is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Human Resource count (activity granularity)	C	A	Number of human resources that are involved in the execution of an activity.	If the event log does not contain cost information, Human Resource count can be used as a proxy for the cost performance of an activity.	***act*** – Event attribute: activity ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable.	MIN
Human Resource count (case granularity)	C	C	Number of human resources that are involved in the execution of a case.	If the event log does not contain cost information, Human Resource count can be used as a proxy for the cost performance of a case.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable.	MIN
Human Resource count (group of cases granularity)	C	G	Number of human resources that are involved in the execution of a group of cases.	If the event log does not contain cost information, Human Resource count can be used as a proxy for the cost performance of a group of cases.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable.	MIN
Expected Human Resource count	C	G	Expected number of human resources that are involved in the execution of a case belonging to a group of cases.	If the event log does not contain cost information, Expected Human Resource count can be used as a proxy for the cost performance of a group of cases.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable.	MIN
Inventory Cost considering single events of activity instances (activity instance granularity)	C	I	Inventory cost associated with an activity instance, measured as the latest recorded value among the events of the activity instance.	Event log in which inventory cost information is recorded.	***actinst*** – Event attribute: activity instance ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Inventory Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	Inventory cost associated with an activity instance, measured as the sum of all values among the events of the activity instance.	Event log in which inventory cost information is recorded.	***actinst*** – Event attribute: activity instance ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN
Inventory Cost (activity granularity)	C	A	Inventory cost associated with all instantiations of an activity in the event log.	Event log in which inventory cost information is recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN
Inventory Cost (case granularity)	C	C	Inventory cost associated with all activity instances of a case.	Event log in which inventory cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN
Inventory Cost (group of cases granularity)	C	G	Inventory cost associated with all activity instances of a group of cases.	Event log in which inventory cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Inventory Cost	C	G	Expected inventory cost associated with all activity instances of a case belonging to a group of cases.	Event log in which inventory cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***ic*** – Event attribute: inventory cost ***type*** – Event attribute: lifecycle type	-	Minimizing inventory cost is desirable.	MIN
Labor Cost and Total Cost Ratio (activity instance granularity)	C	I	Ratio between the labor cost associated with an activity instance and the total cost associated with an activity instance.	Event log in which both labor and total cost information are recorded.	***actinst*** – Event attribute: activity instance ***lc*** – Event attribute: labor cost ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost associated with human resources is desirable.	MIN
Labor Cost and Total Cost Ratio (activity granularity)	C	A	Ratio between the labor cost associated with all instantiations of an activity in the event log, and the total cost associated with all instantiations of an activity in the event log.	Event log in which both labor and total cost information are recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***lc*** – Event attribute: labor cost ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost associated with human resources is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Labor Cost and Total Cost Ratio (case granularity)	C	C	Ratio between the labor cost associated with all activity instances of a case, and the total cost associated with all activity instances of a case.	Event log in which both labor and total cost information are recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost associated with human resources is desirable.	MIN
Labor Cost and Total Cost Ratio (group of cases granularity)	C	G	Ratio between the labor cost associated with all activity instances of a group of cases, and the total cost associated with all activity instances of a group of cases.	Event log in which both labor and total cost information are recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost associated with human resources is desirable.	MIN
Expected Labor Cost and Total Cost Ratio	C	G	Expected ratio between the labor cost associated with all activity instances of a group of cases, and the total cost associated with all activity instances of a group of cases.	Event log in which both labor and total cost information are recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost associated with human resources is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Labor Cost considering single events of activity instances (activity instance granularity)	C	I	Labor cost associated with an activity instance, measured as the latest recorded value among the events of the activity instance.	Event log in which labor cost information is recorded.	***actinst*** – Event attribute: activity instance ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN
Labor Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	Labor cost associated with an activity instance, measured as the sum of all values among the events of the activity instance.	Event log in which labor cost information is recorded.	***actinst*** – Event attribute: activity instance ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN
Labor Cost (activity granularity)	C	A	Labor cost associated with all instantiations of the activity in the event log.	Event log in which labor cost information is recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN
Labor Cost (case granularity)	C	C	Labor cost associated with all activity instances of a case.	Event log in which labor cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Labor Cost (group of cases granularity)	C	G	Labor cost associated with all activity instances of a group of cases.	Event log in which labor cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN
Expected Labor Cost	C	G	Expected labor cost associated with all activity instances of a case belonging to a group of cases.	Event log in which labor cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***lc*** – Event attribute: labor cost ***type*** – Event attribute: lifecycle type	-	Minimizing labor cost is desirable.	MIN
Maintenance Cost (case granularity)	C	C	Maintenance cost associated with a case.	Process where the cost of conducting maintenance is relevant.	***case*** – Event attribute: case id ***mainc*** – Case attribute: maintenance cost	-	Minimizing maintenance cost is desirable.	MIN
Maintenance Cost (group of cases granularity)	C	G	Maintenance cost associated with all cases in a group of cases.	Process where the cost of conducting maintenance is relevant.	***case*** – Event attribute: case id ***mainc*** – Case attribute: maintenance cost	-	Minimizing maintenance cost is desirable.	MIN
Expected Maintenance Cost	C	G	Expected maintenance cost associated with a case belonging to a group of cases.	Process where the cost of conducting maintenance is relevant.	***case*** – Event attribute: case id ***mainc*** – Case attribute: maintenance cost	-	Minimizing maintenance cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Missed Deadline Cost (case granularity)	C	C	Cost for missing deadlines associated with a case.	Process where missing a deadline incurs in additional costs.	***case*** – Event attribute: case id ***mdc*** – Case attribute: missed deadline cost	-	Minimizing missed deadline cost is desirable.	MIN
Missed Deadline Cost (group of cases granularity)	C	G	Cost for missing deadlines associated with all cases in a group of cases.	Process where missing a deadline incurs in additional costs.	***case*** – Event attribute: case id ***mdc*** – Case attribute: missed deadline cost	-	Minimizing missed deadline cost is desirable.	MIN
Expected Missed Deadline Cost	C	G	Expected cost for missing deadlines associated with a case belonging to a group of cases.	Process where missing a deadline incurs in additional costs.	***case*** – Event attribute: case id ***mdc*** – Case attribute: missed deadline cost	-	Minimizing missed deadline cost is desirable.	MIN
Overhead Cost (case granularity)	C	C	Total cost associated with all instantiations of activities that do not have a direct effect on the outcome of a case.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Overhead Cost (group of cases granularity)	C	G	Total cost associated with all instantiations of activities that do not have a direct effect on the outcome of cases in a group of cases.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN
Expected Overhead Cost	C	G	Expected total cost associated with all instantiations of activities that do not have a direct effect on the outcome of a case belonging to a group of cases.	Event log in which activities can be categorized based on whether they directly contribute to the outcome of the process or not.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	***DCl*** – List of direct cost activities	Minimizing total cost is desirable.	MIN
Resource count (activity granularity)	C	A	Number of resources that are involved in the execution of an activity in the event log.	If the event log does not contain cost information, Resource count can be used as a proxy for the cost performance of an activity.	***act*** – Event attribute: activity ***res*** – Event attribute: resource	-	Minimizing the number of resources involved is desirable.	MIN
Resource count (case granularity)	C	C	Number of resources that are involved in the execution of a case.	If the event log does not contain cost information, Resource count can be used as a proxy for the cost performance of a case.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	Minimizing the number of resources involved is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Resource count (group of cases granularity)	C	G	Number of resources that are involved in the execution of a group of cases.	If the event log does not contain cost information, Resource count can be used as a proxy for the cost performance of a group of cases.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	Minimizing the number of resources involved is desirable.	MIN
Expected Resource count	C	G	Expected number of resources that are involved in the execution of a case belonging to a group of cases.	If the event log does not contain cost information, Expected Resource count can be used as a proxy for the cost performance of a group of cases.	***case*** – Event attribute: case id ***res*** – Event attribute: resource	-	Minimizing the number of resources involved is desirable.	MIN
Rework Cost (activity granularity)	C	A	Total cost of all times that an activity is instantiated again, after its first instantiation, in any case of the event log.	Event log that contains cost information where measuring rework is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost of rework is desirable.	MIN
Rework Cost (case granularity)	C	C	Total cost of all times that any activity is instantiated again, after its first instantiation, in a case.	Event log that contains cost information where measuring rework is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost of rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Cost (group of cases granularity)	C	G	Sum of total cost of all times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	Event log that contains cost information where measuring rework is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost of rework is desirable.	MIN
Expected Rework Cost	C	G	Expected total cost of all times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	Event log that contains cost information where measuring rework is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing the cost of rework is desirable.	MIN
Rework Count (activity granularity)	C	A	Number of times that an activity is instantiated again, after its first instantiation, in any case of the event log.	If the event log does not contain cost information, Rework Count can be used as a proxy for the cost performance of an activity.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Count (case granularity)	C	C	Number of times that any activity is instantiated again, after its first instantiation, in a case.	If the event log does not contain cost information, Rework Count can be used as a proxy for the cost performance of a case.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Count (group of cases granularity)	C	G	Number of times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	If the event log does not contain cost information, Rework Count can be used as a proxy for the cost performance of a group of cases.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Expected Rework Count	C	G	Expected number of times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	If the event log does not contain cost information, Expected Rework Count can be used as a proxy for the cost performance of a group of cases.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Percentage (activity granularity)	C	A	Percentage of times that an activity is instantiated again, after its first instantiation, in any case of the event log.	If the event log does not contain cost information, Rework Percentage can be used as a proxy for the cost performance of an activity.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Percentage (case granularity)	C	C	Percentage of times that any activity is instantiated again, after its first instantiation, in a case.	If the event log does not contain cost information, Rework Percentage can be used as a proxy for the cost performance of a case.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Percentage (group of cases granularity)	C	G	Percentage of times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	If the event log does not contain cost information, Rework Percentage can be used as a proxy for the cost performance of a group of cases.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Rework Percentage	C	G	Expected percentage of times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	If the event log does not contain cost information, Expected Rework Percentage can be used as a proxy for the cost performance of a group of cases.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Total Cost considering single events of activity instances (activity instance granularity)	C	I	Total cost associated with an activity instance, measured as the latest recorded value among the events of an activity instance.	Event log in which total cost information is recorded.	***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN
Total Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	Total cost associated with an activity instance, measured as the sum of all values among the events of an activity instance.	Event log in which total cost information is recorded.	***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN
Total Cost (activity granularity)	C	A	Total cost associated with all instantiations of an activity in the event log.	Event log in which total cost information is recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost (case granularity)	C	C	Total cost associated with all activity instances of a case.	Event log in which total cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN
Total Cost (group of cases granularity)	C	G	Total cost associated with all activity instances of a group of cases.	Event log in which total cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN
Expected Total Cost	C	G	Expected total cost associated with all activity instances of a case belonging to a group of cases.	Event log in which total cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	Minimizing total cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost and Lead Time Ratio (activity instance granularity)	C	I	Ratio between the total cost associated with an activity instance and the lead time of an activity instance.	Event log in which the total cost in relation to the total time taken for conducting activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and lead time should be as close to 0 as possible.	MIN
Total Cost and Lead Time Ratio (activity granularity)	C	A	Ratio between the total cost associated with all instantiations of an activity in the event log, and the sum of lead time for all instantiations of an activity in the event log.	Event log in which the total cost in relation to the total time taken for conducting activity instances is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and lead time should be as close to 0 as possible.	MIN
Total Cost and Lead Time Ratio (case granularity)	C	C	Ratio between the total cost of a case and the lead time of a case.	Event log in which the total cost in relation to the total time taken for conducting cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and lead time should be as close to 0 as possible.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost and Lead Time Ratio (group of cases granularity)	C	G	Ratio between the total cost of a group of cases and the lead time of a group of cases.	Event log in which the total cost in relation to the total time taken for conducting the cases of a group of cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and lead time should be as close to 0 as possible.	MIN
Expected Total Cost and Lead Time Ratio	C	G	Ratio between the expected total cost of a case belonging to a group of cases and the expected lead time of a case belonging to a group of cases.	Event log in which the total cost in relation to the total time taken for conducting cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and lead time should be as close to 0 as possible.	MIN
Total Cost and outcome Unit Ratio (activity instance granularity)	C	I	Ratio between the total cost associated with an activity instance and the outcome units associated with an activity instance.	Event log in which the total cost in relation to the outcome units of activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Ratio between total cost and outcome unit count should be as close to 0 as possible.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost and outcome Unit Ratio (activity granularity)	C	A	Ratio between the total cost associated with all instantiations of an activity in the event log, and the outcome units associated with all instantiations of an activity in the event log.	Event log in which the total cost in relation to the outcome units of activity instances is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Ratio between total cost and outcome unit count should be as close to 0 as possible.	MIN
Total Cost and outcome Unit Ratio (case granularity)	C	C	Ratio between the total cost associated with all activity instances of a case, and the outcome units associated with all activity instances of a case.	Event log in which the total cost in relation to the outcome units of cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Ratio between total cost and outcome unit count should be as close to 0 as possible.	MIN
Total Cost and outcome Unit Ratio (group of cases granularity)	C	G	Ratio between the total cost associated with all activity instances of a group of cases, and the outcome units associated with all activity instances of a group of cases.	Event log in which the total cost in relation to the outcome units of the cases of a group of cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Ratio between total cost and outcome unit count should be as close to 0 as possible.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Total Cost and outcome Unit Ratio	C	G	Ratio between the expected total cost of a case belonging to a group of cases, and the expected outcome units associated with all activity instances of a case belonging to a group of cases.	Event log in which the total cost in relation to the outcome units of cases is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Ratio between total cost and outcome unit count should be as close to 0 as possible.	MIN
Total Cost and Service Time Ratio (activity instance granularity)	C	I	Ratio between the total cost associated with an activity instance, and the service time of an activity instance.	Event log in which the total cost in relation to the time taken for conducting activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and service time should be as close to 0 as possible.	MIN
Total Cost and Service Time Ratio (activity granularity)	C	A	Ratio between the total cost associated with all instantiations of an activity, and the sum of the service time of all instantiations of an activity.	Event log in which the total cost in relation to the time taken for conducting activity instances is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and service time should be as close to 0 as possible.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost and Service Time Ratio (case granularity)	C	C	Ratio between the total cost of a case and the service time of a case.	Event log in which the total cost in relation to the time taken for conducting activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and service time should be as close to 0 as possible.	MIN
Total Cost and Service Time Ratio (group of cases granularity)	C	G	Ratio between the total cost of a group of cases, and the service time of a group of cases.	Event log in which the total cost in relation to the time taken for conducting activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and service time should be as close to 0 as possible.	MIN
Expected Total Cost and Service Time Ratio	C	G	Ratio between the expected total cost of a case belonging to a group of cases, and the expected service time of a case belonging to a group of cases.	Event log in which the total cost in relation to the time taken for conducting activity instances is of relevance.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***tc*** – Event attribute: total cost ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Ratio between total cost and service time should be as close to 0 as possible.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Transportation Cost (case granularity)	C	C	Transportation cost associated with a case.	Process where the cost of transportation is relevant.	***case*** – Event attribute: case id ***transc*** – Case attribute: transportation cost	-	Minimizing transportation cost is desirable.	MIN
Transportation Cost (group of cases granularity)	C	G	Sum of transportation cost associated with all cases in a group of cases.	Process where the cost of transportation is relevant.	***case*** – Event attribute: case id ***transc*** – Case attribute: transportation cost	-	Minimizing transportation cost is desirable.	MIN
Expected Transportation Cost	C	G	Expected transportation cost of a case belonging to a group of cases.	Process where the cost of transportation is relevant.	***case*** – Event attribute: case id ***transc*** – Case attribute: transportation cost	-	Minimizing transportation cost is desirable.	MIN
Variable Cost considering single events of activity instances (activity instance granularity)	C	I	Variable cost associated with an activity instance, measured as the latest recorded value among the events of the activity instance.	Event log in which variable cost information is recorded.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN
Variable Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	Variable cost associated with an activity instance, measured as the sum of all values among the events of the activity instance.	Event log in which variable cost information is recorded.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Variable Cost (activity granularity)	C	A	Sum of variable cost of all instantiations of an activity in the event log.	Event log in which variable cost information is recorded.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN
Variable Cost (case granularity)	C	C	Sum of variable cost of all activity instances of a case.	Event log in which variable cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN
Variable Cost (group of cases granularity)	C	G	Sum of variable cost of all activity instances of a group of cases.	Event log in which variable cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Variable Cost	C	G	Expected variable cost of a case belonging to a group of cases.	Event log in which variable cost information is recorded.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***vc*** – Event attribute: variable cost	-	Minimizing variable cost is desirable.	MIN
Warehousing Cost (case granularity)	C	C	Warehousing cost associated with all activity instances of a case.	Process where the cost of warehousing is relevant.	***case*** – Event attribute: case id ***warec*** – Case attribute: warehousing cost	-	Minimizing warehousing cost is desirable.	MIN
Warehousing Cost (group of cases granularity)	C	G	Warehousing cost associated with all activity instances of a group of cases.	Process where the cost of warehousing is relevant.	***case*** – Event attribute: case id ***warec*** – Case attribute: warehousing cost	-	Minimizing warehousing cost is desirable.	MIN
Expected Warehousing Cost	C	G	Expected warehousing cost associated with all activity instances of a case belonging to a group of cases.	Process where the cost of warehousing is relevant.	***case*** – Event attribute: case id ***warec*** – Case attribute: warehousing cost	-	Minimizing warehousing cost is desirable.	MIN

Table B.16: Cost PPI descriptions

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
activity Instance count by Human Resource (activity granularity)	Q	A	Number of times a specific human resource performs an activity in the event log.	Process where individual performance of human resources is relevant.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	***hr*** – the specific human resource	Maximizing the number of activities conducted by the same human resource is desirable.	MAX
activity Instance count by Human Resource (case granularity)	Q	C	Number of times a specific human resource instantiates any activity in a case.	Process where individual performance of human resources is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	***hr*** – the specific human resource	Maximizing the number of activities conducted by the same human resource is desirable.	MAX
activity Instance count by Human Resource (group of cases granularity)	Q	G	Number of times a specific human resource instantiates any activity in a group of cases.	Process where individual performance of human resources is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	***hr*** – the specific human resource	Maximizing the number of activities conducted by the same human resource is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected activity Instance count by Human Resource	Q	G	Expected number of times a specific human resource instantiates any activity in a case belonging to a group of cases.	Process where individual performance of human resources is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource ***type*** – Event attribute: lifecycle type	***hr*** – the specific human resource	Maximizing the number of activities conducted by the same human resource is desirable.	MAX
activity Instance count by Role (case granularity)	Q	C	Number of times that any activity is instantiated by a specific role in a case.	Process where performance of human resources with certain roles is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***role*** – Event attribute: role ***type*** – Event attribute: lifecycle type	***rl*** – the specific role	Maximizing the number of activities conducted by human resources with the specific role is desirable.	MAX
activity Instance count by Role (group of cases granularity)	Q	G	Number of times that any activity is instantiated by a specific role in a group of cases.	Process where performance of human resources with certain roles is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***role*** – Event attribute: role ***type*** – Event attribute: lifecycle type	***rl*** – the specific role	Maximizing the number of activities conducted by the same role is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected activity Instance count by Role	Q	G	Expected number of times that any activity is instantiated by a specific role in a case belonging to a group of cases.	Process where performance of human resources with certain roles is relevant.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***role*** – Event attribute: role ***type*** – Event attribute: lifecycle type	***rl*** – the specific role	Maximizing the number of activities conducted by the same role is desirable.	MAX
Automated Activity count (case granularity)	Q	C	Number of automated activities that occur in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX
Automated Activity count (group of cases granularity)	Q	G	Number of automated activities that occur in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX
Expected Automated Activity count	Q	G	Expected number of automated activities that occur in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX
Automated activity Instance count (case granularity)	Q	C	Number of times that an automated activity is instantiated in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Automated activity Instance count (group of cases granularity)	Q	G	Number of times that an automated activity is instantiated in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX
Expected Automated activity Instance count	Q	G	Expected number of times that an automated activity is instantiated in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Automated activities yield more reliable, higher-quality results.	MAX
Case and Client count Ratio (group of cases granularity)	Q	G	Ratio between the number of cases in a group of cases and the number of distinct clients associated with cases in a group of cases.	Process is executed for different clients.	***case*** – Event attribute: case id ***cli*** – Case attribute: client	-	It is desirable to maximize the number of cases per client.	MAX
Case Count where Activity after Time Frame (group of cases granularity)	Q	G	Number of cases in a group of cases where a specific activity occurs after a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***et*** – the end timestamp of the timeframe	Maximizing the number of cases where the activity occurs after the timeframe is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case Count where Activity before Time Frame (group of cases granularity)	Q	G	Number of cases in a group of cases where a specific activity occurs before a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***st*** – the start timestamp of the timeframe	Maximizing the number of cases where the activity occurs before the timeframe is desirable.	MAX
Case Count where Activity during Time Frame (group of cases granularity)	Q	G	Number of cases in a group of cases where a specific activity occurs within a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***st*** – the start timestamp of the timeframe ***et*** – the end timestamp of the timeframe	Maximizing the number of cases where the activity occurs within the timeframe is desirable.	MAX
Case Count where end activity is A (group of cases granularity)	Q	G	Number of cases in a group of cases where a specific activity is the last one to be instantiated it.	Process is expected to end with specific activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Maximizing the number of cases that end with the specific activity is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case Count where start activity is A (group of cases granularity)	Q	G	Number of cases in a group of cases where a specific activity is the first one to be instantiated it.	Process is expected to start with specific activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Maximizing the number of cases that start with the specific activity is desirable.	MAX
Case Count with Rework (group of cases granularity)	Q	G	Number of cases in a group of cases where there is rework.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing the number of cases with rework is desirable.	MIN
Case Percentage where Activity after Time Frame (group of cases granularity)	Q	G	Percentage of cases in a group of cases where a specific activity occurs after a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***et*** – the end timestamp of the timeframe	Maximizing the number of cases where the activity occurs after the timeframe is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case Percentage where Activity before Time Frame (group of cases granularity)	Q	G	Percentage of cases in a group of cases where a specific activity occurs before a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***st*** – the start timestamp of the timeframe	Maximizing the number of cases where the activity occurs before the timeframe is desirable.	MAX
Case Percentage where Activity during Time Frame (group of cases granularity)	Q	G	Percentage of cases in a group of cases where a specific activity occurs within a specific timeframe.	Event log is associated with a process in which specific timeframes must be taken into consideration.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***a*** – the considered activity ***st*** – the start timestamp of the timeframe ***et*** – the end timestamp of the timeframe	Maximizing the number of cases where the activity occurs within the timeframe is desirable.	MAX
Case Percentage where End Activity is A (group of cases granularity)	Q	G	Percentage of cases in a group of cases where a specific activity is the last instantiated one.	Process is expected to end with specific activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Maximizing the number of cases that end with the specific activity is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Case Percentage where Start Activity is A (group of cases granularity)	Q	G	Percentage of cases in a group of cases where a specific activity is the first one to be instantiated it.	Process is expected to start with specific activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***a*** – the specific activity	Maximizing the number of cases that start with the specific activity is desirable.	MAX
Case Percentage with Missed Deadline (group of cases granularity)	Q	G	Percentage of cases in a group of cases whose latest event occurs after a given deadline.	Event log is associated with a process in which cases should be completed before a defined deadline.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	***val*** – A timestamp indicating the deadline that should not be exceeded	Exceeding the deadline of interest is undesired.	MIN
Case Percentage with Rework (group of cases granularity)	Q	G	Percentage of cases in a group of cases where there is rework.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing the number of cases with rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Client count and Total Cost Ratio (activity granularity)	Q	A	Ratio between the number of distinct clients associated with cases where an activity is instantiated in the event log, and the total cost associated with all instantiations of an activity in the event log.	Process is executed for different clients where the event log contains cost information.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the specific activity for as many clients at the least cost.	MAX
Client count and Total Cost Ratio (group of cases granularity)	Q	G	Ratio between the number of distinct clients associated with cases in a group of cases, and the total cost of a group of cases.	Process is executed for different clients where the event log contains cost information.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the process for as many clients at the least cost.	MAX
Expected Client count and Total Cost Ratio	Q	G	Ratio between the number of distinct clients associated with cases in a group of cases, and the total cost of a group of cases.	Process is executed for different clients where the event log contains cost information.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the process for as many clients at the least cost.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Desired Activity count (case granularity)	Q	C	Number of instantiated activities whose occurrence is desirable in a case.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Desired Activity count (group of cases granularity)	Q	G	Number of instantiated activities whose occurrence is desirable in a group of cases.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Expected Desired Activity count	Q	G	Expected number of instantiated activities whose occurrence is desirable in a case belonging to a group of cases.	Event log that contains optional activities whose occurrence is desirable.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Desl*** – List of desirable activities	Desirable activities should be maximized.	MAX
Human Resource count (activity granularity)	Q	A	Number of human resources that are involved in the execution of an activity in the event log.	Human Resource count can be used as a proxy to measure the quality of the process.	***act*** – Event attribute: activity ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable, as it reduces the possibility of errors due to, e.g., handovers of work.	MIN
Human Resource count (case granularity)	Q	C	Number of human resources that are involved in the execution of a case.	Human Resource count can be used as a proxy to measure the quality of the process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable, as it reduces the possibility of errors due to, e.g., handovers of work.	MIN
Human Resource count (group of cases granularity)	Q	G	Number of human resources that are involved in the execution of cases in a group of cases.	Human Resource count can be used as a proxy to measure the quality of the process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable, as it reduces the possibility of errors due to, e.g., handovers of work.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Human Resource count	Q	G	Expected number of human resources that are involved in the execution of a case belonging to a group of cases.	Human Resource count can be used as a proxy to measure the quality of the process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved is desirable, as it reduces the possibility of errors due to, e.g., handovers of work.	MIN
Non-Automated Activity count (case granularity)	Q	C	Number of non-automated activities that occur in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN
Non-Automated Activity count (group of cases granularity)	Q	G	Number of non-automated activities that occur in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN
Expected Non-Automated Activity count (group of cases granularity)	Q	G	Expected number of non-automated activities that occur in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN
Non-Automated activity Instance count (case granularity)	Q	C	Number of times that a non-automated activity is instantiated in a case.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Non-Automated activity Instance count (group of cases granularity)	Q	G	Number of times that a non-automated activity is instantiated in a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN
Expected Non-Automated activity Instance count	Q	G	Expected number of times that a non-automated activity is instantiated in a case belonging to a group of cases.	Event log that contains automated activities.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Autl*** – List of automated activities	Non-automated activities are prone to errors, which reduces the quality of the process.	MIN
outcome Unit count considering single events of activity instances (activity instance granularity)	Q	I	Outcome units associated with an activity instance, measured as the latest recorded value among the events of the activity instance.	Process where the outcome of activity instances can be measured.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX
outcome Unit count considering the sum of all events of activity instances (activity instance granularity)	Q	I	Outcome units associated with an activity instance, measured as the sum of all values among the events of the activity instance.	Process where the outcome of activity instances can be measured.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
outcome Unit count (activity granularity)	Q	A	Outcome units associated with all instantiations of an activity in the event log.	Process where the outcome of activity instances can be measured.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX
outcome Unit count (case granularity)	Q	C	Outcome units associated with all activity instances in a case.	Process where the outcome of activity instances can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX
outcome Unit count (group of cases granularity)	Q	G	Outcome units associated with all activity instances in a group of cases.	Process where the outcome of activity instances can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected outcome Unit count	Q	G	Expected outcome units associated with all activity instances in a case belonging to a group of cases.	Process where the outcome of activity instances can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***unt*** – Event attribute: outcome units	-	Maximizing outcome unit count is desirable.	MAX
Overall Quality (case granularity)	Q	C	Overall quality associated with the outcome of a case.	Event log contains information that allows measuring overall quality, such as customer satisfaction or a quantifier of the outcome of the process.	***case*** – Event attribute: case id ***qual*** – Case attribute: quality indicator	-	Maximizing the quality of the process is desirable.	MAX
Expected Overall Quality	Q	G	Expected overall quality associated with the outcome of a case belonging to a group of cases.	Event log contains information that allows measuring overall quality, such as customer satisfaction or a quantifier of the outcome of the process.	***case*** – Event attribute: case id ***qual*** – Case attribute: quality indicator	-	Maximizing the quality of the process is desirable.	MAX
Repeatability (case granularity)	Q	C	One minus the ratio between the number of activities that occur in a case, and the number of times that an activity is instantiated in a case.	Process where activity repetition is common.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	It is desirable not to repeat activities.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Repeatability (group of cases granularity)	Q	G	One minus the ratio between the number of activities that occur in a group of cases, and the number of times that an activity is instantiated in a group of cases.	Process where activity repetition is common.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	It is desirable not to repeat activities.	MIN
Expected Repeatability	Q	G	One minus the expected ratio between the number of activities that occur in a case belonging to a group of cases, and the number of times that an activity is instantiated in a case belonging to a group of cases.	Process where activity repetition is common.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id	-	It is desirable not to repeat activities.	MIN
Rework Count (activity granularity)	Q	A	Number of times that an activity is instantiated again, after its first instantiation, in any case of the event log.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Count (case granularity)	Q	C	Number of times that any activity is instantiated again, after its first instantiation, in a case.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Count (group of cases granularity)	Q	G	Sum of the number of times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Expected Rework Count	Q	G	Expected number of times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Count by Value (activity granularity)	Q	A	Number of times that an activity is instantiated again, after it is instantiated a certain number of times, in any case of the event log.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Count by Value (case granularity)	Q	C	Number of times that any activity is instantiated again, after it is instantiated a certain number of times, in a case.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN
Rework Count by Value (group of cases granularity)	Q	G	Number of times that any activity is instantiated again, after it is instantiated a certain number of times, in every case of a group of cases.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN
Expected Rework Count by Value	Q	G	Expected number of times that any activity is instantiated again, after it is instantiated a certain number of times, in a case belonging to a group of cases.	Process where rework recurrently occurs.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework of activities Subset (case granularity)	Q	C	Number of times that any activity belonging to a subset of activities is instantiated again, after its first instantiation, in a case.	Process where rework occurs recurrently, but only the rework of some activities is of interest.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***sub*** – Subset of activities to consider	Minimizing rework is desirable.	MIN
Rework of activities Subset (group of cases granularity)	Q	G	Number of times that any activity belonging to a subset of activities is instantiated again, after its first instantiation, in all case of a group of cases.	Process where rework occurs recurrently, but only the rework of some activities is of interest.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***sub*** – Subset of activities to consider	Minimizing rework is desirable.	MIN
Expected Rework of activities Subset	Q	G	Expected number of times that any activity belonging to a subset of activities is instantiated again, after its first instantiation, in a case belonging to a group of cases.	Process where rework occurs recurrently, but only the rework of some activities is of interest.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***sub*** – Subset of activities to consider	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Percentage (activity granularity)	Q	A	Percentage of times that an activity is instantiated again, after its first instantiation, in any case of the event log.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Percentage (case granularity)	Q	C	Percentage of times that any activity is instantiated again, after its first instantiation, in a case.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Percentage (group of cases granularity)	Q	G	Percentage of times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Rework Percentage	Q	G	Expected percentage of times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Minimizing rework is desirable.	MIN
Rework Percentage by Value (activity granularity)	Q	A	Percentage of times that an activity is instantiated again, after it is instantiated a certain number of times, in any case in the event log.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN
Rework Percentage by Value (case granularity)	Q	C	Percentage of times that any activity is instantiated again, after it is instantiated a certain number of times, in a case.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Percentage by Value (group of cases granularity)	Q	G	Percentage of times that any activity is instantiated again, after it is instantiated a certain number of times, in every case of a group of cases.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN
Expected Rework Percentage by Value	Q	G	Expected percentage of times that any activity is instantiated again, after it is instantiated a certain number of times, in a case belonging to a group of cases.	Process where rework occurs recurrently.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***val*** – Number of times an activity can be instantiated without being considered as rework	Minimizing rework is desirable.	MIN
Rework Time (activity granularity)	Q	A	Sum of the elapsed time for all times an activity is instantiated again, after its first instantiation, in any case in the event log.	Process where measuring the time spent on rework activities is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing the time spent on reworking is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Rework Time (case granularity)	Q	C	Sum of the elapsed time for all times that any activity is instantiated again, after its first instantiation, in a case.	Process where measuring the time spent on rework activities is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing the time spent on reworking is desirable.	MIN
Rework Time (group of cases granularity)	Q	G	Sum of the elapsed time for all times that any activity is instantiated again, after its first instantiation, in every case of a group of cases.	Process where measuring the time spent on rework activities is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing the time spent on reworking is desirable.	MIN
Expected Rework Time	Q	G	Expected sum of the elapsed time for all times that any activity is instantiated again, after its first instantiation, in a case belonging to a group of cases.	Process where measuring the time spent on rework activities is of relevance.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***time*** – Event attribute: timestamp ***type*** – Event attribute: lifecycle type	-	Minimizing the time spent on reworking is desirable.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Successful outcome Unit Count (activity instance granularity)	Q	I	Outcome units associated with an activity instance, after deducting those that were unsuccessfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the number of successful outcome units is desirable.	MAX
Successful outcome Unit Count (activity granularity)	Q	A	Outcome units associated with all instantiations of an activity in the event log, after deducting those that were unsuccessfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the number of successful outcome units is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Successful outcome Unit Count (case granularity)	Q	C	Outcome units associated with all activity instances in a case, after deducting those that were unsuccessfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the number of successful outcome units is desirable.	MAX
Successful outcome Unit Count (group of cases granularity)	Q	G	Outcome units associated with all activity instances in a group of cases, after deducting those that were unsuccessfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the number of successful outcome units is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Successful outcome Unit Count	Q	G	Expected outcome units associated with all activity instances in a case belonging to a group of cases, after deducting those that were unsuccessfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the number of successful outcome units is desirable.	MAX
Successful outcome Unit Percentage (activity instance granularity)	Q	I	Percentage of outcome units associated with an activity instance that are successfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the percentage of successful outcome units is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Successful outcome Unit Percentage (activity granularity)	Q	A	Percentage of outcome units associated with all instantiations of an activity that are successfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the percentage of successful outcome units is desirable.	MAX
Successful outcome Unit Percentage (case granularity)	Q	C	Percentage of outcome units associated with all activity instances in a case that are successfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the percentage of successful outcome units is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Successful outcome Unit Percentage (group of cases granularity)	Q	G	Percentage outcome units associated with all activity instances in a group of cases that are successfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the percentage of successful outcome units is desirable.	MAX
Expected Successful outcome Unit Percentage	Q	G	Expected percentage of outcome units associated with all activity instances of a case belonging to a group of cases that are successfully completed.	Process where the outcome of activity instances, as well as whether they are successful or not, can be measured.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type ***uns*** – Event attribute: unsuccessful outcome units ***unt*** – Event attribute: outcome units	-	Maximizing the percentage of successful outcome units is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Total Cost and Client count Ratio (activity granularity)	Q	A	Ratio between the total cost associated with all instantiations of an activity in the event log, and the number of distinct clients associated with cases where an activity is instantiated in the event log.	Process is executed for different clients, where the event log contains cost information.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the specific activity for as many clients at the least cost.	MIN
Total Cost and Client count Ratio (group of cases granularity)	Q	G	Ratio between the total cost associated with all activity instances of a group of cases, and the number of distinct clients associated with cases in a group of cases.	Process is executed for different clients, where the event log contains cost information.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the process for as many clients at the least cost.	MIN
Expected Total Cost and Client count Ratio	Q	G	Ratio between the total cost associated with all activity instances of a group of cases, and the number of distinct clients associated with cases in a group of cases.	Process is executed for different clients, where the event log contains cost information.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***cli*** – Case attribute: client ***tc*** – Event attribute: total cost ***type*** – Event attribute: lifecycle type	-	It is desirable to execute the process for as many clients at the least cost.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Unwanted Activity count (case granularity)	Q	C	Number of unwanted activities that occur in a case.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Unwanted Activity count (group of cases granularity)	Q	G	Number of unwanted activities that occur in a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Expected Unwanted Activity count	Q	G	Expected number of unwanted activities that occur in a case belonging to a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Unwanted Activity Percentage (case granularity)	Q	C	Percentage of unwanted activities that occur in a case.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Unwanted Activity Percentage (group of cases granularity)	Q	G	Percentage of unwanted activities that occur in a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Expected Unwanted Activity Percentage	Q	G	Expected percentage of unwanted activities that occur in a case belonging to a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Unwanted activity Instance count (case granularity)	Q	C	Number of times that an unwanted activity is instantiated in a case.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Unwanted activity Instance count (group of cases granularity)	Q	G	Number of times that an unwanted activity is instantiated in a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Expected Unwanted activity Instance count	Q	G	Expected number of times that an unwanted activity is instantiated in a case belonging to a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Unwanted activity Instance Percentage (case granularity)	Q	C	Percentage of times that an unwanted activity is instantiated in a case.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Unwanted activity Instance Percentage (group of cases granularity)	Q	G	Percentage of times that an unwanted activity is instantiated in a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN
Expected Unwanted activity Instance Percentage	Q	G	Expected percentage of times that an unwanted activity is instantiated in a case belonging to a group of cases.	Event log that contains activities whose execution is unwanted.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	***Unwl*** – List of unwanted activities	Unwanted activities negatively affect the quality of the process.	MIN

Table B.17: Quality PPI descriptions

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Activity and Role count Ratio (case granularity)	F	C	Ratio between the number of activities that occur in a case, and the number of human resource roles that are involved in the execution of a case.	Process where several roles are involved.	***act*** – Event attribute: activity ***case*** – Event attribute: case id ***role*** – Event attribute: role	-	Minimizing the number of human resource roles involved in the execution of the case is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Activity and Role count Ratio (group of cases granularity)	F	G	Ratio between the number of activities that occur in a group of cases, and the number of human resource roles that are involved in the execution of a group of cases.	Process where several roles are involved.	***act*** – Event attribute: activity ***case*** – Event attribute: case id ***role*** – Event attribute: role	-	Minimizing the number of human resource roles involved in the execution of a group of cases is desirable.	MAX
Expected Activity and Role count Ratio	F	G	Ratio between the expected number of activities that occur in a case belonging to a group of cases, and the expected number of human resource roles that are involved in the execution of a case belonging to a group of cases.	Process where several roles are involved.	***act*** – Event attribute: activity ***case*** – Event attribute: case id ***role*** – Event attribute: role	-	Minimizing the number of human resource roles involved in the execution of a case is desirable.	MAX
activity Instance and Human Resource count Ratio (activity granularity)	F	A	Ratio between the number of times that an activity is instantiated in the event log, and the number of human resources that are involved in the execution of an activity in the event log.	Process where several human resources are involved.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved in the execution of an activity is desirable.	MAX
activity Instance and Human Resource count Ratio (case granularity)	F	C	Ratio between the number of times that any activity is instantiated in a case, and the number of human resources that are involved in the execution of a case.	Process where several human resources are involved.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved in the execution of a case is desirable.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
activity Instance and Human Resource count Ratio (group of cases granularity)	F	G	Ratio between the number of times that any activity is instantiated in a group of cases, and the number of human resources that are involved in the execution of a group of cases.	Process where several human resources are involved.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved in the execution of a group of cases is desirable.	MAX
Expected activity Instance and Human Resource count Ratio	F	G	Ratio between the expected number of times that any activity is instantiated in a case belonging to a group of cases, and the expected number of human resources that are involved in the execution of a case belonging to a group of cases.	Process where several human resources are involved.	***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Minimizing the number of human resources involved in the execution of a case is desirable.	MAX
Client count (activity granularity)	F	A	Number of distinct clients associated with cases where the activity is instantiated in the event log.	Process is executed for different clients.	***act*** – Event attribute: activity ***case*** – Event attribute: case id ***cli*** – Case attribute: client	-	It is desirable to maximize the number of clients.	MAX
Client count (group of cases granularity)	F	G	Number of distinct clients associated with cases in a group of cases.	Process is executed for different clients.	***case*** – Event attribute: case id ***cli*** – Case attribute: client	-	It is desirable to maximize the number of clients.	MAX
Expected Client count	F	G	Ratio between the number of distinct clients associated with cases in a group of cases, and the number of cases belonging to a group of cases.	Process is executed for different clients.	***case*** – Event attribute: case id ***cli*** – Case attribute: client	-	It is desirable to maximize the number of clients.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Directly-Follows Relations and Activity count Ratio (case granularity)	F	C	Ratio between the number of activity pairs where one is instantiated directly after the other in a case, and the number of activities that occur in a case.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX
Directly-Follows Relations and Activity count Ratio (group of cases granularity)	F	G	Ratio between the number of activity pairs where one is instantiated directly after the other in a group of cases, and the number of activities that occur in a group of cases.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX
Expected Directly-Follows Relations and Activity count Ratio	F	G	Ratio between the expected number of activity pairs where one is instantiated directly after the other in a case belonging to a group of cases, and the expected number of activities that occur in a case belonging to a group of cases.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Directly-Follows Relations count (activity granularity)	F	A	Number of activities that have been instantiated directly after the activity of interest in the event log.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX
Directly-Follows Relations count (case granularity)	F	C	Number of activity pairs where one is instantiated directly after the other in a case.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX
Directly-Follows Relations count (group of cases granularity)	F	G	Number of activity pairs where one is instantiated directly after the other in a group of cases.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Directly-Follows Relations count	F	G	Expected number of activity pairs where one is instantiated directly after the other in a case belonging to a group of cases.	Process where several distinct directly-follows relations can occur.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of directly-follows relations is desirable, as it implies a greater capability to address unexpected situations.	MAX
Human Resource count (activity granularity)	F	A	Number of human resources that are involved in the execution of an activity in the event log.	Human Resource count can be used as a proxy to measure the flexibility of a process.	***act*** – Event attribute: activity ***hres*** – Event attribute: human resource	-	Maximizing the number of human resources involved is desirable, as it implies a greater capability to address unexpected situations.	MAX
Human Resource count (case granularity)	F	C	Number of human resources that are involved in the execution of a case.	Human Resource count can be used as a proxy to measure the flexibility of a process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Maximizing the number of human resources involved is desirable, as it implies a greater capability to address unexpected situations.	MAX
Human Resource count (group of cases granularity)	F	G	Number of human resources that are involved in the execution of a group of cases.	Human Resource count can be used as a proxy to measure the flexibility of a process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Maximizing the number of human resources involved is desirable, as it implies a greater capability to address unexpected situations.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Expected Human Resource count	F	G	Expected number of human resources that are involved in the execution of a case belonging to a group of cases.	Human Resource count can be used as a proxy to measure the flexibility of a process.	***case*** – Event attribute: case id ***hres*** – Event attribute: human resource	-	Maximizing the number of human resources involved is desirable, as it implies a greater capability to address unexpected situations.	MAX
Optional Activity count (case granularity)	F	C	Number of optional activities that are instantiated in a case. An activity is considered optional if there is at least one case in the event log where it does not occur.	fProcess with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX
Optional Activity count (group of cases granularity)	F	G	Number of optional activities that are instantiated in a group of cases. An activity is considered optional if there is at least one case in the event log where it does not occur.	Process with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX
Expected Optional Activity count	F	G	Expected number of optional activities that are instantiated in a case belonging to a group of cases. An activity is considered optional if there is at least one case in the event log where it does not occur.	Process with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Optionality (case granularity)	F	C	Ratio between the number of optional activities that are instantiated in a case, and the number of activities that occur in a case. An activity is considered optional if there is at least one case in the event log where it does not occur.	Process with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX
Optionality (group of cases granularity)	F	G	Ratio between the number of optional activities that are instantiated in a group of cases, and the number of activities that occur in a group of cases. An activity is considered optional if there is at least one case in the event log where it does not occur.	Process with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX
Expected Optionality	F	G	Ratio between the expected number of optional activities that are instantiated in a case belonging to a group of cases, and the expected number of activities that occur in a case belonging to a group of cases. An activity is considered optional if there is at least one case in the event log where it does not occur.	Process with multiple activities that are not instantiated in every case.	***act*** – Event attribute: activity ***case*** – Event attribute: case id	-	A greater number of optional activities is desired, as it implies a greater capability to address unexpected situations.	MAX

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Role and Variant count Ratio (group of cases granularity)	F	G	Ratio between the number of human resource roles that are involved in the execution of a case, and the number of variants that are observed in a group of cases.	Process where several roles are involved	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***role*** – Event attribute: role ***type*** – Event attribute: lifecycle type	-	Maximizing the number of roles involved in the execution of different process variants is desirable.	MAX
Role count (case granularity)	F	C	Number of human resource roles that are involved in the execution of a case.	Process where several roles are involved.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	A greater number of roles involved in the execution of the process implies greater flexibility.	MAX
Role count (group of cases granularity)	F	G	Number of human resource roles that are involved in the execution of a group of cases.	Process where several roles are involved.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	A greater number of roles involved in the execution of the process implies greater flexibility.	MAX
Expected Role count	F	G	Expected number of human resource roles that are involved in the execution of a case belonging to a group of cases.	Process where several roles are involved.	***case*** – Event attribute: case id ***role*** – Event attribute: role	-	A greater number of roles involved in the execution of the process implies greater flexibility.	MAX
Variant Case Coverage (case granularity)	F	C	Percentage of cases in the event log that possess the same variant as a given case.	Process with several possible execution variants.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Smaller coverage implies that the case is executed with higher flexibility than other cases in the event log.	MIN

Indicator	D.	G.	Explanation	Potential Use	Req. Attributes	Extra Input	Assumptions	D. Value
Variant Case Coverage (group of cases granularity)	F	G	Percentage of cases in the event log that possess the same variant as any case in a group of cases.	Process with several possible execution variants.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	Smaller coverage implies that the cases in a group of cases have been executed with higher flexibility than other cases in the event log.	MIN
Variant count (group of cases granularity)	F	G	Number of variants that are observed in a group of cases.	Process with several possible execution variants.	***act*** – Event attribute: activity ***actinst*** – Event attribute: activity instance ***case*** – Event attribute: case id ***type*** – Event attribute: lifecycle type	-	A greater number of variants implies greater flexibility.	MAX

Table B.18: Flexibility PPI descriptions

Appendix C. Formal Definitions of PPIs

Let L be an event log, $L = (E, C, \#, \prec)$, let I be the set of activity instances, and let A be the set of activities. Tables C.19, C.20, C.21, C.22, and C.23 present the PPIs that have been defined for different performance dimensions and granularities. Specifically, the indicators are defined at the level of activity instances (I, denoted by I), activities (A, denoted by A), and cases (C, denoted by C for individual cases and G for groups of cases). Each table corresponds to a specific dimension, namely General (G), Time (T), Cost (C), Quality (Q), and Flexibility (F) performance. Each table row presents the indicator name, its performance dimension (D.), its granularity (G.), the required formal parameters (F. Par.), the indicator's function name, and its formal definition.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Activity count (case granularity)	G	C	$c \in C$	$A(c)$	$\lvert act(c)\rvert$
Activity count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$A(g)$	$\lvert \cup_{c \in g} act(c)\rvert$
Expected Activity count	G	G	$g \subseteq C$: $g \neq \varnothing$	$eA(g)$	$\frac{\sum_{c \in g} A(c)}{C(g)}$
activity Instance count (activity granularity)	G	A	$a \in A$	$I(a)$	$\lvert inst(a)\rvert$
activity Instance count (case granularity)	G	C	$c \in C$	$I(c)$	$\lvert inst(c)\rvert$
activity Instance count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$I(g)$	$\sum_{c \in g} I(c)$
Expected activity Instance count	G	G	$g \subseteq C$: $g \neq \varnothing$	$eI(g)$	$\frac{I(g)}{C(g)}$
Case count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$C(g)$	$\lvert g\rvert$
Human Resource count (activity granularity)	G	A	$a \in A$	$HR(a)$	$\lvert hres(a)\rvert$
Human Resource count (case granularity)	G	C	$c \in C$	$HR(c)$	$\lvert hres(c)\rvert$
Human Resource count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$HR(g)$	$\lvert \cup_{c \in g} hres(c)\rvert$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Expected Human Resource count	G	G	$g \subseteq C$: $g \neq \varnothing$	$eHR(g)$	$\frac{\sum_{c \in g} HR(c)}{C(g)}$
Resource count (activity granularity)	G	A	$a \in A$	$R(a)$	$\|res(a)\|$
Resource count (case granularity)	G	C	$c \in C$	$R(c)$	$\|res(c)\|$
Resource count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$R(g)$	$\| \cup_{c \in g} res(c)\|$
Expected Resource count	G	G	$g \subseteq C$: $g \neq \varnothing$	$eR(g)$	$\frac{\sum_{c \in g} R(c)}{C(g)}$
Role count (case granularity)	G	C	$c \in C$	$Role(c)$	$\|role(c)\|$
Role count (group of cases granularity)	G	G	$g \subseteq C$: $g \neq \varnothing$	$Role(g)$	$\| \cup_{c \in g} role(c)\|$
Expected Role count	G	G	$g \subseteq C$: $g \neq \varnothing$	$eRole(g)$	$\frac{\sum_{c \in g} Role(c)}{C(g)}$

Table C.19: General PPI definitions

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Active Time (case granularity)	T	C	$c \in C$	$AT(c)$	$LT(c) - IT(c)$
Expected Active Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eAT(g)$	$\frac{\sum_{c \in g} AT(c)}{C(g)}$
Activity count (case granularity)	T	C	$c \in C$	$A(c)$	$\|act(c)\|$
Activity count (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$A(g)$	$\| \cup_{c \in g} act(c)\|$
Expected Activity count	T	G	$g \subseteq C$: $g \neq \varnothing$	$eA(g)$	$\frac{\sum_{c \in g} A(c)}{C(g)}$
activity Instance count (case granularity)	T	C	$c \in C$	$I(c)$	$\|inst(c)\|$
activity Instance count (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$I(g)$	$\sum_{c \in g} I(c)$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Expected activity Instance count	T	G	$g \subseteq C$: $g \neq \varnothing$	$eI(g)$	$\frac{I(g)}{C(g)}$
Automated Activity count (case granularity)	T	C	$c \in C$	$Auta(c)$	$\|Autl \cap act(c)\|$
Automated Activity count (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$Auta(g)$	$\|Autl \cap (\cup_{c \in g} act(c))\|$
Expected Automated Activity count	T	G	$g \subseteq C$: $g \neq \varnothing$	$eAuta(g)$	$\frac{\sum_{c \in g} Auta(c)}{C(g)}$
Automated activity Instance count (case granularity)	T	C	$c \in C$	$Auti(c)$	$\|\{i \in inst(c) \mid act(i) \in Autl\}\|$
Automated activity Instance count (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$Auti(g)$	$\sum_{c \in g} Auti(c)$
Expected Automated activity Instance count	T	G	$g \subseteq C$: $g \neq \varnothing$	$eAuti(g)$	$\frac{Auti(g)}{C(g)}$
Automated Activity Service Time (case granularity)	T	C	$c \in C$	$AutST(c)$	$\sum_{i \in inst(c), act(i) \in Autl} ST(i)$
Automated Activity Service Time (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$AutST(g)$	$\sum_{c \in g} AutST(c)$
Expected Automated Activity Service Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eAutST(g)$	$\frac{AutST(g)}{C(g)}$
Case count and Lead Time Ratio (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$CLTR(g)$	$\frac{C(g)}{LT(g)}$
Case Count where Lead Time Over value (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{R}$	$CLTOC(g, val)$	$\|\{c \in g \mid LT(c) > val\}\|$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Case Percentage where Lead Time Over value (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{R}$	$CLTOP(g, val)$	$\frac{CLTOC(g,val)}{C(g)}$
Case Percentage with Missed Deadline (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$ $val \in U_{time}$	$CMDP(g, val)$	$\frac{\lvert\{c \in g \mid endt(c) > val\}\rvert}{C(g)}$
Handover count (case granularity)	T	C	$c \in C$	$H(c)$	$\sum_{i \in inst(c)} dres(i)$
Expected Handover count	T	G	$g \subseteq C$: $g \neq \varnothing$	$eH(g)$	$\frac{\sum_{c \in g} H(c)}{C(g)}$
Idle Time (case granularity)	T	C	$c \in C$	$IT(c)$	$\sum_{i \in inst(c)} \begin{cases} \frac{WT(i)}{\lvert concstr(i)\rvert} & if\ prevstr(i) = \varnothing \\ 0 & if\ prevstr(i) \neq \varnothing \end{cases}$
Expected Idle Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eIT(g)$	$\frac{\sum_{c \in g} IT(c)}{C(g)}$
Lead Time (activity instance granularity)	T	I	$i \in I$	$LT(i)$	$ST(i) + WT(i)$
Lead Time (activity granularity)	T	A	$a \in A$	$LT(a)$	$\sum_{i \in inst(a)} LT(i)$
Lead Time (case granularity)	T	C	$c \in C$	$LT(c)$	$endt(c) - startt(c)$
Lead Time (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$LT(g)$	$\underset{c \in g}{max}\, endt(c) - \underset{c \in g}{min}\, startt(c)$
Expected Lead Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eLT(g)$	$\frac{\sum_{c \in g} LT(c)}{C(g)}$
Lead Time and Case count Ratio (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$LTCR(g)$	$\frac{LT(g)}{C(g)}$
Lead Time Deviation from time Limit (case granularity)	T	C	$c \in C$ $val \in \mathbb{R}$	$LTDL(c, val)$	$val - LT(c)$
Expected Lead Time Deviation from time Limit	T	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{R}$	$eLTDL(g, val)$	$\frac{\sum_{c \in g} LTDL(c,val)}{C(g)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Lead Time Deviation from Expectation (case granularity)	T	C	$c \in C$ $val \in \mathbb{R}$	$LTDE(c, val)$	$\lvert val - LT(c) \rvert$
Expected Lead Time Deviation from Expectation	T	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{R}$	$eLTDE(g, val)$	$\frac{\sum_{c \in g} LTDE(c,val)}{C(g)}$
Lead Time from activity A (case granularity)	T	C	$c \in C$ $a \in A$	$LTfA(c, a)$	$\begin{cases} lt(x,y) & for\ any\ x \in fi^{s}(c,a)\ and\ y \in endin(c) \\ \perp & if\ fi^{s}(c,a) = \varnothing \end{cases}$
Expected Lead Time from activity A	T	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$eLTfA(g, a)$	$\frac{\sum_{c \in g} \begin{cases} LTfA(c,a) & if\ fi^{s}(c,a) \neq \varnothing \\ 0 & if\ fi^{s}(c,a) = \varnothing \end{cases}}{\lvert \{c \in g \mid fi^{s}(c,a) \neq \varnothing\} \rvert}$
Lead Time from activity A to activity B (case granularity)	T	C	$c \in C$ $a, b \in A$: $a \neq b$	$LTAB(c, a, b)$	$\begin{cases} lt(x,y) & for\ any\ x \in fi^{s}(c,a)\ and\ y \in fi^{sc}(c,a,b) \\ \perp & if\ fi^{sc}(c,a,b) = \varnothing \end{cases}$
Expected Lead Time from activity A to activity B	T	G	$g \subseteq C$: $g \neq \varnothing$ $a, b \in A$: $a \neq b$	$eLTAB(g, a, b)$	$\frac{\sum_{c \in g} \begin{cases} LTAB(c,a,b) & if\ fi^{sc}(c,a,b) \neq \varnothing \\ 0 & if\ fi^{sc}(c,a,b) = \varnothing \end{cases}}{\lvert \{c \in g \mid fi^{sc}(c,a,b) \neq \varnothing)\} \rvert}$
Lead Time to activity A (case granularity)	T	C	$c \in C$ $a \in A$	$LTtA(c, a)$	$\begin{cases} lt(x,y) & for\ any\ x \in strin(c)\ and\ y \in fi^{c}(c,a) \\ \perp & if\ fi^{c}(c,a) = \varnothing \end{cases}$
Expected Lead Time to activity A	T	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$eLTtA(g, a)$	$\frac{\sum_{c \in g} \begin{cases} LTtA(c,a) & if\ fi^{c}(c,a) \neq \varnothing \\ 0 & if\ fi^{c}(c,a) = \varnothing \end{cases}}{\lvert \{c \in g \mid if\ \ fi^{c}(c,a) \neq \varnothing\} \rvert}$
Service and Lead Time Ratio (activity instance granularity)	T	I	$i \in I$	$SLTR(i)$	$\frac{ST(i)}{LT(i)}$
Service and Lead Time Ratio (activity granularity)	T	A	$a \in A$	$SLTR(a)$	$\frac{ST(a)}{LT(a)}$
Service and Lead Time Ratio (case granularity)	T	C	$c \in C$	$SLTR(c)$	$\frac{ST(c)}{LT(c)}$
Service and Lead Time Ratio (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$SLTR(g)$	$\frac{ST(g)}{LT(g)}$
Expected Service and Lead Time Ratio	T	G	$g \subseteq C$: $g \neq \varnothing$	$eSLTR(g)$	$\frac{ST(g)}{\sum_{c \in g} LT(c)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Service Time (activity instance granularity)	T	I	$i \in I$	$ST(i)$	$ctime(i) - stime(i)$
Service Time (activity granularity)	T	A	$a \in A$	$ST(a)$	$\sum_{i \in inst(a)} ST(i)$
Service Time (case granularity)	T	C	$c \in C$	$ST(c)$	$\sum_{i \in inst(c)} ST(i)$
Service Time (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$	$ST(g)$	$\sum_{c \in g} ST(c)$
Expected Service Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eST(g)$	$\frac{ST(g)}{C(g)}$
Service Time from activity A to activity B (case granularity)	T	C	$c \in C$ $a, b \in A$: $a \neq b$	$STAB^{\times}(c, a, b)$ $\times \in \{s, c, sc, w\}$	$\begin{cases} st^{\times}(x, y) & \textit{for any } x \in \mathit{fi}^{s}(c, a) \textit{ and } y \in \mathit{fi}^{sc}(c, a, b) \\ \perp & \textit{if } \mathit{fi}^{sc}(c, a, b) = \varnothing \end{cases}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses st^{s}; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses st^{c}; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses st^{sc}; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses st^{w}.
Service Time from activity A to activity B (group of cases granularity)	T	G	$g \subseteq C$: $g \neq \varnothing$ $a, b \in A$: $a \neq b$	$STAB^{\times}(g, a, b)$ $\times \in \{s, c, sc, w\}$	$\sum_{c \in g} \begin{cases} STAB^{\times}(c, a, b) & \textit{if } \mathit{fi}^{sc}(c, a, b) \neq \varnothing \\ 0 & \textit{if } \mathit{fi}^{sc}(c, a, b) = \varnothing \end{cases}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses $STAB^{s}$; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses $STAB^{c}$; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses $STAB^{sc}$; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses $STAB^{w}$.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Expected Service Time from activity A to activity B	T	G	$g \subseteq C$: $g \neq \varnothing$ $a, b \in A$: $a \neq b$	$eSTAB^{\times}(g, a, b)$ $\times \in \{s, c, sc, w\}$	$\frac{STAB^{\times}(g,a,b)}{\lvert\{c \in g \mid fi^{sc}(c,a,b) \neq \varnothing\}\rvert}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses $STAB^{s}$; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses $STAB^{c}$; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses $STAB^{sc}$; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses $STAB^{w}$.
Waiting Time (activity instance granularity)	T	I	$i \in I$	$WT(i)$	$\begin{cases} stime(i) - ctime(x) & for\ any\ x \in prev(i) \\ 0 & if\ prev(i) = \varnothing \end{cases}$
Waiting Time (activity granularity)	T	A	$a \in A$	$WT(a)$	$\sum_{i \in inst(a)} WT(i)$
Waiting Time (case granularity)	T	C	$c \in C$	$WT(c)$	$\sum_{i \in inst(c)} WT(i)$
Expected Waiting Time	T	G	$g \subseteq C$: $g \neq \varnothing$	$eWT(g)$	$\frac{\sum_{c \in g} WT(c)}{C(g)}$
Waiting Time from activity A to activity B (case granularity)	T	C	$c \in C$ $a, b \in A$: $a \neq b$	$WTAB^{\times}(c, a, b)$ $\times \in \{s, c, sc, w\}$	$\begin{cases} wt^{\times}(x, y) & for\ any\ x \in fi^{s}(c, a)\ and\ y \in fi^{sc}(c, a, b) \\ \perp & if\ fi^{sc}(c, a, b) = \varnothing \end{cases}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses wt^{s}; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses wt^{c}; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses wt^{sc}; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses wt^{w}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Expected Waiting Time from activity A to activity B	T	G	$g \subseteq C$: $g \neq \varnothing$ $a, b \in A$: $a \neq b$	$eWTAB^{\times}(g, a, b)$ $\times \in \{s, c, sc, w\}$	$\frac{\sum_{c \in g} \begin{cases} WTAB^{\times}(c,a,b) & if\ fi^{sc}(c,a,b) \neq \varnothing \\ 0 & if\ fi^{sc}(c,a,b) = \varnothing \end{cases}}{\lvert\{c \in g \mid fi^{sc}(c,a,b) \neq \varnothing\}\rvert}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{s, c, sc, w\}$. If $\times = s$, then the function considers activity instances that were started within the start and end activity instances, i.e., it uses $WTAB^s$; if $\times = c$, then the function considers activity instances that were completed within the start and end activity instances, i.e., it uses $WTAB^c$; if $\times = sc$, then the function considers activity instances that were either started or completed within the start and end activity instances, i.e., it uses $WTAB^{sc}$; if $\times = {}^{w}$, then the function considers all activity instances that were active within the start and end activity instances, i.e., it uses $WTAB^w$.

Table C.20: Time dimension PPI definitions

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Automated Activity Cost (case granularity)	C	C	$c \in C$	$AutC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c), act(i) \in Autl} TC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Automated Activity Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$AutC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} AutC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses $AutC^{sgl}$; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses $AutC^{sum}$.
Expected Automated Activity Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eAutC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{AutC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses $AutC^{sgl}$; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses $AutC^{sum}$.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Desired Activity count (case granularity)	C	C	$c \in C$	$DAC(c)$	$\lvert Desl \cap act(c) \rvert$
Desired Activity count (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$DAC(g)$	$\lvert Desl \cap (\cup_{c \in g} act(c)) \rvert$
Expected Desired Activity count	C	G	$g \subseteq C$: $g \neq \varnothing$	$eDAC(g)$	$\frac{\sum_{c \in g} DAC(c)}{C(g)}$
Direct Cost (case granularity)	C	C	$c \in C$	$DC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c), act(i) \in DCl} TC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Direct Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$DC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} DC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses DC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses DC^{sum}.
Expected Direct Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eDC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{DC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses DC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses DC^{sum}.
Fixed Cost considering single events of activity instances (activity instance granularity)	C	I	$i \in I$	$FC^{sgl}(i)$	$\begin{cases} \#_{fc}(cpl(i)) & \textit{if } \#_{fc}(cpl(i)) \neq \bot \\ \#_{fc}(str(i)) & \textit{if } \#_{fc}(cpl(i)) = \bot \wedge \#_{fc}(str(i)) \neq \bot \\ \bot & \textit{if } \#_{fc}(cpl(i)) = \bot \wedge \#_{fc}(str(i)) = \bot \end{cases}$
Fixed Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	$i \in I$	$FC^{sum}(i)$	$\begin{cases} \#_{fc}(str(i)) + & \\ \#_{fc}(cpl(i)) & \textit{if } \#_{fc}(cpl(i)) \neq \bot \wedge \#_{fc}(str(i)) \neq \bot \\ \bot & \textit{if } \#_{fc}(cpl(i)) = \bot \vee \#_{fc}(str(i)) = \bot \end{cases}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Fixed Cost (activity granularity)	C	A	$a \in A$	$FC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} FC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses FC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses FC^{sum}.
Fixed Cost (case granularity)	C	C	$c \in C$	$FC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} FC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses FC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses FC^{sum}.
Fixed Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$FC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} FC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses FC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses FC^{sum}.
Expected Fixed Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eFC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{FC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses FC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses FC^{sum}.
Human Resource and Case count Ratio (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$HRCR(g)$	$\frac{HR(g)}{C(g)}$
Human Resource count (activity granularity)	C	A	$a \in A$	$HR(a)$	$\lvert hres(a)\rvert$
Human Resource count (case granularity)	C	C	$c \in C$	$HR(c)$	$\lvert hres(c)\rvert$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Human Resource count (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$HR(g)$	$\lvert \cup_{c \in g} hres(c) \rvert$
Expected Human Resource count	C	G	$g \subseteq C$: $g \neq \varnothing$	$eHR(g)$	$\frac{\sum_{c \in g} HR(c)}{C(g)}$
Inventory Cost considering single events of activity instances (activity instance granularity)	C	I	$i \in I$	$IC^{sgl}(i)$	$\begin{cases} \#_{ic}(cpl(i)) & if\ \#_{ic}(cpl(i)) \neq \perp \\ \#_{ic}(str(i)) & if\ \#_{ic}(cpl(i)) = \perp \wedge \#_{ic}(str(i)) \neq \perp \\ \perp & if\ \#_{ic}(cpl(i)) = \perp \wedge \#_{ic}(str(i)) = \perp \end{cases}$
Inventory Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	$i \in I$	$IC^{sum}(i)$	$\begin{cases} \#_{ic}(str(i)) + \\ \#_{ic}(cpl(i)) & if\ \#_{ic}(cpl(i)) \neq \perp \wedge \#_{ic}(str(i)) \neq \perp \\ \perp & if\ \#_{ic}(cpl(i)) = \perp \vee \#_{ic}(str(i)) = \perp \end{cases}$
Inventory Cost (activity granularity)	C	A	$a \in A$	$IC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} IC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses IC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses IC^{sum}.
Inventory Cost (case granularity)	C	C	$c \in C$	$IC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} IC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses IC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses IC^{sum}.
Inventory Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$IC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} IC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses IC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses IC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{IC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses IC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses IC^{sum}.
Expected Inventory Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eIC^{\times}(g)$ $\times \in \{sgl, sum\}$	
Labor Cost and Total Cost Ratio (activity instance granularity)	C	I	$i \in I$	$LCTCR^{\times}(i)$ $\times \in \{sgl, sum\}$	$\frac{LC^{\times}(i)}{TC^{\times}(i)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl} and TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum} and TC^{sum}.
Labor Cost and Total Cost Ratio (activity granularity)	C	A	$a \in A$	$LCTCR^{\times}(a)$ $\times \in \{sgl, sum\}$	$\frac{LC^{\times}(a)}{TC^{\times}(a)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl} and TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum} and TC^{sum}.
Labor Cost and Total Cost Ratio (case granularity)	C	C	$c \in C$	$LCTCR^{\times}(c)$ $\times \in \{sgl, sum\}$	$\frac{LC^{\times}(c)}{TC^{\times}(c)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl} and TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum} and TC^{sum}.
Labor Cost and Total Cost Ratio (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$LCTCR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{LC^{\times}(g)}{TC^{\times}(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl} and TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum} and TC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{LC^{\times}(g)}{TC^{\times}(g)}$
Expected Labor Cost and Total Cost Ratio	C	G	$g \subseteq C$: $g \neq \varnothing$	$eLCTCR^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl} and TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum} and TC^{sum}.
Labor Cost considering single events of activity instances (activity instance granularity)	C	I	$i \in I$	$LC^{sgl}(i)$	$\begin{cases} \#_{lc}(cpl(i)) & \mathit{if}\ \#_{lc}(cpl(i)) \neq \perp \\ \#_{lc}(str(i)) & \mathit{if}\ \#_{lc}(cpl(i)) = \perp \wedge \#_{lc}(str(i)) \neq \perp \\ \perp & \mathit{if}\ \#_{lc}(cpl(i)) = \perp \wedge \#_{lc}(str(i)) = \perp \end{cases}$
Labor Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	$i \in I$	$LC^{sum}(i)$	$\begin{cases} \#_{lc}(str(i))+ & \\ \#_{lc}(cpl(i)) & \mathit{if}\ \#_{lc}(cpl(i)) \neq \perp \wedge \#_{lc}(str(i)) \neq \perp \\ \perp & \mathit{if}\ \#_{lc}(cpl(i)) = \perp \vee \#_{lc}(str(i)) = \perp \end{cases}$
Labor Cost (activity granularity)	C	A	$a \in A$	$LC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} LC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum}.
Labor Cost (case granularity)	C	C	$c \in C$	$LC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} LC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum}.
Labor Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$LC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} LC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Expected Labor Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eLC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{LC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses LC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses LC^{sum}.
Maintenance Cost (case granularity)	C	C	$c \in C$	$MainC(c)$	$\#_{mainc}(c)$
Maintenance Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$MainC(g)$	$\sum_{c \in g} MainC(c)$
Expected Maintenance Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eMainC(g)$	$\frac{MainC(g)}{C(g)}$
Missed Deadline Cost (case granularity)	C	C	$c \in C$	$MDC(c)$	$\#_{mdc}(c)$
Missed Deadline Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$MDC(g)$	$\sum_{c \in g} MDC(c)$
Expected Missed Deadline Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eMDC(g)$	$\frac{MDC(g)}{C(g)}$
Overhead Cost (case granularity)	C	C	$c \in C$	$OC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c), act(i) \notin DCl} TC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Overhead Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$OC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} OC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses OC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses OC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{OC^{\times}(g)}{C(g)}$
Expected Overhead Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eOC^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses OC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses OC^{sum}.
Resource count (activity granularity)	C	A	$a \in A$	$R(a)$	$\lvert res(a)\rvert$
Resource count (case granularity)	C	C	$c \in C$	$R(c)$	$\lvert res(c)\rvert$
Resource count (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$R(g)$	$\lvert \cup_{c \in g} res(c)\rvert$
Expected Resource count	C	G	$g \subseteq C$: $g \neq \varnothing$	$eR(g)$	$\frac{\sum_{c \in g} R(c)}{C(g)}$
Rework Cost (activity granularity)	C	A	$a \in A$	$RC^{\times}(a)$ $\times \in \{sgl, sum\}$	$TC(a) - \sum_{c \in C} fitc^{\times}(c, a)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses $fitc^{sgl}$; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses $fitc^{sum}$.
Rework Cost (case granularity)	C	C	$c \in C$	$RC^{\times}(c)$ $\times \in \{sgl, sum\}$	$TC(c) - \sum_{a \in A} fitc^{\times}(c, a)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses $fitc^{sgl}$; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses $fitc^{sum}$.
Rework Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$RC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} RC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses RC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses RC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{RC^{\times}(g)}{C(g)}$
Expected Rework Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eRC^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses RC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses RC^{sum}.
Rework Count (activity granularity)	C	A	$a \in A$	$RewC(a)$	$\sum_{c \in C} max(0, count(c, a) - 1)$
Rework Count (case granularity)	C	C	$c \in C$	$RewC(c)$	$\sum_{a \in A} max(0, count(c, a) - 1)$
Rework Count (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$RewC(g)$	$\sum_{c \in g} RewC(c)$
Expected Rework Count	C	G	$g \subseteq C$: $g \neq \varnothing$	$eRewC(g)$	$\frac{RewC(g)}{C(g)}$
Rework Percentage (activity granularity)	C	A	$a \in A$	$RewP(a)$	$\frac{RewC(a)}{I(a)}$
Rework Percentage (case granularity)	C	C	$c \in C$	$RewP(c)$	$\frac{RewC(c)}{I(c)}$
Rework Percentage (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$RewP(g)$	$\frac{RewC(g)}{I(g)}$
Expected Rework Percentage	C	G	$g \subseteq C$: $g \neq \varnothing$	$eRewP(g)$	$\frac{\sum_{c \in g} RewP(c)}{C(g)}$
Total Cost considering single events of activity instances (activity instance granularity)	C	I	$i \in I$	$TC^{sgl}(i)$	$\begin{cases} \#_{tc}(cpl(i)) & if\ \#_{tc}(cpl(i)) \neq \perp \\ \#_{tc}(str(i)) & if\ \#_{tc}(cpl(i)) = \perp \wedge \#_{tc}(str(i)) \neq \perp \\ \perp & if\ \#_{tc}(cpl(i)) = \perp \wedge \#_{tc}(str(i)) = \perp \end{cases}$
Total Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	$i \in I$	$TC^{sum}(i)$	$\begin{cases} \#_{tc}(str(i)) + & \\ \#_{tc}(cpl(i)) & if\ \#_{tc}(cpl(i)) \neq \perp \wedge \#_{tc}(str(i)) \neq \perp \\ \perp & if\ \#_{tc}(cpl(i)) = \perp \vee \#_{tc}(str(i)) = \perp \end{cases}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Total Cost (activity granularity)	C	A	$a \in A$	$TC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} TC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost (case granularity)	C	C	$c \in C$	$TC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} TC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$TC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} TC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Expected Total Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eTC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and Lead Time Ratio (activity instance granularity)	C	I	$i \in I$	$TCLTR^{\times}(i)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(i)}{LT(i)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.

Indicator	D. G. F. Par.	Function Name	Formal Definition
Total Cost and Lead Time Ratio (activity granularity)	C A $a \in A$	$TCLTR^{\times}(a)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(a)}{LT(a)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and Lead Time Ratio (case granularity)	C C $c \in C$	$TCLTR^{\times}(c)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(c)}{LT(c)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and Lead Time Ratio (group of cases granularity)	C G $g \subseteq C$: $g \neq \varnothing$	$TCLTR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(g)}{LT(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Expected Total Cost and Lead Time Ratio	C G $g \subseteq C$: $g \neq \varnothing$	$eTCLTR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(g)}{\sum_{c \in g} LT(c)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and outcome Unit Ratio (activity instance granularity)	C I $i \in I$	$TCUR^{\times}(i)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(i)}{U^{\times}(i)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sum} and U^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{TC^{\times}(a)}{U^{\times}(a)}$
Total Cost and outcome Unit Ratio (activity granularity)	C	A	$a \in A$	$TCUR^{\times}(a)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sum} and U^{sum}. $\frac{TC^{\times}(c)}{U^{\times}(c)}$
Total Cost and outcome Unit Ratio (case granularity)	C	C	$c \in C$	$TCUR^{\times}(c)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sum} and U^{sum}. $\frac{TC^{\times}(g)}{U^{\times}(g)}$
Total Cost and outcome Unit Ratio (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$TCUR^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sum} and U^{sum}. $\frac{TC^{\times}(g)}{U^{\times}(g)}$
Expected Total Cost and outcome Unit Ratio	C	G	$g \subseteq C$: $g \neq \varnothing$	$eTCUR^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost and outcome unit calculations, i.e., it uses TC^{sum} and U^{sum}. $\frac{TC^{\times}(i)}{ST(i)}$
Total Cost and Service Time Ratio (activity instance granularity)	C	I	$i \in I$	$TCSTR^{\times}(i)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Total Cost and Service Time Ratio (activity granularity)	C	A	$a \in A$	$TCSTR^{\times}(a)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(a)}{ST(a)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and Service Time Ratio (case granularity)	C	C	$c \in C$	$TCSTR^{\times}(c)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(c)}{ST(c)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Total Cost and Service Time Ratio (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$TCSTR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(g)}{ST(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Expected Total Cost and Service Time Ratio	C	G	$g \subseteq C$: $g \neq \varnothing$	$eTCSTR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{TC^{\times}(g)}{ST(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Transportation Cost (case granularity)	C	C	$c \in C$	$TransC(c)$	$\#_{transc}(c)$
Transportation Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$TransC(g)$	$\sum_{c \in g} TransC(c)$
Expected Transportation Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eTransC(g)$	$\frac{TransC(g)}{C(g)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Variable Cost considering single events of activity instances (activity instance granularity)	C	I	$i \in I$	$VC^{sgl}(i)$	$\begin{cases} \#_{vc}(cpl(i)) & if\ \#_{vc}(cpl(i)) \neq \perp \\ \#_{vc}(str(i)) & if\ \#_{vc}(cpl(i)) = \perp \wedge \#_{vc}(str(i)) \neq \perp \\ \perp & if\ \#_{vc}(cpl(i)) = \perp \wedge \#_{vc}(str(i)) = \perp \end{cases}$
Variable Cost considering the sum of all events of activity instances (activity instance granularity)	C	I	$i \in I$	$VC^{sum}(i)$	$\begin{cases} \#_{vc}(str(i)) + & \\ \#_{vc}(cpl(i)) & if\ \#_{vc}(cpl(i)) \neq \perp \wedge \#_{vc}(str(i)) \neq \perp \\ \perp & if\ \#_{vc}(cpl(i)) = \perp \vee \#_{vc}(str(i)) = \perp \end{cases}$
Variable Cost (activity granularity)	C	A	$a \in A$	$VC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} VC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses VC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses VC^{sum}.
Variable Cost (case granularity)	C	C	$c \in C$	$VC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} VC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses VC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses VC^{sum}.
Variable Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$VC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} VC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses VC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses VC^{sum}.
Expected Variable Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$e\,VC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{VC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses VC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses VC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Warehousing Cost (case granularity)	C	C	$c \in C$	$WareC(c)$	$\#_{warec}(c)$
Warehousing Cost (group of cases granularity)	C	G	$g \subseteq C$: $g \neq \varnothing$	$WareC(g)$	$\sum_{c \in g} WareC(c)$
Expected Warehousing Cost	C	G	$g \subseteq C$: $g \neq \varnothing$	$eWareC(g)$	$\frac{WareC(g)}{C(g)}$

Table C.21: Cost dimension PPI definitions

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
activity Instance count by Human Resource (activity granularity)	Q	A	$a \in A$ $hr \in U_{hres}$	$IbyHR(a, hr)$	$\lvert\{i \in inst(a) \mid hres(i) = hr\}\rvert$
activity Instance count by Human Resource (case granularity)	Q	C	$c \in C$ $hr \in U_{hres}$	$IbyHR(c, hr)$	$\lvert\{i \in inst(c) \mid hres(i) = hr\}\rvert$
activity Instance count by Human Resource (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $hr \in U_{hres}$	$IbyHR(g, hr)$	$\sum_{c \in g} IbyHR(c, hr)$
Expected activity Instance count by Human Resource	Q	G	$g \subseteq C$: $g \neq \varnothing$ $hr \in U_{hres}$	$eIbyHR(g, hr)$	$\frac{IbyHR(g,hr)}{C(g)}$
activity Instance count by Role (case granularity)	Q	C	$c \in C$ $rl \in U_{role}$	$IbyRole(c, rl)$	$\lvert\{i \in inst(c) \mid role(i) = rl\}\rvert$
activity Instance count by Role (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $rl \in U_{role}$	$IbyRole(g, rl)$	$\sum_{c \in g} IbyRole(c, rl)$
Expected activity Instance count by Role	Q	G	$g \subseteq C$: $g \neq \varnothing$ $rl \in U_{role}$	$eIbyRole(g, rl)$	$\frac{IbyRole(g,rl)}{C(g)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Automated Activity count (case granularity)	Q	C	$c \in C$	$Auta(c)$	$\lvert Autl \cap act(c) \rvert$
Automated Activity count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$Auta(g)$	$\lvert Autl \cap (\cup_{c \in g} act(c)) \rvert$
Expected Automated Activity count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eAuta(g)$	$\frac{\sum_{c \in g} Auta(c)}{C(g)}$
Automated activity Instance count (case granularity)	Q	C	$c \in C$	$Auti(c)$	$\lvert \{i \in inst(c) \mid act(i) \in Autl\} \rvert$
Automated activity Instance count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$Auti(g)$	$\sum_{c \in g} Auti(c)$
Expected Automated activity Instance count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eAuti(g)$	$\frac{Auti(g)}{C(g)}$
Case and Client count Ratio (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$CCliR(g)$	$\frac{C(g)}{Cli(g)}$
Case Count where Activity after Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $et \in U_{time}$	$CApastTFC(g, a, et)$	$\lvert \{c \in g \mid \exists i \in inst(c, a)[stime(i) \geq et]\} \rvert$
Case Count where Activity before Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $st \in U_{time}$	$CAdueTFC(g, a, st)$	$\lvert \{c \in g \mid \exists i \in inst(c, a)[stime(i) \leq st]\} \rvert$
Case Count where Activity during Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $st, et \in U_{time}$: $st < et$	$CAinTFC(g, a, st, et)$	$\lvert \{c \in g \mid \exists i \in inst(c, a)[st \leq stime(i) \wedge stime(i) \leq et]\} \rvert$
Case Count where end activity is A (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$CendAC(g, a)$	$\lvert \{c \in g \mid \exists i \in endin(c)[act(i) = a]\} \rvert$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Case Count where start activity is A (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$CstartAC(g, a)$	$\lvert\{c \in g \mid \exists i \in strin(c)[act(i) = a]\}\rvert$
Case Count with Rework (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$CRewC(g)$	$\lvert\{c \in g \mid RewC(c) > 0\}\rvert$
Case Percentage where Activity after Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $et \in U_{time}$	$CApastTFP(g, a, et)$	$\frac{CApastTFC(g,a,et)}{C(g)}$
Case Percentage where Activity before Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $st \in U_{time}$	$CAdueTFP(g, a, st)$	$\frac{CAdueTFC(g,a,st)}{C(g)}$
Case Percentage where Activity after Time Frame (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$ $st, et \in U_{time}$: $st < et$	$CAinTFP(g, a, st, et)$	$\frac{CAinTFC(g,a,st,et)}{C(g)}$
Case Percentage where End Activity is A (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$CendAP(g, a)$	$\frac{CendAC(g,a)}{C(g)}$
Case Percentage where Start Activity is A (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $a \in A$	$CstartAP(g, a)$	$\frac{CstartAC(g,a)}{C(g)}$
Case Percentage with Missed Deadline (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $val \in U_{time}$	$CMDP(g, val)$	$\frac{\lvert\{c \in g \mid endt(c) > val\}\rvert}{C(g)}$
Case Percentage with Rework (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$CRewP(g)$	$\frac{CRewC(g)}{C(g)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Client count and Total Cost Ratio (activity granularity)	Q	A	$a \in A$	$CliTCR^{\times}(a)$ $\times \in \{sgl, sum\}$	$\frac{Cli(a)}{TC^{\times}(a)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Client count and Total Cost Ratio (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$CliTCR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{Cli(g)}{TC^{\times}(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Expected Client count and Total Cost Ratio	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eCliTCR^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{Cli(g)}{TC^{\times}(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Desired Activity count (case granularity)	Q	C	$c \in C$	$DAC(c)$	$\lvert Desl \cap act(c) \rvert$
Desired Activity count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$DAC(g)$	$\lvert Desl \cap (\cup_{c \in g} act(c)) \rvert$
Expected Desired Activity count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eDAC(g)$	$\frac{\sum_{c \in g} DAC(c)}{C(g)}$
Human Resource count (activity granularity)	Q	A	$a \in A$	$HR(a)$	$\lvert hres(a) \rvert$
Human Resource count (case granularity)	Q	C	$c \in C$	$HR(c)$	$\lvert hres(c) \rvert$
Human Resource count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$HR(g)$	$\lvert \cup_{c \in g} hres(c) \rvert$
Expected Human Resource count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eHR(g)$	$\frac{\sum_{c \in g} HR(c)}{C(g)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Non-Automated Activity count (case granularity)	Q	C	$c \in C$	$NAutA(c)$	$\lvert act(c) \setminus Autl \rvert$
Non-Automated Activity count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$NAutA(g)$	$\lvert (\cup_{c \in g} act(c)) \setminus Autl \rvert$
Expected Non-Automated Activity count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$NAutA(g)$	$\frac{\sum_{c \in g} NAutA(c)}{C(g)}$
Non-Automated activity Instance count (case granularity)	Q	C	$c \in C$	$NAutI(c)$	$\lvert \{i \in inst(c) \mid act(i) \notin Autl\} \rvert$
Non-Automated activity Instance count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$NAutI(g)$	$\sum_{c \in g} NAutI(c)$
Expected Non-Automated activity Instance count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eNAutI(g)$	$\frac{NAutI(g)}{C(g)}$
outcome Unit count considering single events of activity instances (activity instance granularity)	Q	I	$i \in I$	$U^{sgl}(i)$	$\begin{cases} \#_{unt}(cpl(i)) & \mathit{if}\ \#_{unt}(cpl(i)) \neq \bot \\ \#_{unt}(str(i)) & \mathit{if}\ \#_{unt}(cpl(i)) = \bot \wedge \\ & \#_{unt}(str(i)) \neq \bot \\ \bot & \mathit{if}\ \#_{unt}(cpl(i)) = \bot \wedge \\ & \#_{unt}(str(i)) = \bot \end{cases}$
outcome Unit count considering the sum of all events of activity instances (activity instance granularity)	Q	I	$i \in I$	$U^{sum}(i)$	$\begin{cases} \#_{unt}(str(i)) + & \mathit{if}\ \#_{unt}(cpl(i)) \neq \bot \wedge \\ \#_{unt}(cpl(i)) & \#_{unt}(str(i)) \neq \bot \\ \bot & \mathit{if}\ \#_{unt}(cpl(i)) = \bot \vee \\ & \#_{unt}(str(i)) = \bot \end{cases}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
outcome Unit count (activity granularity)	Q	A	$a \in A$	$U^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} U^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses U^{sum}.
outcome Unit count (case granularity)	Q	C	$c \in C$	$U^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{e \in inst(c)} U^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses U^{sum}.
outcome Unit count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$U^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in g} U^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses U^{sum}.
Expected outcome Unit count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eU^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{U^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses U^{sum}.
Overall Quality (case granularity)	Q	C	$c \in C$	$OQ(c)$	$\#_{qual}(c)$
Expected Overall Quality	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eOQ(g)$	$\frac{\sum_{c \in g} OQ(c)}{C(g)}$
Repeatability (case granularity)	Q	C	$c \in C$	$Rep(c)$	$1 - \frac{A(c)}{I(c)}$
Repeatability (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$Rep(g)$	$1 - \frac{A(g)}{I(g)}$
Expected Repeatability	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eRep(g)$	$1 - \frac{\sum_{c \in g} A(c)}{I(g)}$
Rework Count (activity granularity)	Q	A	$a \in A$	$RewC(a)$	$\sum_{c \in C} max(0, count(c, a) - 1)$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Rework Count (case granularity)	Q	C	$c \in C$	$RewC(c)$	$\sum_{a\in A} max(0, count(c,a) - 1)$
Rework Count (group of cases granularity)	Q	G	$g \subseteq C: g \neq \varnothing$	$RewC(g)$	$\sum_{c\in g} RewC(c)$
Expected Rework Count	Q	G	$g \subseteq C: g \neq \varnothing$	$eRewC(g)$	$\frac{RewC(g)}{C(g)}$
Rework Count by Value (activity granularity)	Q	A	$a \in A$ $val \in \mathbb{N}$	$RewCV(a, val)$	$\sum_{c\in C} max(0, count(c,a) - val)$
Rework Count by Value (case granularity)	Q	C	$c \in C$ $val \in \mathbb{N}$	$RewCV(c, val)$	$\sum_{a\in A} max(0, count(c,a) - val)$
Rework Count by Value (group of cases granularity)	Q	G	$g \subseteq C: g \neq \varnothing$ $val \in \mathbb{N}$	$RewCV(g, val)$	$\sum_{c\in g} RewCV(c, val)$
Expected Rework Count by Value	Q	G	$g \subseteq C: g \neq \varnothing$ $val \in \mathbb{N}$	$eRewCV(g, val)$	$\frac{RewCV(g,val)}{C(g)}$
Rework of activities Subset (case granularity)	Q	C	$c \in C$ $sub \subseteq A: sub \neq \varnothing$	$RewSub(c, sub)$	$\sum_{a\in sub} max(0, count(c,a) - 1)$
Rework of activities Subset (group of cases granularity)	Q	G	$g \subseteq C: g \neq \varnothing$ $sub \subseteq A: sub \neq \varnothing$	$RewSub(g, sub)$	$\sum_{c\in g} RewSub(c, sub)$
Expected Rework of activities Subset	Q	G	$g \subseteq C: g \neq \varnothing$ $sub \subseteq A: sub \neq \varnothing$	$eRewSub(g, sub)$	$\frac{RewSub(g,sub)}{C(g)}$
Rework Percentage (activity granularity)	Q	A	$a \in A$	$RewP(a)$	$\frac{RewC(a)}{I(a)}$
Rework Percentage (case granularity)	Q	C	$c \in C$	$RewP(c)$	$\frac{RewC(c)}{I(c)}$
Rework Percentage (group of cases granularity)	Q	G	$g \subseteq C: g \neq \varnothing$	$RewP(g)$	$\frac{RewC(g)}{I(g)}$
Expected Rework Percentage	Q	G	$g \subseteq C: g \neq \varnothing$	$eRewP(g)$	$\frac{\sum_{c\in g} RewP(c)}{C(g)}$
Rework Percentage by Value (activity granularity)	Q	A	$a \in A$ $val \in \mathbb{N}$	$RewPV(a, val)$	$\frac{RewCV(a,val)}{I(a)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Rework Percentage by Value (case granularity)	Q	C	$c \in C$ $val \in \mathbb{N}$	$RewPV(c, val)$	$\frac{RewCV(c,val)}{I(c)}$
Rework Percentage by Value (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{N}$	$RewPV(g, val)$	$\frac{RewCV(g,val)}{I(g)}$
Expected Rework Percentage by Value	Q	G	$g \subseteq C$: $g \neq \varnothing$ $val \in \mathbb{N}$	$eRewPV(g)$	$\frac{\sum_{c \in g} RewPV(c,val)}{C(g)}$
Rework Time (activity granularity)	Q	A	$a \in A$	$RT(a)$	$LT(a) - \sum_{c \in C} \mathit{filt}(c, a)$
Rework Time (case granularity)	Q	C	$c \in C$	$RT(c)$	$LT(c) - \sum_{a \in A} \mathit{filt}(c, a)$
Rework Time (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$RT(g)$	$\sum_{c \in g} RT(c)$
Expected Rework Time	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eRT(g)$	$\frac{RT(g)}{C(g)}$
Successful outcome Unit Count (activity instance granularity)	Q	I	$i \in I$	$SUC^{\times}(i)$ $\times \in \{sgl, sum\}$	$\begin{cases} U^{\times}(i) - \#_{uns}(cpl(i)) & \text{if } \#_{uns}(cpl(i)) \neq \bot \\ U^{\times}(i) & \text{if } \#_{uns}(cpl(i)) = \bot \end{cases}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses U^{sum}.
Successful outcome Unit Count (activity granularity)	Q	A	$a \in A$	$SUC^{\times}(a)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(a)} SUC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum}.
Successful outcome Unit Count (case granularity)	Q	C	$c \in C$	$SUC^{\times}(c)$ $\times \in \{sgl, sum\}$	$\sum_{i \in inst(c)} SUC^{\times}(i)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Successful outcome Unit Count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$SUC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\sum_{c \in G} SUC^{\times}(c)$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum}.
Expected Successful outcome Unit Count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eSUC^{\times}(g)$ $\times \in \{sgl, sum\}$	$\frac{SUC^{\times}(g)}{C(g)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum}.
Successful outcome Unit Percentage (activity instance granularity)	Q	I	$i \in I$	$SUP^{\times}(i)$ $\times \in \{sgl, sum\}$	$\frac{SUC^{\times}(i)}{U^{\times}(i)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum} and U^{sum}.
Successful outcome Unit Percentage (activity granularity)	Q	A	$a \in A$	$SUP^{\times}(a)$ $\times \in \{sgl, sum\}$	$\frac{SUC^{\times}(a)}{U^{\times}(a)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum} and U^{sum}.
Successful outcome Unit Percentage (case granularity)	Q	C	$c \in C$	$SUP^{\times}(c)$ $\times \in \{sgl, sum\}$	$\frac{SUC^{\times}(c)}{U^{\times}(c)}$ The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum} and U^{sum}.

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
					$\frac{SUC^{\times}(g)}{U^{\times}(g)}$
Successful outcome Unit Percentage (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$SUP^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUC^{sgl} and U^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUC^{sum} and U^{sum}.
					$\frac{\sum_{c \in g} SUP^{\times}(c)}{C(g)}$
Expected Successful outcome Unit Percentage	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eSUP^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for outcome unit calculations, i.e., it uses SUP^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for outcome unit calculations, i.e., it uses SUP^{sum}.
					$\frac{TC^{\times}(a)}{Cli(a)}$
Total Cost and Client count Ratio (activity granularity)	Q	A	$a \in A$	$TCCliR^{\times}(a)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
					$\frac{TC^{\times}(g)}{Cli(g)}$
Total Cost and Client count Ratio (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$TCCliR^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
					$\frac{TC^{\times}(g)}{Cli(g)}$
Expected Total Cost and Client count Ratio	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eTCCliR^{\times}(g)$ $\times \in \{sgl, sum\}$	The $\times$ indicates that the PPI can take multiple forms, in this case, $\times \in \{sgl, sum\}$. If $\times = sgl$, then the function considers single events of activity instances for cost calculations, i.e., it uses TC^{sgl}; if $\times = sum$, then the function considers the sum of all events of activity instances for cost calculations, i.e., it uses TC^{sum}.
Unwanted Activity count (case granularity)	Q	C	$c \in C$	$UAC(c)$	$\lvert Unwl \cap act(c) \rvert$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Unwanted Activity count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$UAC(g)$	$\lvert Unwl \cap (\cup_{c \in g} act(c)) \rvert$
Expected Unwanted Activity count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$UAC(g)$	$\frac{\sum_{c \in g} UAC(c)}{C(g)}$
Unwanted Activity Percentage (case granularity)	Q	C	$c \in C$	$UAP(c)$	$\frac{UAC(c)}{A(c)}$
Unwanted Activity Percentage (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$UAP(g)$	$\frac{UAC(g)}{A(g)}$
Expected Unwanted Activity Percentage	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eUAP(g)$	$\frac{\sum_{c \in g} UAP(c)}{C(g)}$
Unwanted activity Instance count (case granularity)	Q	C	$c \in C$	$UIC(c)$	$\lvert \{i \in inst(c) \mid act(i) \in Unwl\} \rvert$
Unwanted activity Instance count (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$UIC(g)$	$\sum_{c \in g} UIC(c)$
Expected Unwanted activity Instance count	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eUIC(g)$	$\frac{UIC(g)}{C(g)}$
Unwanted activity Instance Percentage (case granularity)	Q	C	$c \in C$	$UIP(c)$	$\frac{UIC(c)}{I(c)}$
Unwanted activity Instance Percentage (group of cases granularity)	Q	G	$g \subseteq C$: $g \neq \varnothing$	$UIP(g)$	$\frac{UIC(g)}{I(g)}$
Expected Unwanted activity Instance Percentage	Q	G	$g \subseteq C$: $g \neq \varnothing$	$eUIP(g)$	$\frac{\sum_{c \in g} UIP(c)}{C(g)}$

Table C.22: Quality dimension PPI definitions

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Activity and Role count Ratio (case granularity)	F	C	$c \in C$	$ARoleR(c)$	$\frac{A(c)}{Role(c)}$
Activity and Role count Ratio (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$ARoleR(g)$	$\frac{A(g)}{Role(g)}$
Expected Activity and Role count Ratio	F	G	$g \subseteq C$: $g \neq \varnothing$	$eARoleR(g)$	$\frac{\sum_{c \in g} A(c)}{\sum_{c \in g} Role(c)}$
activity Instance and Human Resource count Ratio (activity granularity)	F	A	$a \in A$	$IHRR(a)$	$\frac{I(a)}{HR(a)}$
activity Instance and Human Resource count Ratio (case granularity)	F	C	$c \in C$	$IHRR(c)$	$\frac{I(c)}{HR(c)}$
activity Instance and Human Resource count Ratio (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$IHRR(g)$	$\frac{I(g)}{HR(g)}$
Expected activity Instance and Human Resource count Ratio	F	G	$g \subseteq C$: $g \neq \varnothing$	$eIHRR(g)$	$\frac{I(g)}{\sum_{c \in g} HR(c)}$
Client count (activity granularity)	F	A	$a \in A$	$Cli(a)$	$\lvert\{\#_{cli}(c) \mid c \in C[a \in act(c)]\}\rvert$
Client count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$Cli(g)$	$\lvert\{\#_{cli}(c) \mid c \in g\}\rvert$
Expected Client count	F	G	$g \subseteq C$: $g \neq \varnothing$	$eCli(g)$	$\frac{Cli(g)}{C(g)}$
Directly-Follows Relations and Activity count Ratio (case granularity)	F	C	$c \in C$	$DFRAR(c)$	$\frac{DFR(c)}{A(c)}$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Directly-Follows Relations and Activity count Ratio (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$DFRAR(g)$	$\frac{DFR(g)}{A(g)}$
Expected Directly-Follows Relations and Activity count Ratio	F	G	$g \subseteq C$: $g \neq \varnothing$	$eDFRAR(g)$	$\frac{\sum_{c \in g} DFR(c)}{\sum_{c \in g} A(c)}$
Directly-Follows Relations count (activity granularity)	F	A	$a \in A$	$DFR(a)$	$\cup_{i \in inst(a)} \{act(i') \mid i' \in next(i))\}\|$
Directly-Follows Relations count (case granularity)	F	C	$c \in C$	$DFR(c)$	$\|dfrel(c)\|$
Directly-Follows Relations count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$DFR(g)$	$\| \cup_{c \in g} dfrel(c)\|$
Expected Directly-Follows Relations count	F	G	$g \subseteq C$: $g \neq \varnothing$	$eDFR(g)$	$\frac{\sum_{c \in g} DFR(c)}{C(g)}$
Human Resource count (activity granularity)	F	A	$a \in A$	$HR(a)$	$\|hres(a)\|$
Human Resource count (case granularity)	F	C	$c \in C$	$HR(c)$	$\|hres(c)\|$
Human Resource count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$HR(g)$	$\| \cup_{c \in g} hres(c)\|$
Expected Human Resource count	F	G	$g \subseteq C$: $g \neq \varnothing$	$eHR(g)$	$\frac{\sum_{c \in g} HR(c)}{C(g)}$
Optional Activity count (case granularity)	F	C	$c \in C$	$OptA(c)$	$\|\{a \in act(c) \mid \exists c' \in C[a \notin act(c')]\}\|$

Indicator	D.	G.	F. Par.	Function Name	Formal Definition
Optional Activity count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$OptA(g)$	$\|\{a \in act(c) \mid c \in g \wedge \exists c' \in C[a \notin act(c')]\}\|$
Expected Optional Activity count	F	G	$g \subseteq C$: $g \neq \varnothing$	$eOptA(g)$	$\frac{\sum_{c \in g} OptA(c)}{C(g)}$
Optionality (case granularity)	F	C	$c \in C$	$Opt(c)$	$\frac{OptA(c)}{A(c)}$
Optionality (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$Opt(g)$	$\frac{OptA(g)}{A(g)}$
Expected Optionality	F	G	$g \subseteq C$: $g \neq \varnothing$	$eOpt(g)$	$\frac{\sum_{c \in g} OptA(c)}{\sum_{c \in g} A(c)}$
Role and Variant count Ratio (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$RoleVR(g)$	$\frac{Role(g)}{V(g)}$
Role count (case granularity)	F	C	$c \in C$	$Role(c)$	$\|role(c)\|$
Role count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$Role(g)$	$\|\cup_{c \in g} role(c)\|$
Expected Role count	F	G	$g \subseteq C$: $g \neq \varnothing$	$eRole(g)$	$\frac{\sum_{c \in g} Role(c)}{C(g)}$
Variant Case Coverage (case granularity)	F	C	$c \in C$	$VCC(c)$	$\frac{\|\{c' \in C \| \exists tr \in trace(c')[tr \in trace(c)]\}\|}{\|C\|}$
Variant Case Coverage (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$VCC(g)$	$\frac{\|\{c \in C \| \exists tr \in trace(c)[tr \in variants(g)]\|}{\|C\|}$
Variant count (group of cases granularity)	F	G	$g \subseteq C$: $g \neq \varnothing$	$V(g)$	$\|variants(g)\|$

Table C.23: Flexibility dimension PPI definitions